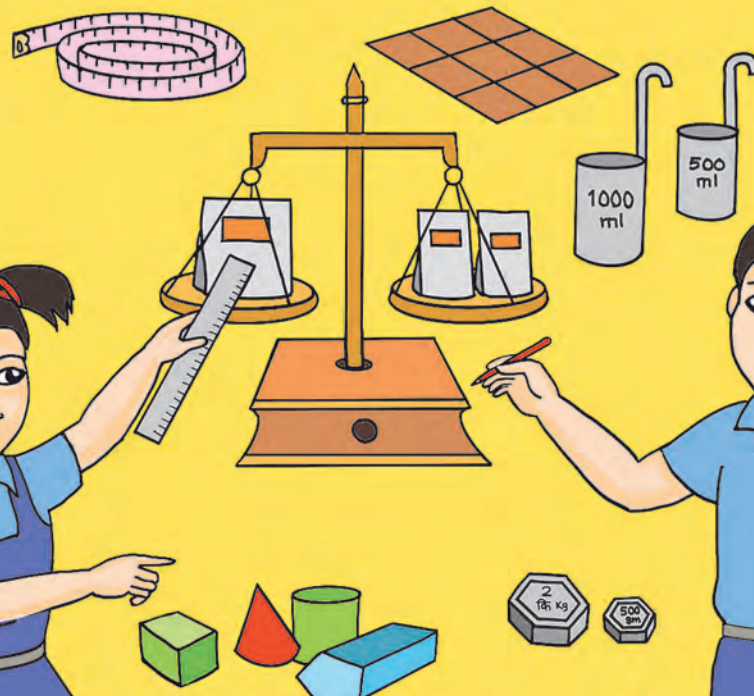




MATHEMATICS

STANDARD FOUR



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गणित इयत्ता चौथी (इंग्रजी माध्यम)



Tele MANAS Program

Tele MANAS (Tele Mental Health Assistance and Networking Across States)
(14416 / 1800-89-14416)

Under the National Mental Health Program, the centrally sponsored scheme Tele MANAS is being implemented in the state of Maharashtra from October 2022. Under the Tele MANAS Program, free 24×7 integrated and comprehensive mental health services are provided through the helpline number 14416. This includes treatment and counselling by experts for various mental problems. The information provided by you after calling the Tele MANAS helpline remains completely confidential, so students, parents and teachers should feel free to call the helpline number and tell them about their problems and doubts and get guidance on it. Tell others to take help from this facility as well.

Manodarpan

The Manodarpan Helpline (8448440632) is a free telephone service that has been made available since July 2020. It is provided by the Ministry of Education through the Manodarpan Cell at the National Council of Educational Research and Training (NCERT) to offer psychological guidance and support to students, parents and teachers. This service is available every day of the week from 8:00 AM to 8:00 PM. Students, parents, and teachers can call this helpline to receive the necessary guidance and counseling.

This textbook aligns with the curriculum and syllabus developed by the State Council of Educational Research and Training (SCERT), Maharashtra, Pune-30 and has been verified by them. The Coordination Committee, established under Government Resolution by GR No.: Abhyas-2116 / (Pra.Kra. 43/16) SD-4 dated April 25, 2016, approved this textbook in its meeting held on April 1, 2026 for use from academic year 2026-27 onwards.

Mathematics

Standard Four



**Maharashtra State Bureau of Textbook Production
and Curriculum Research, Pune - 411004**



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First Edition:2026

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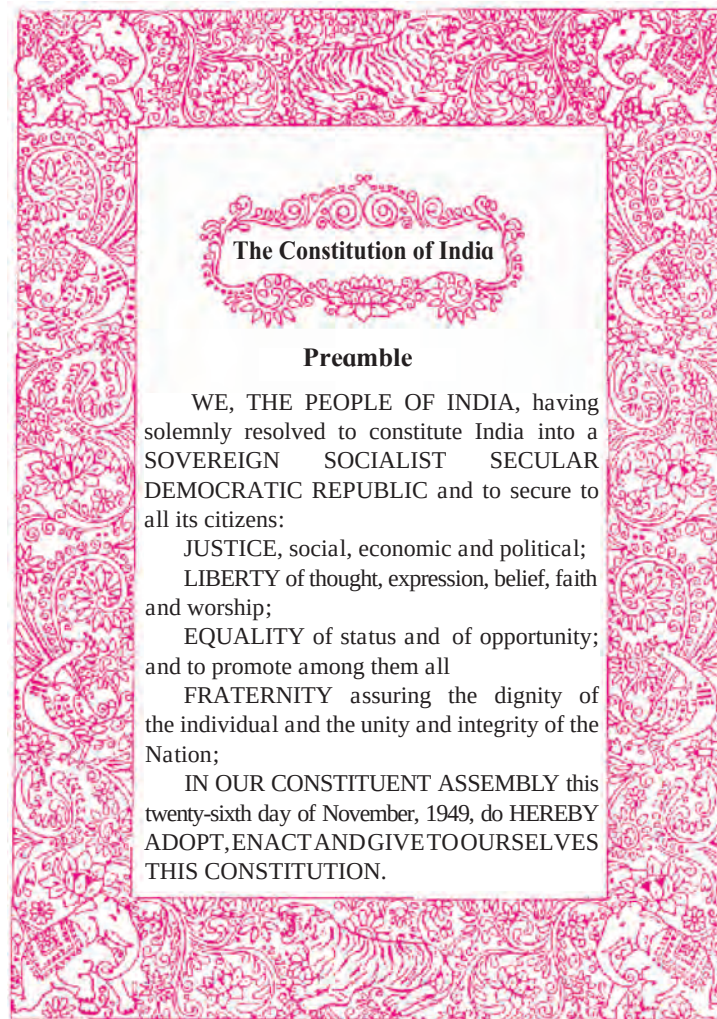
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The Constitution of India

Part IV-A

Fundamental Duties of Citizens

Article 51A

Fundamental Duties - It shall be the duty of every citizen of India -

- (a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem.
- (b) to cherish and follow the noble ideals that inspired our freedom struggle.
- (c) to uphold and protect the sovereignty, unity and integrity of India.
- (d) to defend the country and render national service when called upon.
- (e) to promote harmony and the spirit of common brotherhood amongst all people, transcending religious, linguistic, regional or sectional diversities; to renounce practices derogatory to the dignity of women.
- (f) to value and preserve the rich heritage of our composite culture.
- (g) to protect and improve the natural environment including forests, lakes, rivers and wildlife, and to have compassion for living creatures.
- (h) to develop scientific temper, humanism and the spirit of inquiry and reform.
- (i) to safeguard public property and abjure violence.
- (j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement.
- (k) to provide opportunities for education to one's child or ward between the age of six and fourteen years.

NATIONAL ANTHEM

Jana-gana-mana-adhināyaka jaya hē
Bhārata-bhāgya-vidhātā,

Panjāba-Sindhu-Gujarāta-Marāthā
Drāvīda-Utkala-Banga

Vindhya-Himāchala-Yamunā-Gangā
uchchala-jaladhi-taranga

Tava subha nāmē jāgē, tava subha āsisa māgē,
gāhē tava jaya-gāthā,

Jana-gana-mangala-dāyaka jaya hē
Bhārata-bhāgya-vidhātā,

Jaya hē, Jaya hē, Jaya hē,
Jaya jaya jaya, jaya hē.

PLEDGE

India is my country. All Indians
are my brothers and sisters.

I love my country, and I am proud
of its rich and varied heritage. I shall
always strive to be worthy of it.

I shall give my parents, teachers
and all elders respect, and treat
everyone with courtesy.

To my country and my people,
I pledge my devotion. In their
well-being and prosperity alone lies
my happiness.

राज्यगीत

जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा ॥धृ॥

भिती न आम्हा तुझी मुळी ही गडगडणाऱ्या नभा
अस्मानाच्या सुलतानीला जबाब देती जिभा
सह्याद्रीचा सिंह गर्जतो शिव शंभू राजा
दरी दरीतून नाद गुंजला महाराष्ट्र माझा
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा ॥१॥

काळ्या छातीवरी कोरली, अभिमानाची लेणी
पोलादी मनगटे खेळती, खेळ जीवघेणी
दारिद्र्याच्या उन्हात शिजला
निढळाच्या घामाने भिजला
देश गौरवासाठी झिजला

दिल्लीचे ही तख्त राखितो, महाराष्ट्र माझा
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा
जय जय महाराष्ट्र माझा ॥२॥

Preface

Dear children, teachers, and parents,
I extend my warm greetings to you all.

It gives me great pleasure to present this Mathematics textbook for Class 4, prepared in accordance with the National Education Policy 2020. As per the State Curriculum Framework 2024, it has been suggested that for the preparatory stage from Class 3 to 5, textbooks should be simple and engaging, and the content should not merely be in the form of instructions, but should also provide children with opportunities to work on their own.

While preparing this textbook, the emotional world and preferences of children have been fully considered. Keeping in mind the interest of children in Class 4, attractive pictures have been included. From a mathematical perspective, each unit begins with the use of various objects and situations from the child's emotional world and their surroundings to enrich the content. Various mathematical concepts have been explained clearly in simple language. To ensure children get opportunities to practice and to reinforce understanding, instructions have been provided for teachers. Efforts have been made to make the textbook more engaging through puzzles, Maths Games, and similar activities.

The pictures and examples in this textbook are child-friendly and, moreover, carefully chosen to maintain social balance. An effort has been made to give due priority to all sections of society in the selection of illustrations. The Panchakosha development and 21st century skills mentioned in the National Education Policy 2020 have been integrated into various units of the textbook. The examples in this textbook have been designed to maintain a progressive level of difficulty and diversity. The content of the textbook is aligned with developmental areas, curricular goals and objectives, competencies, and learning outcomes. Activities, games, and stories have been incorporated to make learning enjoyable for children.

Best wishes to all for a bright future!

Pune

Date: 19 April 2026

Indian Solar Date: Vaishakh Shu.2

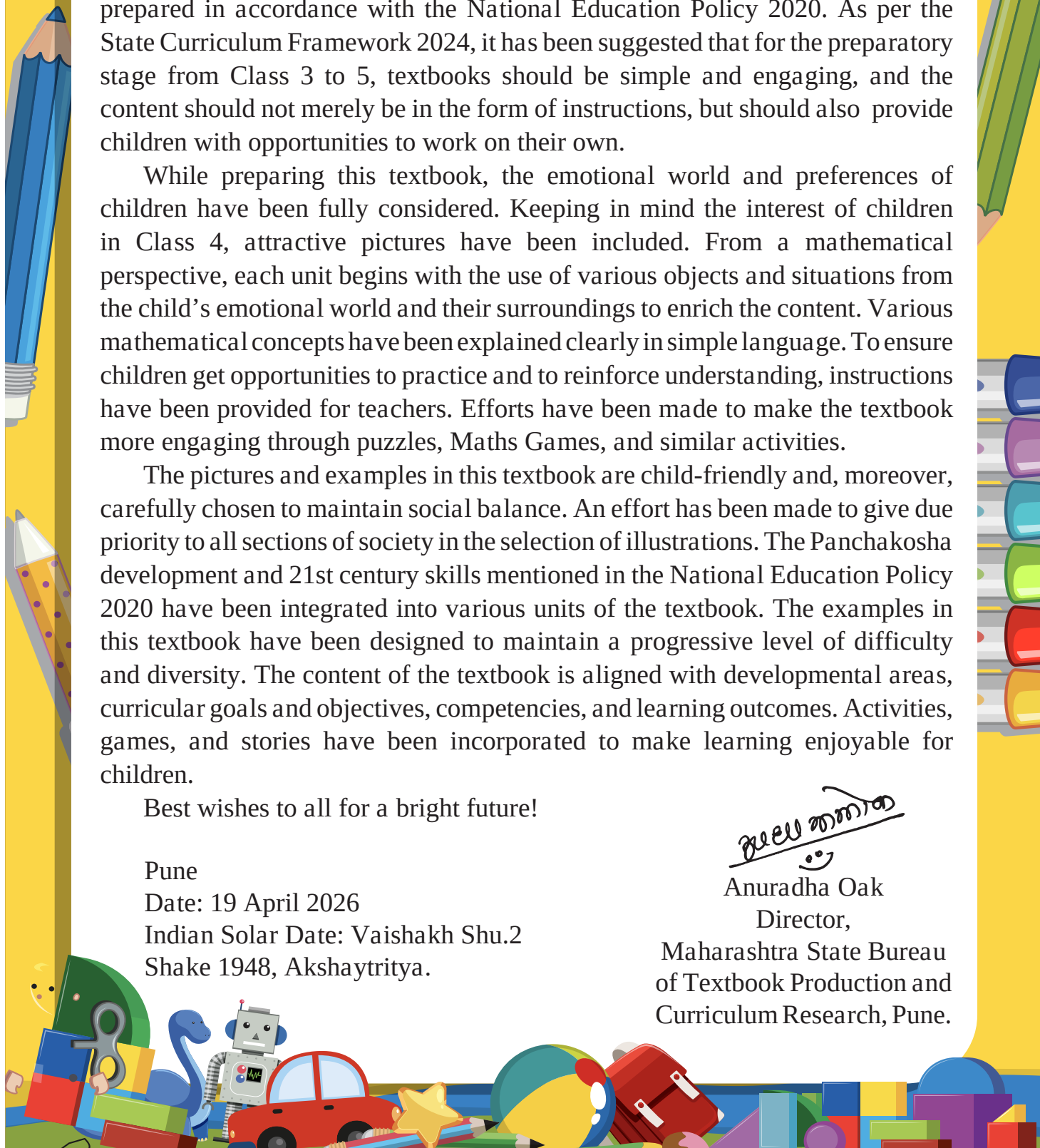
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Anuradha Oak

Director,

Maharashtra State Bureau
of Textbook Production and
Curriculum Research, Pune.



Curricular Goals, Competencies and Learning Outcomes

Learning Competencies	Learning Outcomes	Page Number
Curricular Goal (CG1): Develop understanding/number sense of numbers (counting numbers and fractions). Represent whole numbers using the Indian place value system. Understand and perform the four basic operations on whole numbers. Identify and recognize patterns in number sequences.		
C-1.1 : Uses place value system in the Indian Number System to read and represent numbers, identifies whole numbers, compares them, and reads large numbers.	› Understands the concept of a thousand, expresses the number thousand in various ways. › Recognizes numbers upto 9999. Writes numbers in words. › Tells the face value and place value of digits in a number. › Compares numbers. › Forms numbers according to instructions and arranges them in ascending and descending order. › Indicates curiosity/awareness about the value/use of the number Zero. › Differentiates and classifies even and odd numbers.	1, 2 3, 4, 5, 6 7, 8 9, 10, 11, 12, 13, 14 5, 7, 8, 9, 11, 12 5, 6, 7, 8, 12, 16 15, 16, 17, 115
C-1.2 : Represents commonly used parts of a whole (half, quarter, three-quarters) such as $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$ etc. in fractional form, compares them. (e.g. $\frac{1}{2}$, $\frac{1}{4}$ fractions are shown as parts of whole numbers and their position is shown on a number line)	› Identifies and displays the whole number for its suggested fractional part. › Explains mixed fractions. Uses mixed fractions in daily life. › Writes mixed fractions. › Arranges mixed fractions in a diagrammatic manner. › Compares mixed fractions that are represented in the form of pictures, objects, and numbers.	37 to 41 38 to 41 38 to 41 38 to 41
C-1.3 : Performs operations on numbers, understands and applies their inter-relationships. For example, understands the relationship between addition and multiplication (tables up to at least 10) and uses the four basic operations on numbers to solve everyday problems.	› Forms examples of addition and subtraction. Arranges and solves them. (Answers expected upto 9999) › Forms number bonds upto 100. › Performs transactions upto Rs. 9999. › Forms examples based on multiplication and solves them. To multiply a number upto three digits by a number upto two digits. › Solves and creates division examples. (Dividing a number upto three digits by a two digits) › Explains the concepts of dividend, divisor, quotient, and remainder. › Solves mixed examples.	27, 28, 29 30, 31, 32, 33, 34 29, 30, 31, 32, 33 75 to 83 76 to 78 84, 85
C-1.4 : Identifies, describes, and extends patterns in numbers like odd numbers, even numbers, squares, cubes, powers of two and ten, and Virahanka Fibonacci Numbers.	› Recognizes even and odd numbers. › Creates and extends patterns based on odd and even numbers	15, 16, 17 115, 116, 118
Curricular Goal (CG-2): Analyse the features and properties of 2D and 3D geometric shapes. Specify their positions and describe spatial relationships. Identify and create symmetrical shapes.		

C-2.1 : Identifies, compares, and analyses features of 2D and 3D shapes. Develops vocabulary to describe their shapes and properties.	› Identifies various angles in the surroundings. › Classifies given angles as right angles, acute angles, and obtuse angles. › Compares two-dimensional and three-dimensional shapes. › Explains the concepts of two-dimensional and three-dimensional shapes. › Establishes correlation between two-dimensional and three-dimensional shapes. › Recognizes various shapes and lists. Counts edges, corners and surfaces. › Gives reasoning for the properties while identifying specific two-dimensional shapes. › Finds the two-dimensional shapes that are hidden in other two-dimensional shapes and establishes correlations. For example - $\square =$ › Acquires the skill of freehand drawing of two-dimensional shapes with the help of a pencil. › Draws a freehand circle. Acquires the skill of drawing a circle with the help of tools. › Estimates the circular shape.	18, 19 19, 20 18 21, 22 21, 22 20, 23 23, 24 20, 21 23 26 26
C-2.2: Describes location and position using both spoken language and mathematical vocabulary and understands the concept of a map.	› Identifies the top view, side view, and front view of three-dimensional objects in the surroundings. › Creates appropriate instructions for going to public places. › Checks the map of a public place and discusses whether the design is appropriate or inappropriate. › Creates a visual map (<i>Najri Nakasha</i>) of familiar places. › Places objects in the correct place according to the instructions given in the context.	62, 63 65 66 65 64, 67
C-2.3: Identifies and creates symmetry (translational, reflectional and rotational) in familiar two-dimensional and three-dimensional shapes.	› Recognizes symmetry in various three-dimensional shapes. › Identifies / draws reflective symmetry with the help of mirror reflection.	87 88 to 90
C-2.4: Identifies patterns in two-dimensional and three-dimensional shapes, describes and explains them.	› Identifies the patterns in three-dimensional shapes. › Creates a sample of a pattern that extends with the help of objects and pictures › Creates own patterns using various shapes. › Estimates the tiling for a given area. Verifies the same. › Creates tiling with two different shapes for a given area.	110 to 112 117, 118 111, 112 111, 114 113, 114
Curricular Goal CG-3: To understand the measurable properties of objects, to measure length, weight, area, distance, volume, and time using both non-standard and standard units, and to understand the system of unit of measurement and the process of measurement.		
C-3.2 : Uses appropriate units and tools to measure quantities like length, perimeter, time, weight, capacity, and volume.	› Uses appropriate standardized units and tools to measure a given length. › Estimates length and distance using standard units. › Measures the length of the edges of a closed two-dimensional shape. (Measures the perimeter)	43, 44 44, 45, 47 92

	› Tells the time indicated in a 12-hour clock › Solves examples based to time in everyday life. Solves puzzles. › Uses appropriate standardized units and tools to weigh the object. › Verifies the relationship between the units gram and kilogram. › Interprets the meaning of the words used in daily life, for kilo - a quarter, half, three quarters, one and a quarter kilo and one and a half kilo and correctly converts the units. › Experiences the volume of shapes like cube, cuboid using unit length. › Uses standardized units and tools to measure the capacity of the given liquid.	102, 103, 107, 108 102, 103, 104, 109 51, 54, 56 52 54, 56 101 102, 105
C-3.3 : Converts units between measurement systems, such as centimeters to meters.	› Recognizes the need for unit conversion. › Converts units of measurement between systems. › Divides units of measurement into parts and recognizes them.	45, 46, 47 45 to 48, 52, 53, 104 45 to 48, 52, 53, 104
C-3.4 : Understands the definition of squares and rectangles and the formula for calculating their area in terms of length and width.	› Understands what it means to cover the area. › Calculates the area of simple geometric shapes (rectangle, square) taking a given shape as a unit. › Estimates and verifies how many units will be needed to cover a given area	95 to 98 97 to 100 95, 99, 100
C-3.5 : Estimates and verifies length, distance, time, perimeter, area (regular and irregular shapes), weight and volume using appropriate standard units.	› Estimates the length of an object, the distance between two places etc. Then verifies using actual measurement. › Classifies and compares objects based on length. › Estimates the weight of an object, then verifies it by actual measurement and compares the estimated weight with the actual weight. › Estimates the capacity/volume of a liquid in a container and then verifies it by actual measurement. › Estimates the perimeter of a rectangle or square given the length of its sides and also verifies by actual measurement	45, 46, 47 53, 54, 55 101, 102 99, 100
C-3.6 : Verifies the fact that shapes with same area can have different perimeters, and that shapes with the same perimeter can have different areas.	› Classifies shapes (rectangles and squares) that have the same area among the given shapes. › Classifies given shapes with the same perimeter and calculates the area of those shapes › Discovers that shapes with the same perimeter have different areas.	99, 100 99, 100 99, 100
C-3.7 : Checks for conservation of the measurable attributes related to length and applies it in daily life.	› The measure of length of the object remains the same, though measured by holding it in any direction horizontally, vertically or slanting. › Evaluates the conservation of volume of a liquid using standard units.	46 102, 103
Curricular Goal (CG-4): Develop problem-solving skills to solve mathematical puzzles and real-life problems, and thereby initiate the development of computational thinking.		

C-4.1 : Solves puzzles involving whole numbers with one or more operations and examples based on daily life. This includes word problems, recreational puzzles such as creating a magic square, etc.	› Analyzes the given information and divides it into small stages/steps. › Finds the minimum stages/steps to solve a puzzle/ action/problem. › Reaches the final answers or conclusion with the help of the small steps. › Solves recreational puzzles based on this process. (e.g. mental number recognition, discussion / argumentation with other students in pairs/groups on methods/options for solving puzzles /actions / problems. Understands and enjoys other methods also). › Creates such types of puzzles.	3, 6, 106 3, 64, 67, 86, 126
C-4.2 : Lists all possible permutations and combinations by systematically counting as per the given condition in a simple situation. (e.g., forming a committee of two people from a group of five people.)	› Tells all the possibilities of creating/colouring Rangoli/Arrangement/Patterns/Maps/Pictures etc. according to the given conditions. › Creates all possible combinations of given numbers/ letters/colours by systematically counting them according to given conditions › Arranges given colours/flowers/numbers/letters/ objects etc. according to the given condition (Permutations).	120 to 124 125, 126 120, 122, 124, 125
C-4.3 : Selects appropriate methods and tools as per context for operations on whole numbers. e.g., mental computation, estimation, or using paper and pencil.	› Selects appropriate actions/methods/tools to solve problems in daily life and draws conclusions.	27 to 34, 37 to 42 68 to 86
Curricular Goal (CG-5): Knows and appreciates the development of the Indian decimal place value system, which is now used worldwide.		
C-5.1 : Understands the development of zero in India and the history of the Indian place value system used for writing numbers, its spread across the world, and comprehends its modern impact on our lives and all sciences.	› Appreciates the contribution of Indian mathematicians in the development of zero. › Compares Indian value number systems with other number systems. › Knows the features of the Indian number system.	7, 8 , 12 7, 8, 9
Curricular Goal (CG-6): Develop skills to collect information from daily life experiences, organize it visually, estimate, and use it in daily life.		
C-6.1 : Collects information, describes it and infers meaning, uses pictures/symbols for representing information. Understands scale, prepares pictographs according to scale.	› Creates a chart of the collected information using different pictures. › Understands what is proportion/scale that is needed to create a pictorial chart and hence draws the conclusion or interprets it. › Creates a pictorial chart according to the given scale.	57, 58, 59 60 61



While handling the textbook...

Teachers, Parents and Facilitators should -

- Use materials from the surroundings and from the Maths Kit to clarify all mathematical concepts, activities, projects, practicals and Maths Fun given in the textbook.
- Reinforce concepts by having students do some textbook activities individually, in pairs, or in groups.
- The book includes some new concepts. Expand on them at your level. Study additional contexts based on these concepts. The expansion of specific concepts should be done by teachers at their level as per their teaching plan. For that, guidelines have been provided in the box titled 'For Teachers'.
- Provide students opportunities to apply acquired knowledge and encourage them to think about it.
- Each unit includes learning outcomes. Provide learning experiences to develop outcomes accordingly.
- Instructions in the textbook are guidelines. Make appropriate changes to your teaching-learning method considering local conditions, dialects and local context.
- Instead of just solving examples or questions, students should get higher-level opportunities to create examples or questions themselves. This has been considered at some places in the textbook.
- Try to integrate other subjects while giving learning experiences based on math concepts and principles.
- Emojis have been used in the textbook to prominently highlight the interaction between the teacher and students. Examples: Let's Recall, Let's Understand, I Understood, Do the Activity, Fun with Maths.
- In the process of teaching and learning for special children use sign language as well as complimentary method. Provide opportunities to these students to express themselves in any possible way.

* Index *

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My Country

FOR FREE DISTRIBUTION

The thousand pillared temple of 'Sawir Kabada Basadi' is one of the famous 18 Jain temples in the city of Muddbidri in the state of Karnataka in India. This temple is an excellent example of unique architectural style. This temple is made up of a **thousand** pillars. Each pillar is famous for its attractive carvings. This temple was built by Devaraya Wadiyar, the ruler of Vijayanagar in the 15th century.



Introducing Thousands

Soham: Tai, Ajay and I were playing a game of naming and writing the number next to the given number. We had a lot of fun while playing.

Tai: What fun?

Soham: When I said 100, he said 101 and wrote the number. When I said 599, he said 600 and he wrote that number. But the funny thing is, I said 999 the largest three digit number.

Tai: Then?

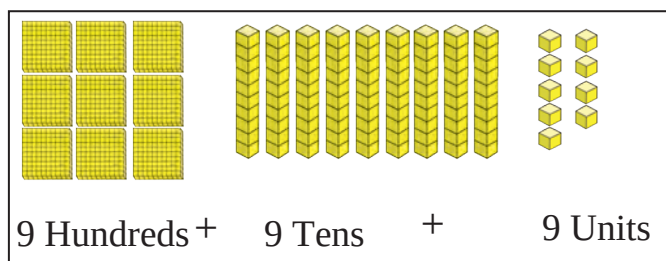
Soham: He said 999 plus 1 but he couldn't write the number. So I won.

Tai: Oh, that number is one thousand.

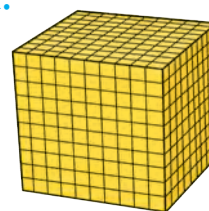
Soham: Tai, I went to a mall yesterday to buy a bat. There, the price of a bat was one thousand rupees.



☉ Let's see how the number 'One thousand' is formed.



and =



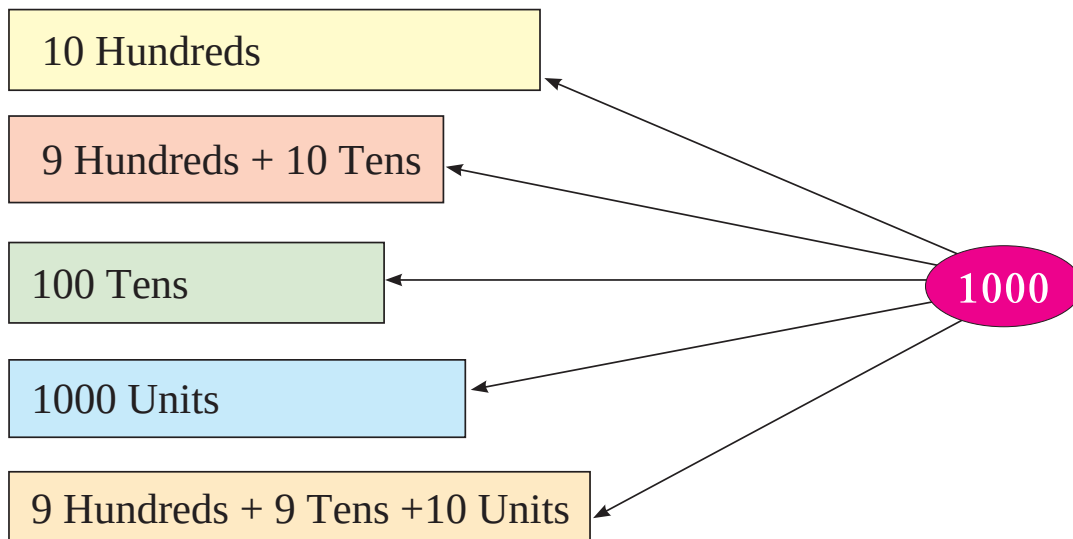
= 1000 units-
form one
cube.

$$\begin{array}{r} 999 \\ + 1 \\ \hline 1000 \end{array}$$

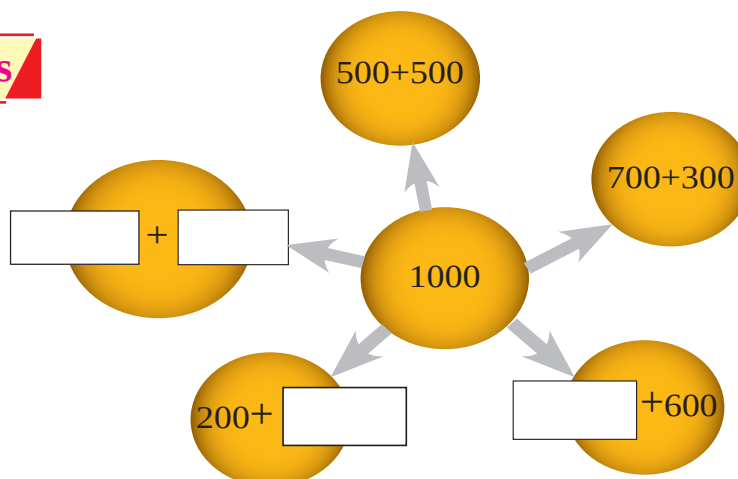
$$999 + 1 = 1000$$

Th	H	T	U
1	1	1	
	9	9	9
			1
1	0	0	0

☉ 'One thousand' in various forms.

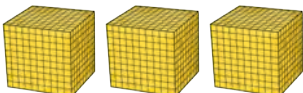
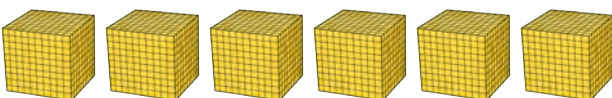
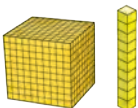
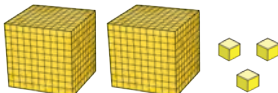
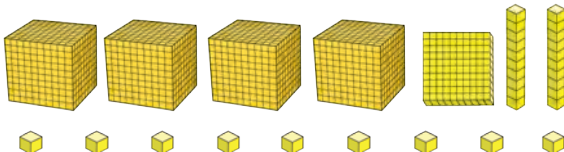


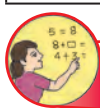
Maths through games



For teachers: Explain to the students how thousand is formed using hundreds board, tens sticks/column, loose blocks or similar material from Math Kit.

☉ Identifying, reading, writing of four-digit numbers

Symbols	No.	Th	H	T	U	Read as
	1000	1	0	0	0	One thousand
	3000	3	0	0	0	Three thousand
	6000	6	0	0	0	Six thousand
	1010	1	0	1	0	One thousand and ten
	2003	2	0	0	3	Two thousand and three
	4129	4	1	2	9	Four thousand one hundred and twenty nine



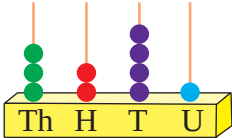
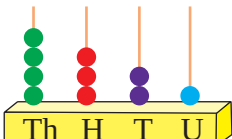
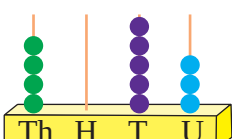
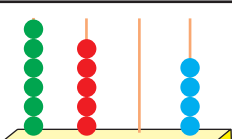
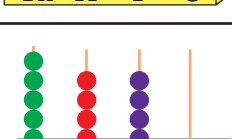
Fun with Maths

- Pick any 3 different digits from 0 to 9? 7, 5, 6
- Form the greatest number using the given digits. 765
- Form the smallest number using the given digits. - 567
- Carry out subtraction of these numbers.
$$\begin{array}{r} 765 \\ - 567 \\ \hline 198 \end{array}$$
- Add this difference to the number obtained after swapping the digits of hundreds and ones place of the difference.
$$\begin{array}{r} 198 \\ + 891 \\ \hline 1089 \end{array}$$
- See what number you get as the answer.
- Repeat this activity by considering different digits, observe the fun in maths each time. See your answer is 1089 every time. Ask this puzzle to your friends and you give the answer.



For Teachers : Students should be given to handle materials from Maths kit. Take practice of forming four digit numbers. Provide information about vehicle numbers in this context.

☉ Complete the following table by identifying the numbers shown by the symbols.

Sr. No.	Symbols	Number in figures	Number in words
1			
2			
3			
4			
5			

Exercise

(1) Write the given numbers in figures.

Sr. No.	Number in words	Th	H	T	U	Number in figures
1	Three thousand five hundred and forty nine					
2	Two thousand seven hundred fifty six					
3	Nine thousand nine hundred ninety nine					
4	Seven thousand and twenty seven					
5	Eight thousand and eight					



For Teacher : While writing a four digit number, first write the digit in thousands place (excluding 0) and then write the appropriate digit from 0 to 9 in the hundreds, tens and unit's place in that order. Student should be given practice of reading and writing four digit numbers. Similarly they should be introduced to the smallest and the largest 4 digit number.

(2) Complete the table.

Sr. No.	Number (Various forms)	No.in figures	Number in words
1	2 Thousands+3 Hundreds+4 Tens +5 Units		
2	5 Thousands+3 Hundreds+8 Tens		
3	6 Thousands+9 Hundreds+8 Units		
4		5150	
5		6789	
6		9007	
7			Five thousand five hundred and one
8			Four thousand one hundred and fifteen

(3) Write the given number as instructed.

Number in figures	Number in words
1100	
1001	
.....	One thousand ten
.....	One thousand

Number in figures	Number in words
6087	
6807	
6780	
.....	Six thousand eight hundred and seventy

(4) Match the pairs appropriately

Number in figures

Number in words

1) 7453

Seven thousand four hundred thirty five

2) 7534

Seven thousand three hundred and fifty four

3) 7435

Seven thousand four hundred fifty three

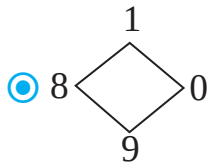
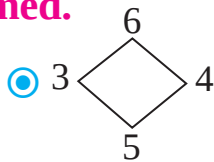
4) 7354

Seven thousand five hundred and thirty four

(5) For the following numbers if the number in words is wrong then correct it and write. If correct put a ✓.

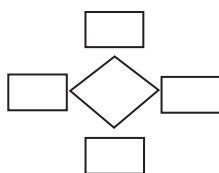
Number in figures	Number in words	Correct number in words
1234	One thousand three hundred and twenty four	
2314	Three thousand two hundred and fourteen	
9009	Nine thousand nine	
5055	Five thousand and fifty	
7777	Seven thousand seven hundred seventy seven	

(6) Using each digit only once, write all the four-digit numbers formed.



.....					
.....					
.....					
.....					
.....					
.....					

Write any four digits of your choice in the boxes. Using each digit only once, write all the four-digit numbers formed.



.....					
.....					
.....					
.....					
.....					
.....					

(7) Write different four-digit numbers formed after interchanging the digits in ones, tens, hundreds and thousands place of the number given on number plate.

Th	H	T	U	In words	
4	1	5	3	Four thousand one hundred fifty three	
					
					
					

Th	H	T	U	In words	
7	9	0	8	Seven thousand eighty nine	
					
					
					



Maths through games

- Have students prepare four sets of digit cards from 0 to 9.
- Teacher should tell a four-digit number.
- The student who forms the four-digit number correctly and immediately, will say 'I got'.
- The student who says 'I got' first, will win the game.



For Teachers : Take practice of forming maximum four-digit numbers using the given four digits. If numbers like 0819, 0189, 0918 are formed, bring it to students notice that these numbers are three-digit numbers.

Let's identify the number

I have two tens, three units, five hundreds and four thousands.

Who am I?

Soham: Four thousands is 4000, five hundreds is 500, two tens is 20 and three units is 3. This number is 'Four thousand five hundred and twenty three'.

Sonali: As per place value, it is arranged as follows.

Thousands	Hundreds	Tens	Units
4	5	2	3

$4000 + 500 + 20 + 3 = 4523$ which is four thousand five hundred twenty three.

- A number has two thousands, three hundreds and five units, identify the number.

Thousands	Hundreds	Tens	Units
2	3	0	5

There is nothing in the tens place of the number, so zero is written in that place. This number is 2305.

Expanded form of the number



Let's Understand

Let's write expanded form of the number 7456

$7 \quad 4 \quad 5 \quad 6$
 $\swarrow \quad \searrow \quad \swarrow \quad \searrow$
 $7000 + 400 + 50 + 6$
 Seven thousand Four hundred Fifty Six

Seven thousands $\longrightarrow 7000$
 Four hundreds $+ \longrightarrow 400$
 Five tens $+ \longrightarrow 50$
 Six units $+ \longrightarrow 6$

Seven thousand four hundred and fifty six

7456

7 Th
 + 4 H
 + 5 T
 + 6 U

7456

$7456 = 7000 + 400 + 50 + 6$ (Expanded form). From the expanded form $7000 + 400 + 50 + 6$, the number can be written as 7456

The expanded form of 4089 is as follows :

4000
 $+ 000$ (Here there is nothing in
 $+ 80$ Hundreds place so let us
 $+ 9$ write 0)

 4089

$4089 = 4000 + 0 + 80 + 9$

Write the expanded form for 5151

5151			
Th	H	T	U
5	0	0	0
+	1	0	0
+		5	0
+			1
5	1	5	1

$5000 + 100 + 50 + 1 = 5151$



For Teachers : Bring student's attention towards the change in the place value of a digit due to presence of zero in the number.



Do the Activity

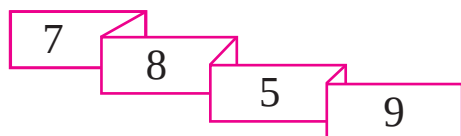
Let's make an educational aid to show a four digit number and its expanded form using a folded strip. Take a paper strip. Mark 10 equal parts, fold along the markings on the strip.



☉ Using this paper strip, show the expanded form of the number 7859.

On the strip made, write the numbers as follows and fold along the bold lines as shown in the picture.

7	000	+	8	00	+	5	0	+	9
---	-----	---	---	----	---	---	---	---	---



Now the number 7859 is seen

7	8	5	9
---	---	---	---

The number is seen on the folded strip, and, its expanded form is seen on the opened strip.

☉ Make the expanded form for the following numbers using folded strips.

☉ 4835

☉ 1909

☉ 4444

☉ 9162

☉ 6067



I Understood

In the number 05, 50, 500, 5000 the place value of 0 is zero. In any number, if there is a zero at any place its place value is always zero.

Exercise

(1) Write the following numbers in expanded form.

- ☉ 4372 =
- ☉ 8092 =
- ☉ 5005 =
- ☉ 5454 =
- ☉ 6666 =



For Teachers :

1

10

100

1000

As shown above, get 36 arrow cards made and cut from students for the digits 1 to 9. Using these arrow cards from Maths Kit, get the practice done for face value and place value.

(2) Write the numbers from their expanded forms.

- ⦿ $4000 + 500 + 90 + 3 =$
- ⦿ $6000 + 80 + 4 =$
- ⦿ $8000 + 600 + 2 =$
- ⦿ $4000 + 400 + 40 + 4 =$

(3) Write the numbers from the given expanded form.

- ⦿ $9 \text{ Th} + 7 \text{ H} + 6 \text{ T} + 5 \text{ U}$
- ⦿ $6 \text{ Th} + 3 \text{ H} + 0 \text{ T} + 4 \text{ U}$
- ⦿ $5 \text{ H} + 4 \text{ Th} + 0 \text{ U} + 3 \text{ T}$
- ⦿ $5 \text{ U} + 7 \text{ T} + 0 \text{ H} + 8 \text{ Th}$
- ⦿ $5 \text{ U} + 5 \text{ H} + 5 \text{ Th} + 5 \text{ T}$

(4) Write the place value for the underlined digit in the given numbers.

- ⦿ 2345
- ⦿ 3150
- ⦿ 9392
- ⦿ 7777
- ⦿ 4708
- ⦿ 5454



Revision

⦿ Complete the following table.

Number before	Number	Number after
499	500	501
3082	3083	3084
	5700	5701
	6789	

Number before	Number	Number after
	1000	
	2071	
	7304	
	7500	

⦿ From the following numbers, ○ the numbers between 3556 and 4556.

3655 3535 4356 4655 3565

⦿ Write the appropriate numbers in the boxes on the number line.



Comparing Numbers



Let's Understand

- Let us decide, which is the smaller or greater number out of 6381 and 6391.



To compare four - digit numbers, first we look at the digit in the thousands place. The greater the digit in the thousands place, greater is the number.

Both 6381 and 6391 have 6 in the thousands place, hence the digits in the thousands place are equal.

Now let us see the digit in hundreds place. Both 6381 and 6391 have 3 in the hundreds place. Hence the digits in the hundreds place are equal.

Now we need to look at the digit in the tens place.

The number 6381 has the digit 8 in the tens place.

The number 6391 has the digit 9 in the tens place.

8 is smaller than 9.

Therefore 6381 is smaller than 6391.

- Between 5461 and 4561, 4561 is the smaller number.
- Between 4331 and 4231, 4331 is the greater number.
- The numbers 7772 and 7772 are equal.

Compare 6381, 6391
In these numbers
6 thousands = 6 thousands
3 hundreds = 3 hundreds
8 tens < 9 tens
Therefore $6381 < 6391$

Exercise

(1) Write the before and after numbers appropriately in the boxes

- ← 2024 →
- ← 3000 →

(2) Use $<$, $>$ sign correctly to compare numbers. In the last boxes, look at the sign and form your own example.

- 4569 5239
- 2068 2089
- $>$
- 7864 7861
- 4005 5004
- $<$

(3) Considering the sign, fill appropriate digits in the blank spaces of the four-digit numbers.

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5 ... 3 < 568 ...

773 ... < 8 ... 2 ...

... 2 ... 1 > 4 ... 3 ...

63 ... 9 > 63 ...

8 ... 6 < 86 ...

91 ... > 9 ...

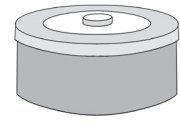
(4) Read the prices of the following objects. Write the correct prices according to the order of signs in the boxes given below them.



₹ 1980



₹ 1099



₹ 1860

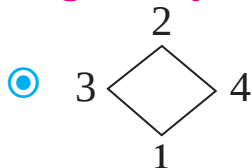
□ < □ < □

□ > □ > □

(5) Complete the following table.

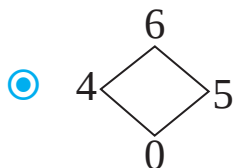
Numbers	Smaller number	Greater number	Numbers	Smaller number	Greater number
4123, 3969			6464, 5397		
2442, 2424			6070, 8089		
8417, 8407			9999, 999		
9456, 9457			7776, 7775		

(6) Form the smallest and the greater number using the following digits only once. For the last example, take digits on your own.



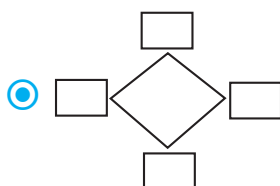
smallest number using the given digits

greatest number using the given digits



smallest number using the given digits

greatest number using the given digits



smallest number using the digits written

greatest number using the digits written

(7) From the grid, find the number as instructed and circle it.

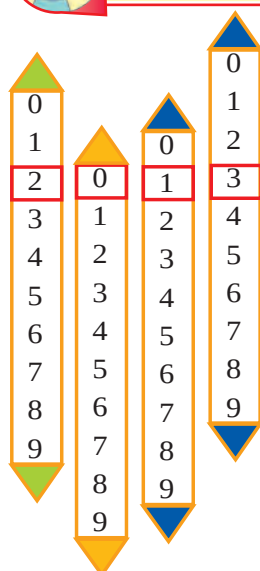
For example Nine thousand one hundred sixty two.

1	2	3	4	8	0	3	9
5	7	2	0	2	5	7	6
2	5	7	6	0	3	8	7
1	9	1	6	2	2	2	2
0	5	0	1	0	1	1	1
1	3	0	1	2	1	1	1
9	4	8	3	6	1	1	1

- 1) Eight thousand three hundred sixty one.
- 2) Find a four-digit number with all four digits same.
- 3) Find the smallest three-digit number in the grid
- 4) Find the largest three-digit number in the grid
- 5) Find the number greater than five thousand and less than five thousand two hundred.



Maths through games



Prepare sliding strips as shown in the adjacent pictures.
Make number strips as per instructions.

	Procedure	Number obtained
1	Show the number after adding 1 to 1969	1970
2	Show the number after adding 10 to 1971	
3	Show the number after taking away 20 from 1975	
4	Show the number after decreasing 2 from 5734	
5	Show the number after adding 111 to 4444	

History of Zero

India has made an extremely important contribution to the discovery of zero and decimal system and its development. This is seen as a revolutionary step in the history of mathematics and its progress from ancient times. In ancient times, Indian mathematicians developed the decimal system in which zero holds great significance. Thus from the time of Aryabhata, the use of zero as a number and the decimal system became widespread in Indian society.

In 499 CE, Aryabhata clearly used zero and the place value system in this work 'Aryabhatiya' in the section on mathematics. In 628 CE, in Brahmasphutasiddhanta, Brahmagupta treated zero as a distinct number and explained the rules of addition, subtraction and multiplication involving it. The concept of zero travelled from India to the Arab world and then spread to Europe. This gave a major impetus to the development of modern mathematics, algebra, computer science and science. The discovery of zero is an invaluable gift India gave the world. Indian mathematicians not only discovered zero but also formulated its mathematical rules, enabling today's modern mathematical system.

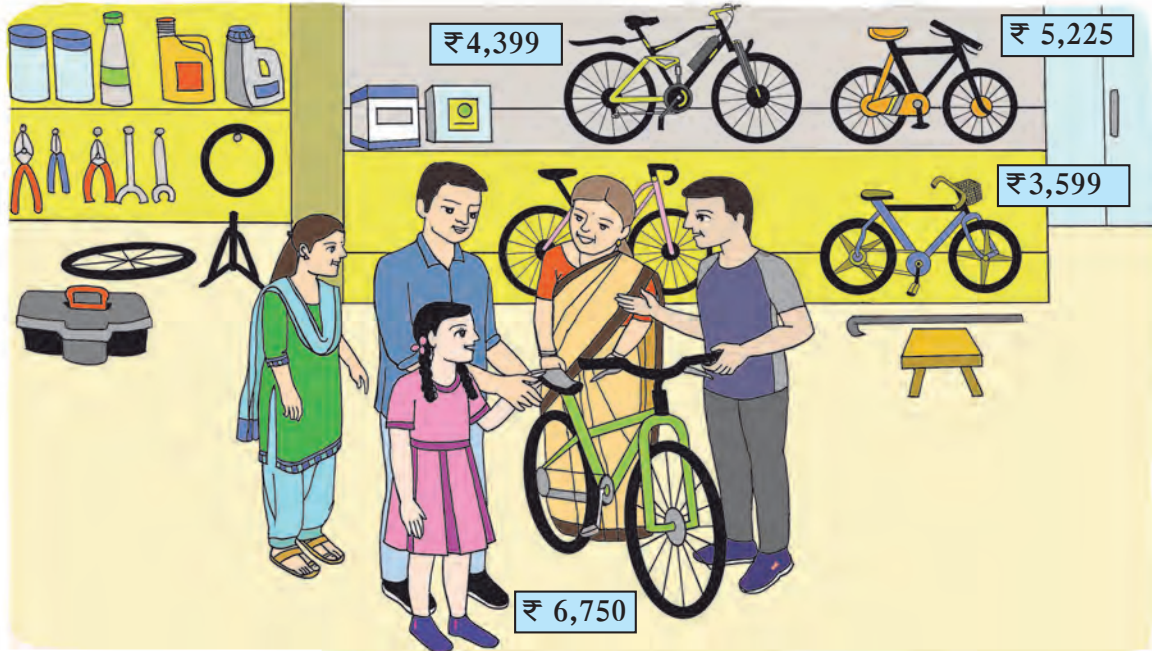


For Teachers : 1) Ask students to make chart as above and form various four-digit numbers 2) Let students create four strips as shown above for playing a game, then rearrange them to make four-digit numbers.

Ascending and Descending order of numbers



Let's Understand



Samruddhi went to a shop with her parents to buy a bicycle. There she saw bicycles of different prices.

In figures

In words

Bicycle with the highest price :

Bicycle with the lowest price :

Write the prices of bicycles in ascending order :,,

Write the prices of bicycles in descending order :,,

Exercise

(1) Arrange the following numbers in ascending and descending order.

Numbers	Ascending order	Descending order
2537, 4537, 3537		
5472, 6792, 4392		
4005, 3005, 7005		

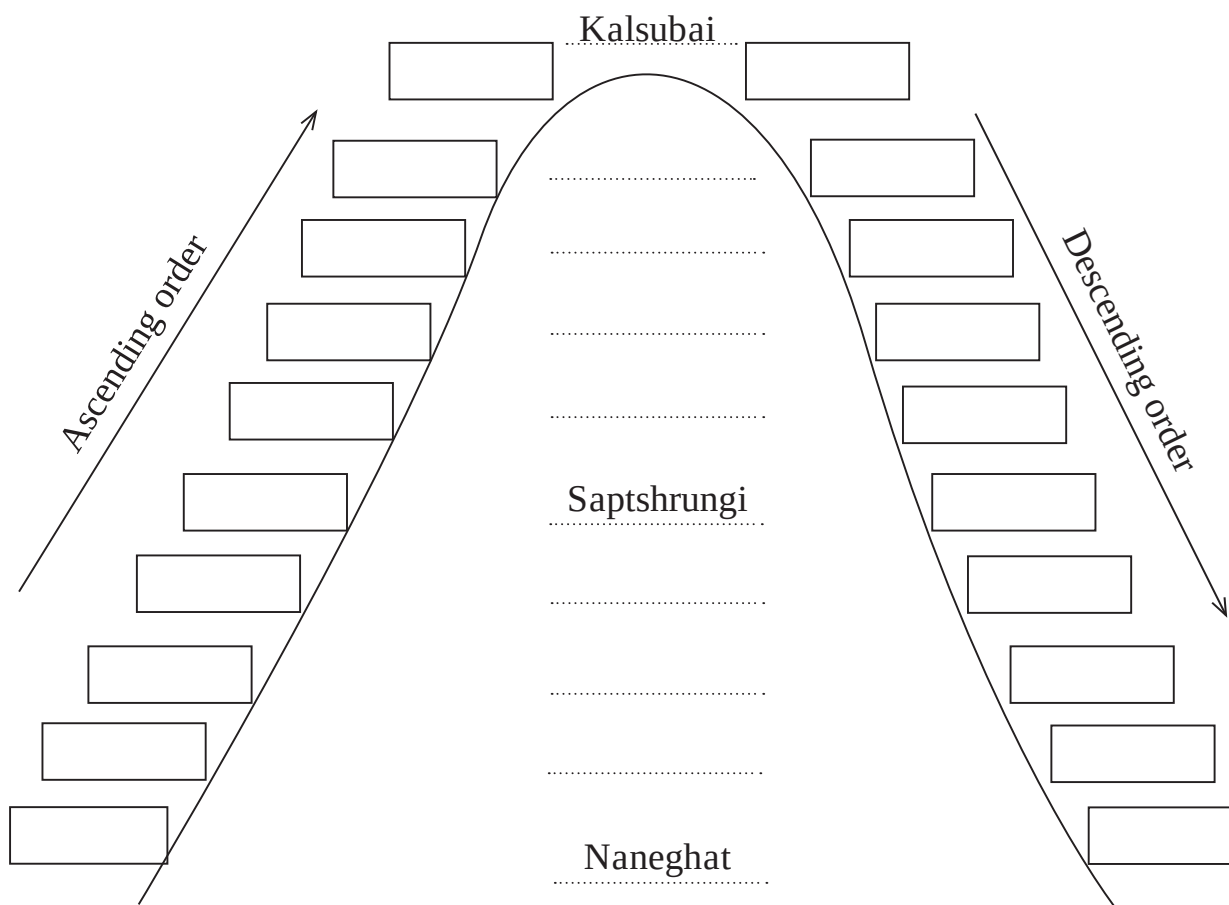
(2) Take any three four-digit numbers of your own choice. Arrange them in descending and ascending order.

Numbers	Descending order	Ascending order
.....		
.....		

(3) In the table below, the altitudes of a few places in Maharashtra are given. Arrange these altitudes in ascending and descending order.

Sr. No.	Places	Altitude (in m)
1)	Salher	1567
2)	Rajgad	1395
3)	Kalsubai	1646
4)	Panhala	1100
5)	Naneghat	860
6)	Saptshrungi	1416
7)	Mahabaleshwar	1488
8)	Chikhaldara	1115
9)	Harishchandragad	1424
10)	Toranmaal	1150

Reference : (A.B. Savadi, The Mega State Maharashtra)



Even numbers and Odd numbers



Let's Understand

I am giving you a few flowers. Put these flowers in groups of two each. Tell me how many flowers are left after making groups.



9 Flowers with Snehal	6 Flowers with Sudhir	8 Flowers with Sameer	4 Flowers with Javed	7 Flowers with George	5 Flowers with Ninad

Snehal: From the 9 flowers with me, I made four groups of 2 flowers each and 1 flower is left.

Sudhir: With the 6 flowers I had, I made 3 groups of 2 flowers but not left with any flower.

Teacher: Which are the numbers that leave no flower after putting them in groups of two?

Snehal: 6, 8, 4

Teacher: Which are the numbers that leave one flower after putting them in groups of two?

Ninad: 9, 7, 5

Teacher: The numbers that leave no flower after making groups of two are called **Even** Numbers and the numbers that leave 1 flower after making groups of two are called **Odd** Numbers.

Distributing large number of objects is time consuming. Distributing a number of objects into 2 equal groups is same as dividing the number by 2. So dividing the given number by 2 can determine whether the number is even or odd.



Let's Discuss

Colour the boxes of the numbers in the table below that are divisible by 2. All these are even numbers. What difference do you notice between the numbers that form groups of two and the numbers that form groups of two but along with a remainder ? Discuss.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Teacher : Numbers that do not leave any remainder when divided by two are even numbers. For example 2, 4, 6, 8, 10, 24 are even numbers. Numbers that leave a remainder of 1 when divided by two are odd numbers. For example, the numbers 3, 5, 7, 9, 11 are odd numbers. Instead of dividing each time, we can determine whether the numbers are even or odd by observing them.

- Digits in the units place of even numbers : 0, 2, 4, 6, 8
- Digits in the units place of odd numbers : 1, 3, 5, 7, 9



Let's Remember

2, 4, 6, 8 are even numbers. If a number has zero in units place, then the number is even.

Even numbers when divided by two, leave a remainder of zero.

Odd numbers when divided by two, leave a remainder of one.

Exercise

- (1) Determine and write whether the given number is an even number or odd number from the digit in the units place.

36, 67, 63, 30, 44, 51, 88, 69, 83, 96, 71, 92, 112, 243, 502

Even numbers :

Odd numbers :

- (2) Take beads as per the numbers given below and group them into pairs. From this, decide if the number is an even number or odd number.

15, 12, 14, 13, 27, 10, 17, 18

Even numbers :

Odd numbers :

- (3) For the numbers given below, ☐ the odd numbers and ☐ the even numbers.

7, 1, 8, 9, 38, 13, 24, 31, 76,

70, 87, 92, 45, 72, 56, 29, 66, 104,

127, 136, 139, 221, 105, 144, 999, 1000, 1111



Fun with Maths



Archana: Shubham, today using maths, I will tell you how many brothers and sisters you have.

Shubham: How is it?

Archana: Do the maths calculations as I suggest.



- 1) First of all, think about the number of siblings you have.
- 2) Now add 2 to it
- 3) Multiply the answer by 2.
- 4) Now add 1 to it.
- 5) Multiply the answer by 5.
- 6) Add the number of sisters to that answer.
- 7) Subtract 25 from the answer you got.
- 8) Now you must have got a two digit number.
- 9) In that number, the first digit is the number of brothers and second digit is the number of sisters.

Shubham: Oh really ! It's perfect!



Angles

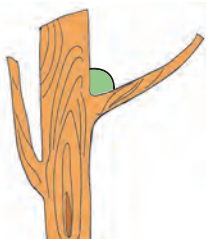


Let's Understand

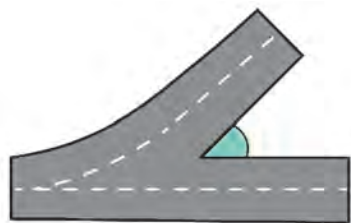
- Look around for the angles formed at different places in your surroundings.



Angle between the two fingers



Angle between the branches of a tree



Where two roads meet, an angle is formed.



An angle is formed between the two blades of a pair of scissors



Angle made by hand



Angle formed in a yoga pose.



An angle is formed between the two hands of a clock.



An angle is formed between the seat and brackets of a chair.



Let's Remember

Wherever two lines meet; an angle is formed.



For Teachers : Discuss with students, the places where we see angles formed in nature, in surroundings.



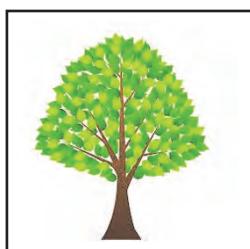
Do the Activity

Activity-1: Mark the places where an angle is formed, in the pictures of the letters below.

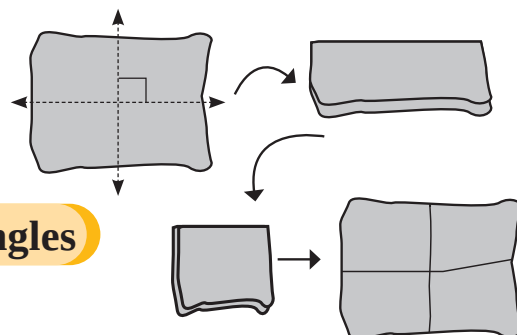
L V M T Z म त

Activity-2 Look around your classroom, observe where angles are made and write it.,,

Activity-3 Look at the pictures below. Observe where angles are made and mark them.



Activity - 4 Fold a paper to make a squared corner as shown in the figure.



Types of Angles



Let's Understand

Verify the types of angles using the squared corner that you made.

1) Right angle :

A squared corner is a **right angle**.



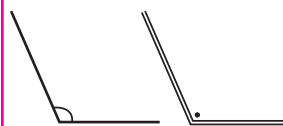
2) Acute angle :

An angle smaller than the squared corner is an **acute angle**.



3) Obtuse angle :

An angle greater than the squared corner is an **obtuse angle**.



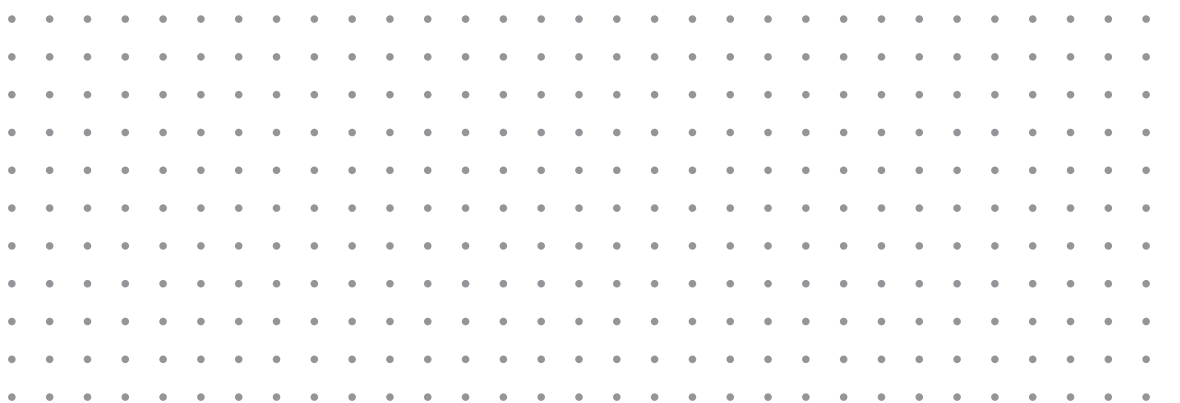
In this way, using the squared corner, you can check whether a given angle is a **right angle**, **acute angle** or **obtuse angle**.

Exercise

(1) Using the squared corner, identify and write the type of the following angles.



(2) Using the dots given below, draw a right angle, acute angle and obtuse angle.

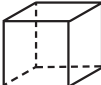
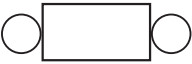
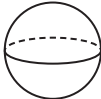
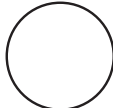


Three-dimensional and Two-dimensional shapes



Let's Understand

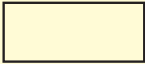
◎ Observe the three-dimensional shapes below and complete the table.

3-D shape	Number of vertices	Number of faces	Number of edges	Related 2-D shape
				
				
				
				
				

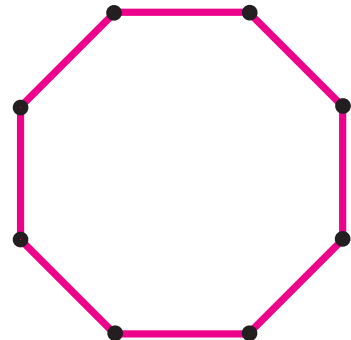
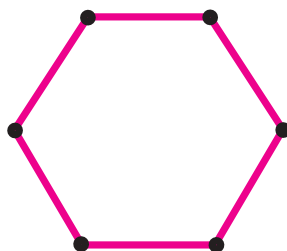
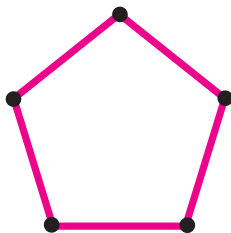
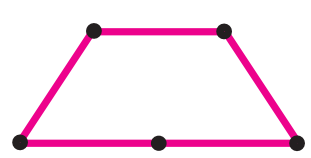
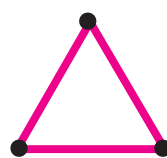
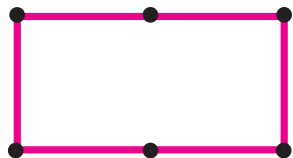
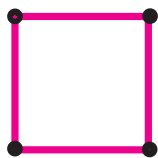
☉ Elements of two-dimensional shapes

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Observe the 2-D shapes and complete the table.

2-D shapes	Number of vertices	Number of sides	Number of angles
			
			
			
			
			
			

☉ The following 2-D shapes are made using Jodo Straws. Count the number of vertices, sides, and angles and then identify the correlation between them.



Features of 2D shapes



Let's Understand

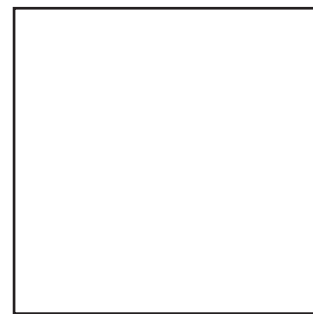
1) Rectangle

- A rectangle has four vertices.
- It has four sides.
- It has four angles.
- The opposite sides of a rectangle are equal in length.
- All four angles of a rectangle are right angles.



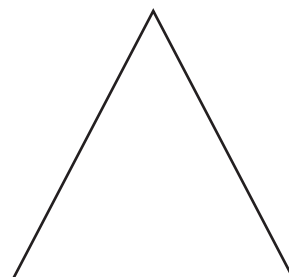
2) Square

- A square has four vertices.
- It has four sides.
- It has four angles.
- All four sides of a square are equal in length.
- All four angles of a square are right angles.



3) Triangle

- A triangle has three vertices.
- It has 3 sides.
- It has 3 angles.
- **A two-dimensional shape has only one face.**
- **A two-dimensional shape can be placed completely on a flat surface.**



Do the Activity

- Try making a triangle with 3 sticks of different lengths.
- Note that 3 sticks don't always form a triangle.
- Try to make different triangles with 4 sticks of different lengths.
- At the most how many squares can be made with 12 sticks of equal length ? Try it out.



For Teachers : Discusses the relationship and difference between 3-D and 2-D shapes. With a rectangular piece of paper, show that opposite sides are equal. With this activity, bring student's attention towards this feature.

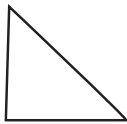


Fun with Maths

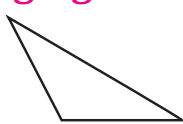
An ant when left on a 2D or 3D object, whether is visible moving till the end, while watching from a point will clarify the concept of 2D and 3D.

Exercise

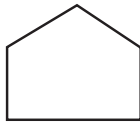
(1) Using the squared corner, write the type of angles shown in the following figures.



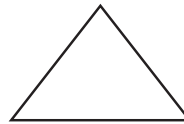
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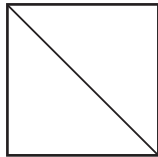


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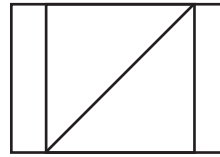
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(2) Find the different 2D shapes hidden in the following 2D figures and write their names

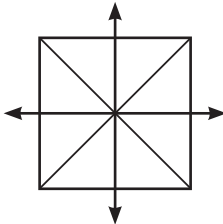


One square
Two triangles

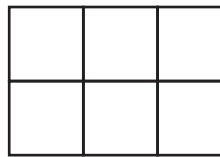
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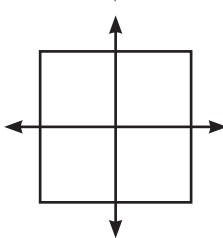
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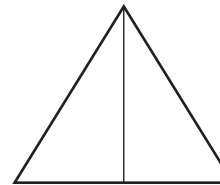
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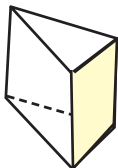
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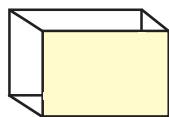
Let's Understand

Prisms and Pyramids

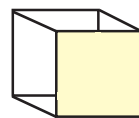
◎ Prism : Some prisms are shown below, make observations about vertical faces.



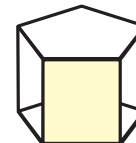
Triangular prism



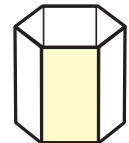
Rectangular prism (cuboid)



Square prism (cube)



Pentagonal prism

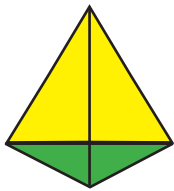


Hexagonal prism

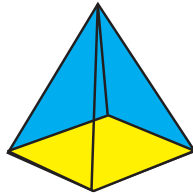
Identify the correlation between the prisms and their names. The top and bottom faces of the prism are identical. The vertical faces of all the prisms are rectangles.

Pyramids

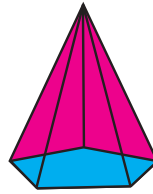
Those 3D shapes whose bases are of different shapes, vertical faces are **triangular** and they meet at a single point at the top, are called pyramids. Some pyramids are shown below. Their type is decided from the shape at the base.



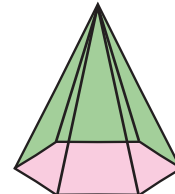
Triangular pyramid



Square pyramid



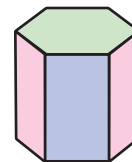
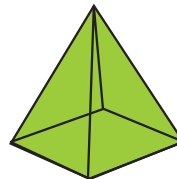
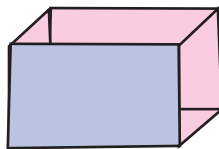
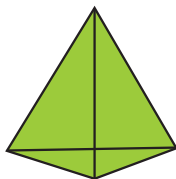
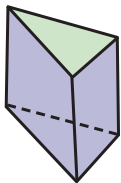
Pentagonal pyramid



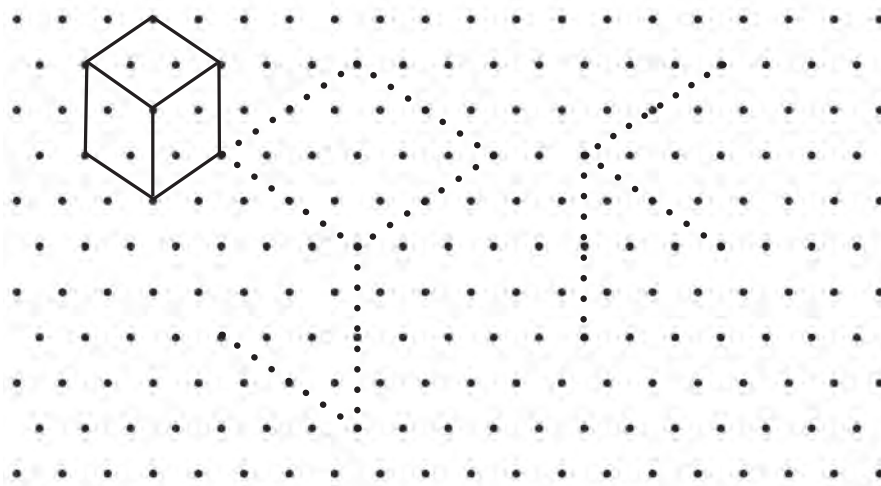
Hexagonal pyramid

Exercise

(1) Observe the 3D shape below, identify whether it is a prism or pyramid and write its number of vertices, edges and faces in your notebook.

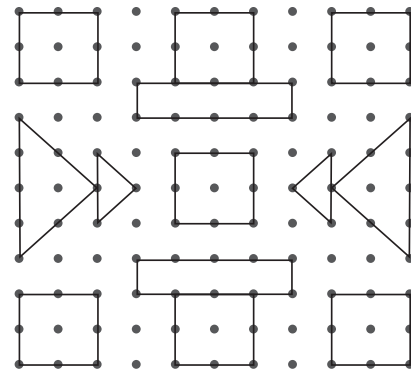
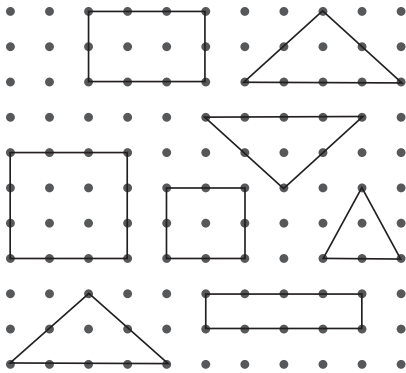


(2) Complete the following figures.

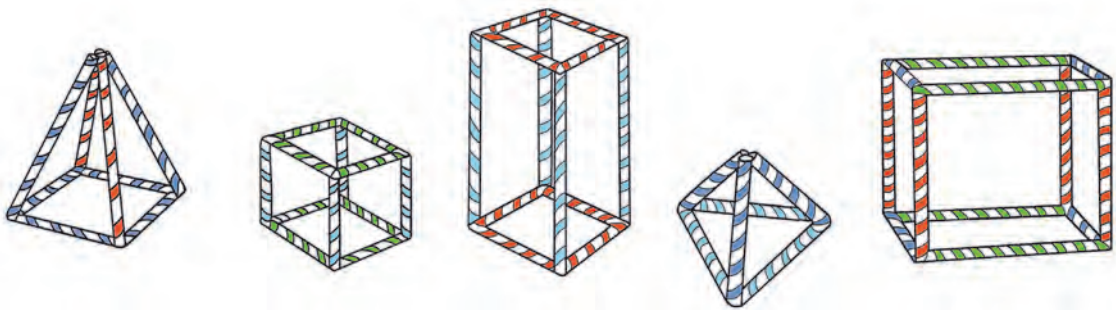


For Teachers : Use examples to explain the difference between prism and pyramid with examples explain the properties, difference and relationship between prism and pyramid. Guide students for making greeting cards using these shapes and drawing 2D shapes in rangolis.

- (3) Identify the shapes formed by joining the dots in the figures. Colour triangles red, squares blue and rectangles green.



- (4) Using Jodo Straws make different prisms and pyramids. Some sample models are shown below. Try making them.



- (5) Draw a triangle, square, rectangle and circle on a paper. Cut them out, paste them in your notebook to obtain a picture as shown and colour it.



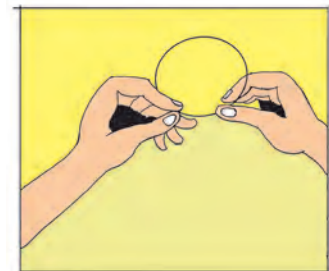
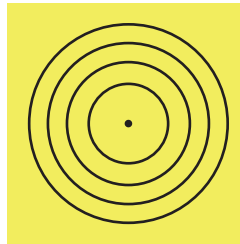
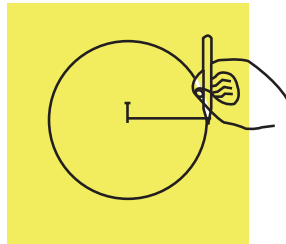
For Teachers : Encourage students to make rectangle, square and triangle using string or geoboard. Guide them to create figures with different shapes which develop logical thinking, creativity and aesthetic sense.



Do the Activity

1) Draw a circle, using a string and chalk.

As shown in the figure, take a piece of cardboard in square shape. Tie one end of thread to the nail and fix it in the centre of the cardboard. Tie the other end of the string to a chalk and rotate the chalk around the nail to draw a circle. Using the same centre, draw several circles of different measures (concentric circles)

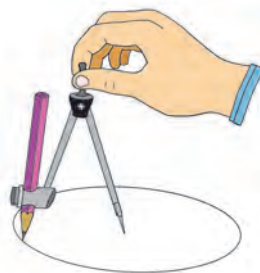


2) Make a circular shape from a piece of wire.

Take a piece of wire. Bend it into a circle as shown.

3) Draw circle using a bangle as shown in the figure.

Use a bangle to trace a circle. Using compass set the appropriate measure and draw circles of different sizes.



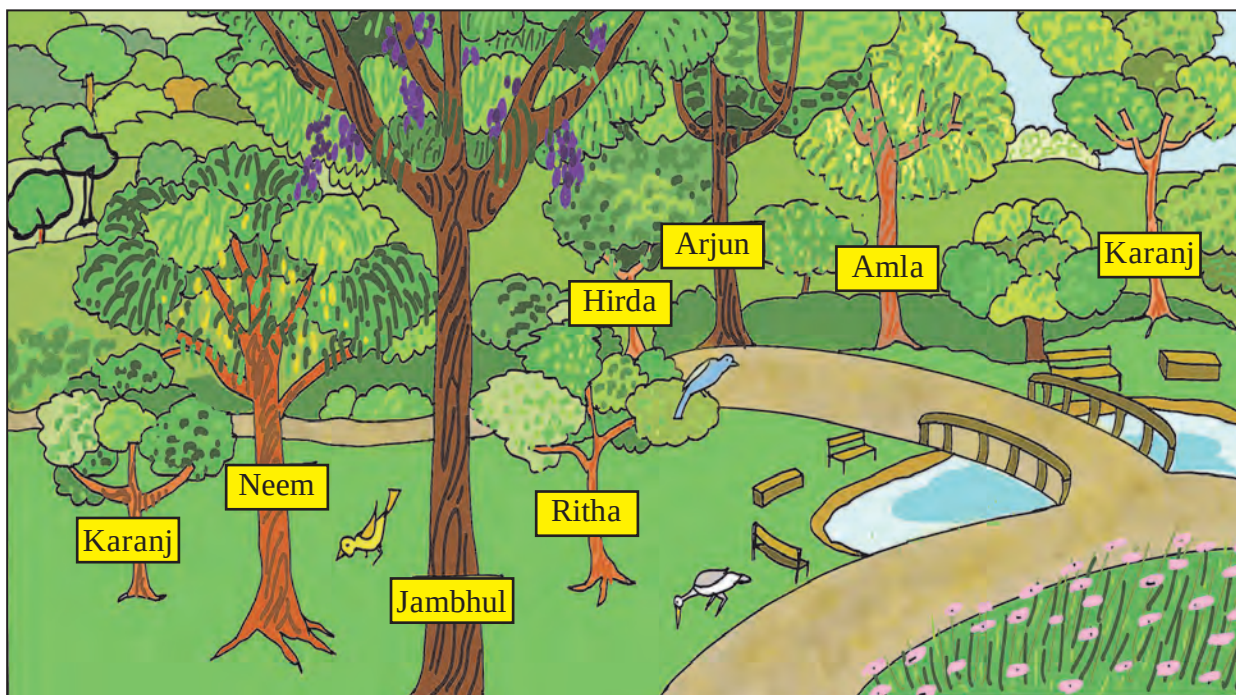
4) Draw circles of different sizes on a page of the notebook as shown. Cut them out and join them to make the given figure.



For Teachers : Different figures are drawn using a string and a stick attached to it. Show these figures to students, guide students to draw circle using different lids and coins. From this emphasise on guessing the size of circle. Give them practice of this. Develop creativity and aesthetic sense associated with circles, concentric circles, intersecting circles and designs.



Let's Recall



In a medicinal plant garden, there are 470 amla, 499 hirda, 749 behada, 275 khair, 350 ritha, 625 arjun, 225 neem, 999 jambhul and 649 karanj trees.

- ⦿ How many amla and hirda trees are there in all?
- ⦿ How many trees of karanj and neem altogether?
- ⦿ How many more jambhul trees are there than arjun trees ?
- ⦿ How many less ritha trees are there than arjun trees ?

Addition

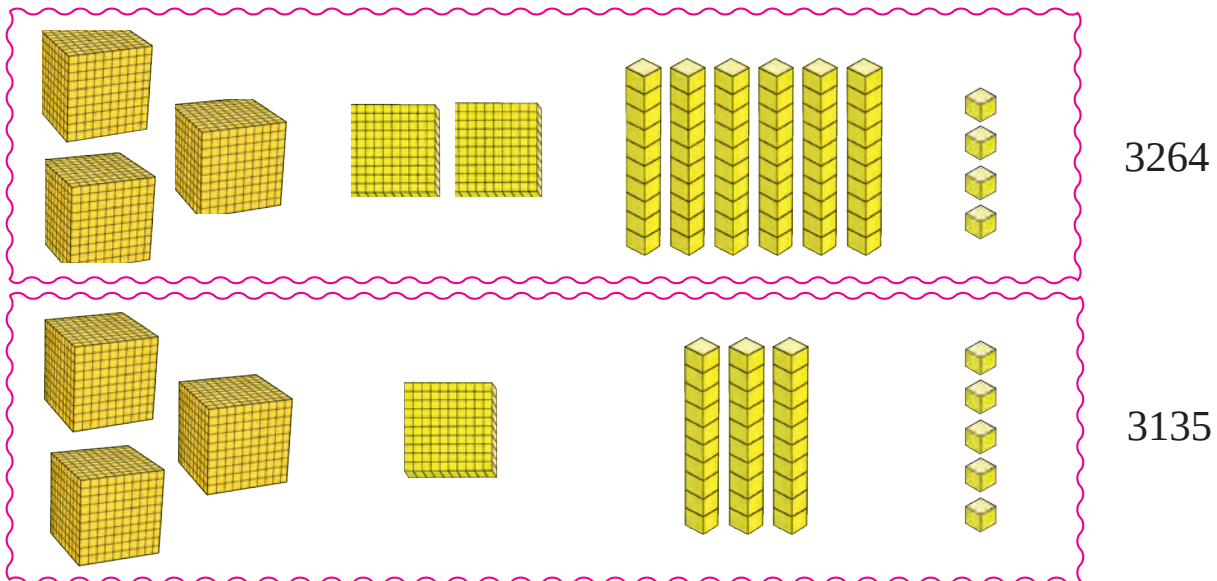


Let's Understand

- ⦿ On the first day of sugarcane harvesting season, the sugar factory produced 3,264 sacks of sugar and on the second day it produced 3,135 sacks of sugar. How many sacks of sugar are produced altogether on both the days?

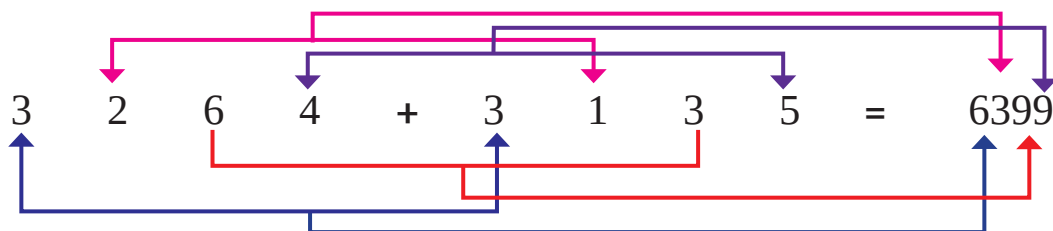
☉ To find the total sacks of sugar produced we need to add.

Pictorial addition model :



Just as we add three-digit numbers, we can add four-digit numbers.

	Th	H	T	U
number of sacks made on first day	3	2	6	4
+ number of sacks made on second day	+ 3	1	3	5
Total sacks made in two days	6	3	9	9



☉ In January 2025, the Regional Transport office registered 3547 four-wheeler vehicles in Kolhapur district and 2695 in Satara district. What is the total number of four-wheelers registered in all in both districts in the month of January 2025 ?

number of vehicles registered in Kolhapur
number of vehicles registered in Satara

Total number of vehicles registered in both districts is **6242** vehicles.

Th	H	T	U
1	1	1	
3	5	4	7
+ 2	6	9	5
6	①2	①4	①2
6	2	4	2

3547
+ 2695
6242

Exercise

(1) Solve the following.

Th	H	T	U
8	7	5	4
+ 1	2	3	5

Th	H	T	U
6	4	0	3
+ 2	0	5	4

Th	H	T	U
5	7	2	0
+ 1	1	2	2
+ 3	1	4	7

Th	H	T	U
7	5	4	6
+ 1	3	6	8

Th	H	T	U
4	9	8	5
+ 1	2	4	7

Th	H	T	U
3	7	5	8
+ 2	1	8	4
+ 3	4	0	3

(2) Arrange vertically and solve.

⦿ $3683 + 2315$

⦿ $4792 + 4206$

⦿ $5768 + 3001$

⦿ $5098 + 3354$

⦿ $5869 + 3452$

⦿ $1267 + 378 + 435$

(3) Solve the following word problems

- ⦿ A factory produced 2354 toys on first day. On the second day 2123 toys were produced. How many total toys were produced on both the days?
- ⦿ In Pokharani village, there are 5134 males and 4765 females. What is the total population of the village ?
- ⦿ The balance of Hirakani women's saving bank account is ₹6718. To this amount ₹2845 is added as the current month's saving. What is the total amount in that account ?
- ⦿ In a government godown, there were 599 sacks of rice. 1888 more sacks of rice were brought in. How many total sacks of rice were there in the godown ?
- ⦿ Children from Sangavi collected funds of ₹3950 in the first week and ₹5980 in the second week for Farmer's Relief Funds. What is the total amount collected?
- ⦿ A school library had 3154 books. The school purchased 1230 more books and received 1140 books as gift. How many total books are there in the library ?



For Teachers : Above examples on addition should be discussed and solved by the students.

Framing an addition word problem from the given information and solving it



Let's Understand

Example : Given Information :

men voters 4167 women voters 4284

Word problem can be framed as follows.

- ⦿ In Sarvade village there are 4167 men voters and 4284 women voters. How many total voters are there in Sarvade village ?

Sarvade village has total 8451 voters.

Th	H	T	U
	1	1	
4	1	6	7
+ 4	2	8	4
8	4	5	1

men voters
women voters
Total voters

Exercise

(1) Frame addition word problems from the given information and solve.

- ⦿ School Saving's Bank
last month's saving ₹ 5869
current month's saving ₹ 3085
- ⦿ Company workers
men workers 1560
women workers 1479
- ⦿ Nursery
mango saplings 4272
chickoo saplings 1864
coconut saplings 2456
- ⦿ price of shirt ₹ 1275
price of trousers ₹ 1899

Subtraction



Let's Understand

- ⦿ For Annual school gathering ₹ 9850 were collected, out of which ₹ 8430 were spent, what is the amount left ?

₹ 1420 left now

Th	H	T	U
9	8	5	0
- 8	4	3	0
1	4	2	0

amount deposited
amount spent
amount left

- ⦿ The population of Songaon is 7436. Out of that 3765 are men, then what is the number of women ?

TH	H	T	U
6	13 ③	13	
7 - 3	4 7	3 6	6 5
3	6	7	1

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Total population
number of men

Songaon has 3671 women

number of women

Exercise

(1) Subtract.

TH	H	T	U
5	4	8	7
- 2	1	6	5

TH	H	T	U
6	8	9	0
- 4	5	7	0

TH	H	T	U
8	2	0	3
- 7	1	0	2

TH	H	T	U
4	6	8	2
- 2	7	6	5

TH	H	T	U
5	8	6	4
- 3	9	7	8

TH	H	T	U
8	0	4	1
-			
1	0	8	9

(2) Arrange vertically and subtract.

⦿ 7456 - 997

⦿ 6479 - 3589

⦿ 7561 - 3943

⦿ 5444 - 2666

⦿ 5400 - 2464

⦿ 8004 - 3108

(3) Solve the following word problems.

- ⦿ The Vidyamandir, Mangoli school collected ₹ 9990 for the picnic, out of which ₹8870 were spent. What is the amount left ?
- ⦿ Palasgaon Primary School received 1950 kg rice. Out of this 1840 kg rice was used for nutri meal. How many kilo grams of rice is left?
- ⦿ The electricity meter reading at Snehal's printing press was

6	4	4	3
---	---	---	---

 on 1st July. On 1st August the meter reading was

7	5	6	5
---	---	---	---

 How many units were used in the month of July ?

- There are 7475 saplings in Vanashree Nursery. Vanraj Nursery has 5967 saplings. How many more saplings does Vanashree Nursery have than Vanraj Nursery ?
- Shamrao received ₹9750 as the milk payment for 10 days from the milk collection centre. He spent ₹5920 on cattle food and other things. What is the amount left with Shamrao ?
- Kanya Vidyalaya collected ₹8972 as flood relief fund. Kumar Vidyamandir collected ₹7964. What is the difference in both fund amounts ?

Frame a subtraction word problem from the given information and solve it



Let's Understand

- Given Information : altitude of Kanchanjunga peak : 8586 metre.

Word Problem : altitude of Kalsubai peak : 1646 metre

- The altitude of Kanchanjunga peak is 8586 metre. The altitude of Kalsubai peak is 1646 metre. Which peak has more altitude ? By how much ?

TH	H	T	U
7	15		
8 - 1	5 6	8 4	6 6
6	9	4	0

altitude of Kanchanjunga
altitude of Kalsubai
difference in altitudes

Kanchanjunga peak is higher than Kalsubai peak by 6940 metres.

Exercise

(1) Frame subtraction word problems from given information and solve.

- Strength of Bololi Centre School • boys 3124 • girls 1113.
- Pranjal's saving account • account balance ₹9442, amount withdrawn ₹3550
- Navegaon Vidyamandir • rice received 4241 kg
• cooked rice 3187 kg.
- Amount with Soham ₹9450. Amount he spent on shopping ₹6865.



For Teachers : Give information to students regarding different peaks in India and Maharashtra. Put them into habit of keeping carry over in mind.

(2) Solve using correlation between addition and subtraction.

6724 + = 9867

- 2794 = 6432

- 3872 = 4777

- = 2345

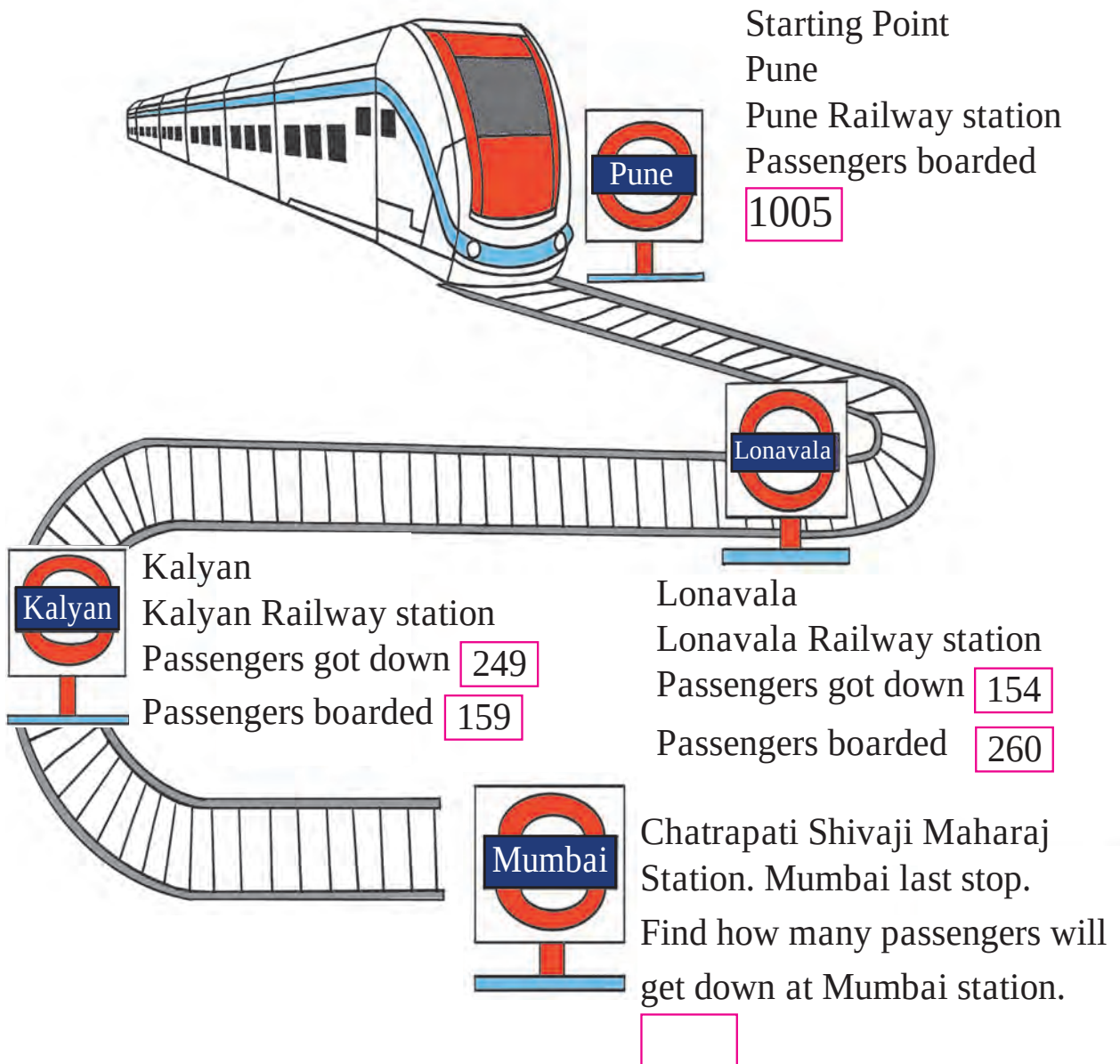
+ 2563 = 7409

5450 - = 2650

+ = 9098

- =

(3) A passenger train starts from Pune for Mumbai. It stops at Lonavala Station first and then at Kalyan station. The number of passengers who boarded and got down is as given in the picture. Solve the following example, from this information.

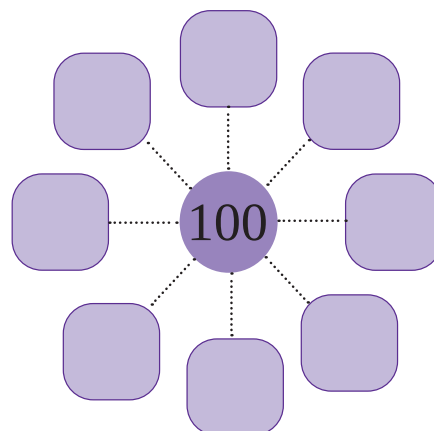
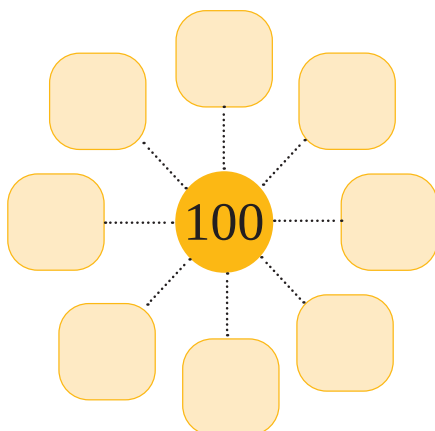
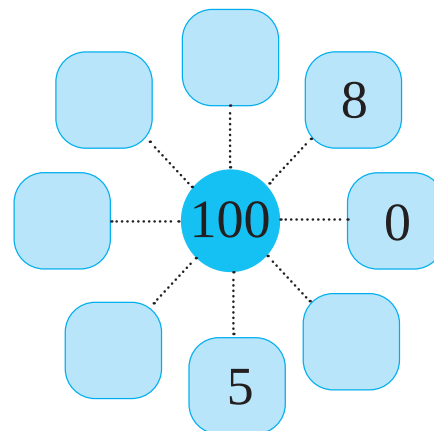
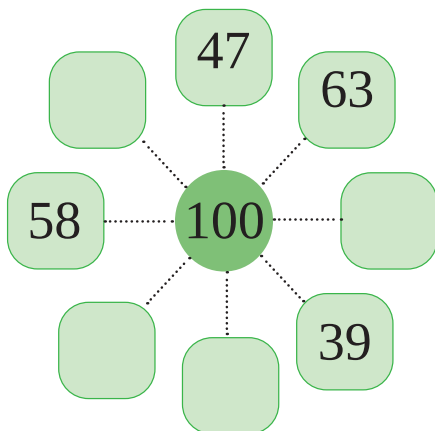
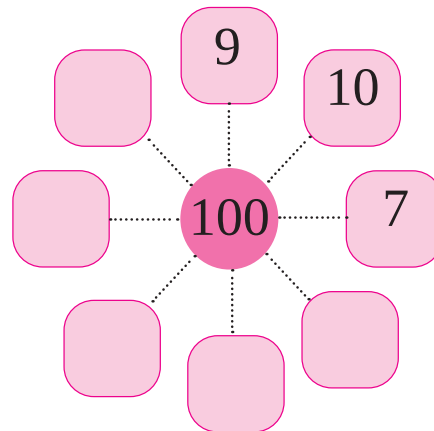
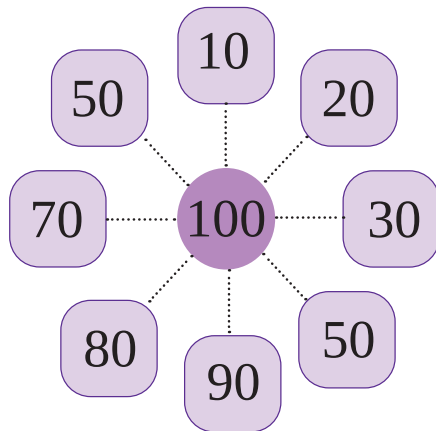


Number bonds



Let's Understand

- Let us make number bonds for 100.
- Fill the boxes of other examples as shown in the given sample.



For Teachers : Teachers should get number bonds till 100 made from students and give them enough practice for it. Make them understand how to use these bonds for addition and subtraction.



Let's Understand

Kaparekar Constant

- Take any four-digit number with distinct digits. For example the number 3524
Number obtained by arranging the digits of the number in descending order = 5432
Number obtained by arranging the digits of the number in ascending order = 2345

- Subtract smaller number from greater number

$$\begin{array}{r} 5432 \\ - 2345 \\ \hline 3087 \end{array}$$

number after arranging digits in descending order : 5432
number after arranging digits in ascending order : 2345

Now let us arrange the digits in the number 3087 in descending order and ascending order.

- Subtract smaller number from greater number

$$\begin{array}{r} 8730 \\ - 0378 \\ \hline 8352 \end{array}$$

number obtained by arranging digits of the answer 3087 in descending order.
number obtained by arranging digits of the answer 3087 in ascending order.

- Subtract smaller number from greater number

$$\begin{array}{r} 8532 \\ - 2358 \\ \hline 6174 \end{array}$$

number obtained by arranging digits of the answer 8352 in descending order.
number obtained by arranging digits of the answer 8352 in ascending order.

$$\begin{array}{r} 7641 \\ - 1467 \\ \hline 6174 \end{array}$$

number obtained by arranging digits of the answer 6174 in descending order.
number obtained by arranging digits of the answer 6174 in ascending order.

Again the number 6174 is obtained.

Repeating this procedure on 6174 any number of times we get the same answer 6174. Note this fact.

Indian Knowledge System

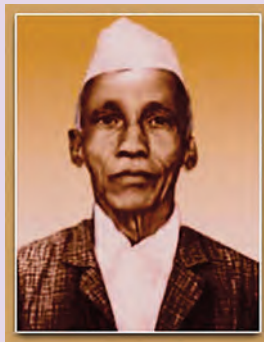
Indian mathematician Dattatraya Ramchandra Kaparekar made a remarkable observation in 1949. Take any four-digit number (in which all digits are not same) and arrange the digits of that number in descending and ascending order and subtract the smaller number from the larger number formed. Repeat this process again and again. After doing this a few times, the remainder of the subtraction is 6174, which is known as the Kaparekar Constant.



For Teachers: Ask students to find the constant for 3 digit number.

Dattatray Ramchandra Kaparekar

Dattatray Ramchandra Kaparekar was born on 17th January in 1905 at Dahanu in Maharashtra. Since school days, he used to enjoy number world than other studies. Solving maths puzzles, mental maths, playing with numbers were his habits due to which he was called 'Number Friend'.



After completing graduation from Pune's Fergusson college, he worked as a teacher in Dahanu, Thane, Nashik. During his college time, he received Rangler R.P. Paranjape Maths Award. He was appointed as Honorary member of Indian Mathematical Society. He used to present his research work every year in this council.

His discovery of constant in Maths subject is famous. 6174 is Kaparekar constant which can be obtained after a certain mathematical procedure. Similarly, Kaparekar discovered 495, a three digit constant. He continued his research with unwavering love for numbers and perseverance. Similarly he also discovered 'Harshad Numbers', 'Vijay Numbers' and 'Magic Square'. He received special recognition for his discoveries at the international level. He died on 4th July 1986 at Nashik.



Let's Discuss

- 1) $5465 + 3245$ and $3357 + 5353$, which of these numbers is greater?
- 2) $3756 + 2534$ and $8022 - 1732$, which of these numbers is greater ?
- 3) Rahul says, $7000 - 2865 = 5235$, is his answer correct ? Check.
- 4) Frame a subtraction word problem, using four-digit numbers.



Do the Activity

Make a tangram.

- Method of making a 5 piece tangram

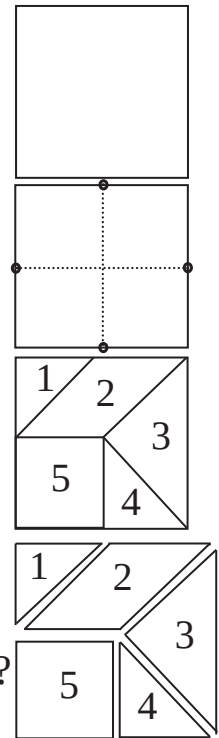
Take a square sheet of paper.

Fold it to find the centre, mark the midpoints of all four sides and draw lines from the centre to these midpoints as shown.

Cut along the lines shown in the figure to get five pieces.

- Using the tangram you made, find the answer for the following questions.

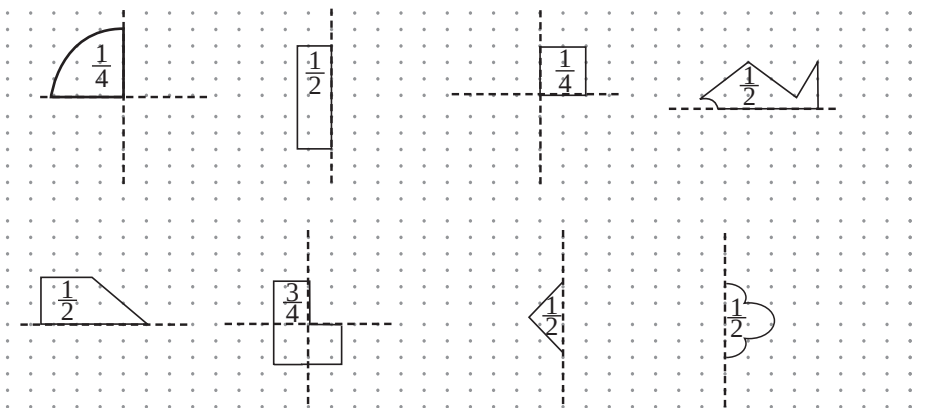
- How many triangles are there in your tangram?
- Are all the triangles same?
- Can you join two triangles to make a square?
- Can you join two triangles to make a larger triangle?
- How many squares are there? How many quadrilaterals?



Do the Activity

- Play a game in groups of making different shapes and designs with the tangram pieces.
- Discuss with friends, siblings and parents how to show various fractions using the tangram. For example piece 5 is $\frac{1}{4}$ of the whole tangram.

- Complete the figure using fractions shown in the following figures.



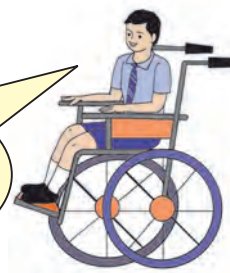
For Teachers : Ask students to make tangrams from paper or cardboard. Colour the pieces as per their wish and discuss concepts like a whole, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, quarter, half, double.



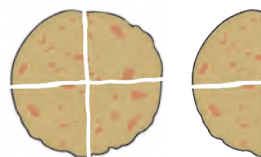
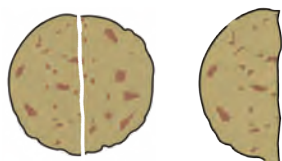
Mixed Fractions

Raju, I made chapati for you. Come, eat hot chapati.

The chapati was very soft. I ate half chapati more. I enjoyed my meal today!



Let's show how much chapati Raju ate in the form of a picture.



One whole chapati and half a chapati
= one and half chapati

$$1 \text{ and } \frac{1}{2} = \frac{3}{2} = 1\frac{1}{2}$$

One whole chapati and half a chapati

$$\frac{4}{4} \text{ and } \frac{2}{4} = \frac{6}{4} = 1\frac{2}{4}$$

One whole is divided into two equal parts.
Therefore denominator is written as 2.

One whole is divided into four equal parts.
Therefore denominator is written as 4.

A fraction larger than a whole is called a mixed fraction.

One and a quarter, two and a quarter, three and a quarter

Common name	Picture form	Notation	Read as
One and quarter		$1\frac{1}{4}$	One and 1 upon 4, One and 1 by 4
Two and quarter		$2\frac{1}{4}$	Two and one upon four
Three and quarter		$3\frac{1}{4}$	Three and one upon four

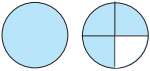
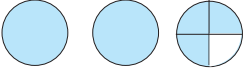
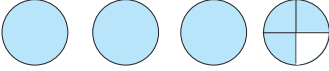
One and half, two and half, three and half

Common name	Picture form	Notation	Read as
One and half		$1\frac{1}{2}$	One and one upon two
Two and half		$2\frac{1}{2}$	Two and one upon two
Three and half		$3\frac{1}{2}$	Three and one upon two



For Teachers : Use concrete objects (eg. chalk, leaves) to reinforce the concept of 'one and half' in fraction.

Quarter to two, Quarter to three, Quarter to four

Common name	Picture form	Notation	Read as
Quarter to two		$1\frac{3}{4}$	One and three upon four
Quarter to three		$2\frac{3}{4}$	Two and three upon four
Quarter to four		$3\frac{3}{4}$	Three and three upon four

☉ Complete the table.

The number of times $\frac{1}{4}$ is taken, has been given. From that show the diagramatic presentation and write its common name.

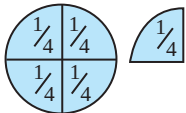
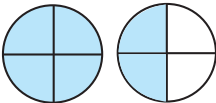
How many times $\frac{1}{4}$	Diagramatic Presentation	Common Name	Notation (fraction)	Numerator/Denominator
2 times				
3 times				
4 times				
5 times				
6 times				
7 times				
8 times				
9 times				



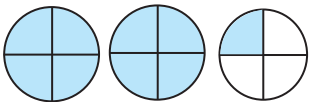
For Teachers : Student should be given practice representing fractions through pictures and objects, common name, reading and writing fractions. Students should be given examples with pictures for identifying fractions with denominator like 3,5,8 etc.

Exercise

(1) Show the given fraction in diagramatic form. Write in numerical form and in words.

In words	Fraction in diagramatic form	How many wholes?	How many parts?	Read as, write as	Common Name
5 quarters		1 whole	$\frac{1}{4}$ part	$1\frac{1}{4}$ 1 and one upon four	One and a quarter
6 quarters					
7 quarters					
3 halves					
3 halves and 1 quarter					

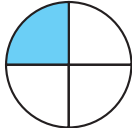
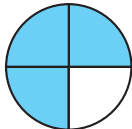
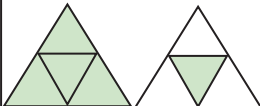
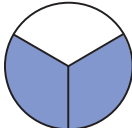
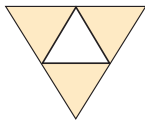
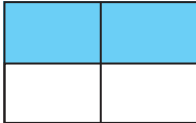
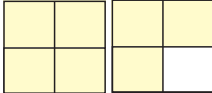
(2) Complete the following table.

Sr. No.	What to show?	Diagramatic form	Mixed Fraction	Means how much ?
1.	One and half days	 one day half day	$1\frac{1}{2}$	24 hours + 12 hours = 36 hours
2.	Two and quarter hours			2hrs + quarter hours 120min + 15min = 135 min.
3.	Three and half years			 months
4.	One and quarter dozen			 pieces

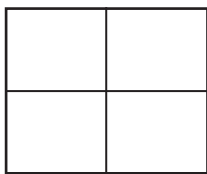


For Teachers : While converting fraction given in diagramatic form to mixed number, bring students' attention towards the number of parts one whole has been made into.

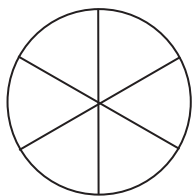
(3) Observe the given diagrams and complete the table.

Diagram	Shaded Part		Unshaded Part		Compare using diagram <, >, =
	Notation	Read as	Notation	Read as	
	$\frac{1}{4}$	One upon four	$\frac{3}{4}$	three upon four	$\frac{1}{4} < \frac{3}{4}$
					
					
	$1\frac{1}{4}$		$\frac{3}{4}$		
					
					
					
					

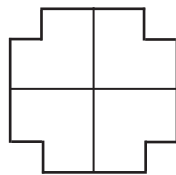
(4) Colour the indicated parts for the given figures.



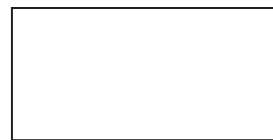
$\frac{2}{4}$ parts



$\frac{4}{6}$ parts



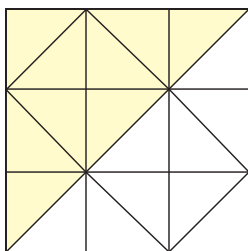
$\frac{1}{4}$ parts

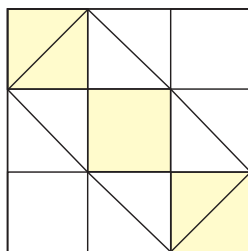


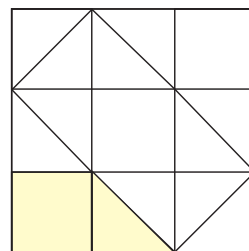
$\frac{3}{4}$ parts

Can you colour $\frac{1}{8}$ parts ?

(5) For the given figures write the coloured part in form of fractions.



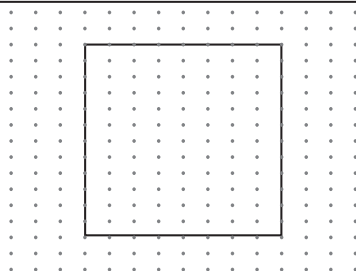




(6) ○ the part of the given set that matches the fraction.

Fraction	Set
$\frac{1}{2}$	
$\frac{3}{4}$	
$\frac{3}{5}$	
$\frac{2}{4}$	

(7) Colour $\frac{3}{4}$ of the given shape



For Teachers : Motivate the students to draw and colour the ○, □, ▭, △ in pairs or groups to show fractions like one and a quarter, one and half, two and a quarter. As a project, list down examples of fractions like one and quarter, one and half, one and three quarters around you after discussing with parents, friends etc.

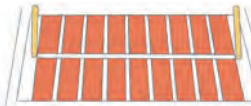


Let's Discuss

To measure the length of the objects on the school ground, correct units have to be used. Decide which units you will use to measure their length.



Write down the units you would use to measure the length of the objects in the following table.

1.		length of the bat
2.		width of the bat
3.		height of the stumps
4.		length of the stick with ghungaroo
5.		length of the spear
6.		distance between two kho-kho pillars

- A discussion should be held in the class about the difficulties that can arise if the distance between the poles of the kho-kho pillars is measured with a measuring scale.

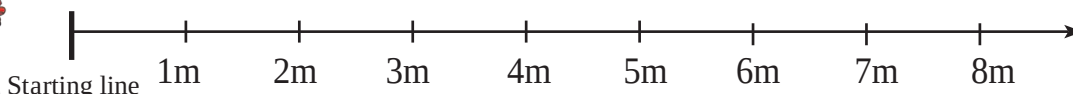


For Teachers : With the help of questions use of units to measure length should be discussed with students. Opportunities should be available to students to measure length, breadth, height, depth, width and distance using material from Maths Kit.



Do the Activity

Children should throw the ball about six metres away from the starting line. Using a chalk, mark the spot where the ball lands first. Using a measuring tape measure the actual distance and record it in the table. **(Put a tick ✓ in the approximate distance column.)**



Sr. No.	names of students	Approximate distance			Actual distance measured
		more than 6 m	less than 6m	6 m	
1					
2					
3					
4					
5					



Let's Remember

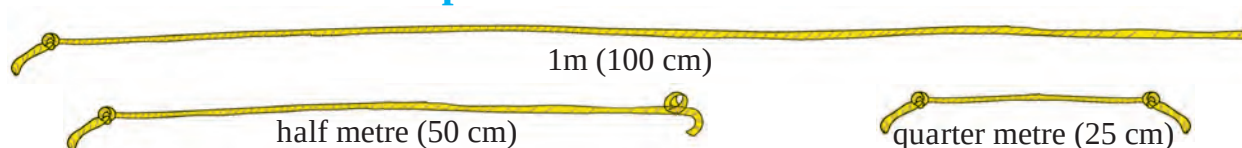
In short **Metre** is written as '**m**' and **centimetre** is written as '**cm**'.



Let's Recall

We have made a metre rope in standard 3. On that metre rope, markings are done like $\frac{1}{2}$ means half metre, $\frac{1}{4}$ means quarter metres.

☉ **Observe the metre rope and discuss.**



- ☉ Half metre put together twice equals one meter.
- ☉ One quarter metre put together four times equals one metre.
- ☉ One quarter metre taken three times equals three quarters of a metre.
- ☉ Half metre and one quarter metre together makes three quarters of a metre.
- ☉ One quarter metre and three quarters of a metre joined together equals one metre in length.



For Teachers : Play long jump game. Ask students to first write estimated length and then measure the actual distance. How correct is the approximation as the actual? From this the skill of correct approximation should be developed.

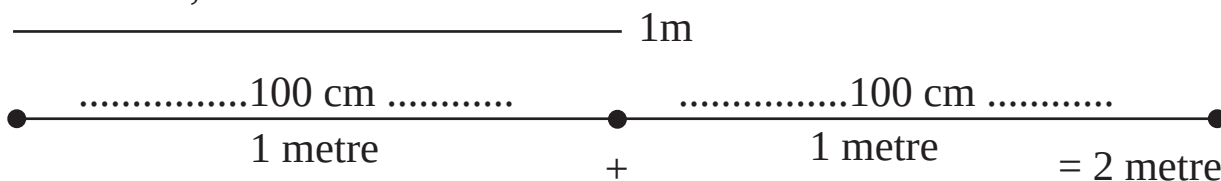
Metre-Centimetre



Let's Understand

☉ Convert metres into centimetres.

We know, $1\text{m} = 100\text{ cm}$



If two ropes of length 100 cm each are joined together, their total length is 200 cm (2 metres).

☉ Let's understand the conversion of metres to centimetres.

1 m = 100 cm

2 m = $2 \times 100\text{ cm} = 200\text{ cm}$

3 m = $3 \times 100\text{ cm} = 300\text{ cm}$

4 m = $4 \times 100\text{ cm} = 400\text{ cm}$

Exercise

☉ Write the correct answer.

☉ $5\text{ m} = 5 \times 100\text{ cm} = \dots\text{ cm}$

☉ $\dots\text{ m} = 8 \times 100\text{ cm} = \dots\text{ cm}$

☉ $7\text{ m} = 7 \times \dots\text{cm} = \dots\text{ cm}$

☉ $\dots\text{ m} = 6 \times 100\text{ cm} = \dots\text{ cm}$

☉ $9\text{ m} = 9 \times \dots\text{ cm} = \dots\text{ cm}$

☉ $15\text{ m} = 15 \times 100\text{ cm} = \dots\text{ cm}$

☉ $\dots = \dots \times \dots\text{ cm} = 1700\text{ cm}$

☉ $19\text{ m} = \dots \times \dots\text{ cm} = \dots\text{ cm}$

☉ Estimate the length and breadth of your classroom. Then actually measure the length and breadth and complete the table.

Sr. No.	Students Name	Approximate distance(Units)		Actual distance (Units)	
		length	breadth	length	breadth
1					
2					
3					
4					
5					



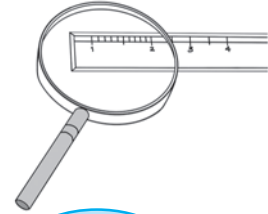
For Teachers : Ask students to make a rope 1m long. Ask to cut the rope into two equal parts. Then cut each of those parts into two equal parts, making four. When four parts are joined end to end, the length is 1m. Ask to verify this. Also, ask students to estimate and measure the lengths of objects like tables, verandas and doors. Let them measure each others height.

Centimetre-Millimetre



Do the Activity

☉ Measure the length of the die, pencil and eraser.



The length of the above die is one centimetre



The length of the pencil is six centimetres



The length of the eraser is greater than two centimetres but less than three centimetres



I noticed that 0 to 1 cm has been divided into ten equal parts

0 1 2 3 4 5 6 7 8 9 10
| | | | | | | | | |
0 1



Let's Understand

☉ Each one centimetre unit is divided into 10 small, equal parts.

That means, every centimetre unit is divided into 10 small equal parts.



When one centimetre is divided into 10 equal parts, each of these small parts measures one millimeter in length, thus, **1 centimetre = 10 millimetre**. The length of an eraser is 2 cm 7mm, it can also be said as 27 mm.



☉ Conversion of centimetre to millimetre.

● $1 \text{ cm} = 1 \times 10 \text{ mm} = 10 \text{ mm}$

● $2 \text{ cm} = 2 \times 10 \text{ mm} = 20 \text{ mm}$.

● $3 \text{ cm} = 3 \times 10 \text{ mm} = 30 \text{ mm}$.

● $4 \text{ cm} = 4 \times 10 \text{ mm} = 40 \text{ mm}$.



Let's Remember

1 centimetre = 10 millimetres

Exercise

(1) Convert centimetres into millimetres.

- ⦿ 5 cm = $5 \times \dots\dots$ mm = $\dots\dots$ mm.
- ⦿ 7 cm = $7 \times \dots\dots$ mm = $\dots\dots$ mm.
- ⦿ $\dots\dots$ cm = $\dots\dots \times 10$ mm = 90 mm.
- ⦿ 12 cm = $\dots\dots \times 10$ mm = $\dots\dots$ mm.

(2) Estimate the lengths of the objects given in the table and then measure and write their actual lengths.

Item Name	Approximately			Actual Measurement
	more than 1cm	less than 1cm	1 cm	
grain of wheat				
length of rajma seed				
length of a chilly				
thickness of a notebook				
thickness of a compass box				
thickness of a floor tile				



Let's Think

- ⦿ What things would you measure in millimetres? For example, the length of a tooth, the thickness of a book etc.



Let's Observe

- ⦿ Observe the picture alongside and discuss.



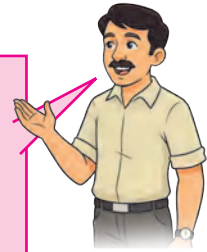
For Teachers : Explain to students using various examples, that when the length of objects is small, it is measured in millimetres.

Kilometre-Metre



Sir, the signboard reads 30km, 23km, 24km, 4km. What does this 'km' stand for?

'km' stands for kilometre. Kilometre (km) is the unit used to measure the distance between two places.



According to the given signboard, the distance from Jamb (Budruk) to Mukhed is 23km. The unit kilometre is used to measure the distance between two villages or cities. **'Kilometre'** is written in short as **'km'**.

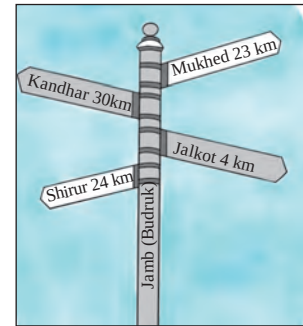
Observe the figure and fill in the blanks:

Distance from Jamb (Budruk) to Shirur = km

Distance from Mukhed to Shirur = km

Distance from Jamb (Budruk) to Kandhar = km

Distance from Jamb (Budruk) to Jalkot = km



Let's Remember

- 1 km = 1000 m
- Half km = $\frac{1}{2}$ km = 500 m
-km = $\frac{1}{4}$ km = 250 m
- Three quarters km = $\frac{3}{4}$ km = 750 m

Conversion of kilometres to metres.

- | | |
|--|---------------------------------------|
| 1 km = 1×1000 m = 1000 m | 7 km = $\times 1000$ m = m |
| 2 km = 2×1000 m = 2000 m | 4 km = $\times 1000$ m = m |
| 3 km = 3×1000 m = 3000 m | 5 km = $\times 1000$ m = m |
| ... km = $\times$ m = 6000 m | km = 8×1000 m = m |

Exercise

(1) Convert the following.

- | | |
|----------------------|----------------------|
| 25 metres = cm | 15 m = cm |
| 5 cm = mm | 4 km = m |
| 7 km 500m = m | 9 m 50 cm = cm |



For Teachers : Initiate a discussion among the students regarding how and where units such as millimetres, centimetres, metres and kilometres are used to measure distance (length).

(2) Match the pairs

⦿ Select the correct answer from group 'B' and write it in front of Group A in the space given below.

Group A

- 1) 3 km (.....)
- 2) 17 cm (.....)
- 3) 19 m (.....)
- 4) 50 mm (.....)
- 5) 12 m (.....)
- 6) 3 cm (.....)

Group B

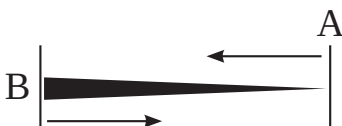
- 1) 170 mm
- 2) 1200 cm
- 3) 3000 m
- 4) 30 mm
- 5) 1900 cm
- 6) 5 cm



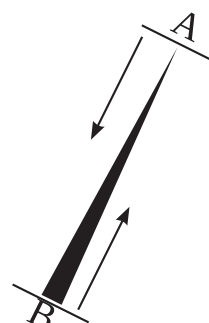
Fun with Maths



Vertical



Horizontal



Slanting

Measure the length of the given objects using appropriate measuring tools by positioning the objects vertically, horizontally or slanting and record your observations.

Object	length of the object held vertically	length of the object held horizontally	length of the object held slanting
pen			
pencil			
stick			
book			



I Understood

The length of any object remains the same, regardless of whether it is measured vertically, horizontally or at a slant using standard units.



Let's Discuss

- 1) Shubhangi has ropes measuring 10m, 6m, and 1m in length. Write down how she can use them to measure distance of 15m and 5m?
- 2) How many bricks would be needed if the bricks of 25 cm each, are placed end to end on the road of length 5 km? How to find this?
- 3) A stick is cut into 10 equal parts. Each part is 30 cm long. What is the total length of the stick in metres?

Indian Knowledge System

Verse 5 of 'Lilavati' written by the Indian mathematician Bhaskaracharya, explains the units used in ancient times to measure distance.

*8 jav =1 Angul, 24 Angul =1 Hasta, 4 Hast =1 Dand,
2000 Dand =1 Krosha, 4 Krosha =1 Yojan*

Neeraj Chopra

Neeraj Chopra, is an Indian Javelin thrower who won India's first gold medal in Javelin throw at the Tokyo 2020 Olympics on 7 th August 2021 with a throw of 87.58 metres.

Neeraj Chopra is the first track and field athlete to win a gold medal. He was born on 24 December 1997 in Panipat, Haryana. He was honoured with the Padma Shri award by the Government of India in 2022.



For Teachers : Give more information about Neeraj Chopra's achievements to students.

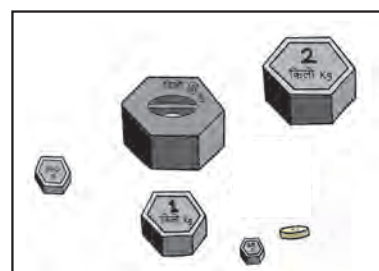
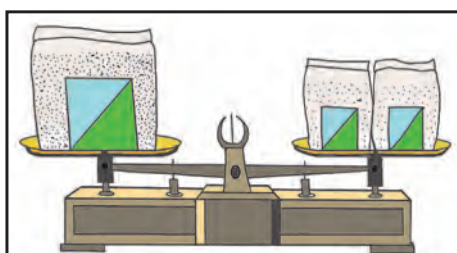
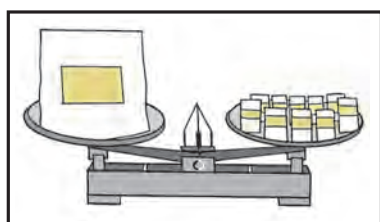
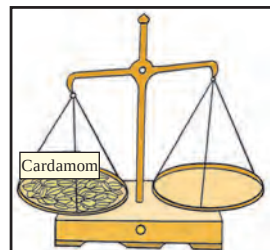
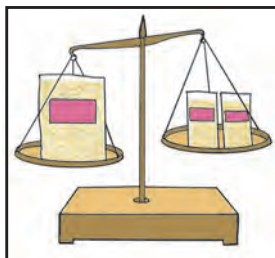
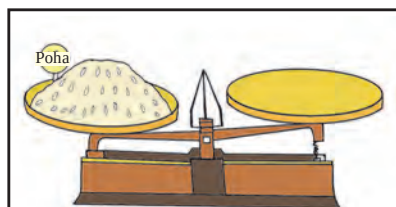
Weight

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Let's Understand

Observe the following pictures and discuss. Use appropriate weights and measures to balance the weighing pans.



There is 3 kg of puffed rice in one pan, and I am going to use a 2 kg weight and 1 kg weight to weigh it.

$2 \text{ kilograms} + 1 \text{ kilogram} = 3 \text{ kilograms}.$



I will use 3 weights of 1 kilogram for it.
 $1 \text{ kilogram} + 1 \text{ kilogram} + 1 \text{ kilogram} = 3 \text{ kilograms}.$



For that, even when I use 1 weight of 2 kilograms and two weights of 500 grams, still the weighing pans will balance

$2 \text{ kilograms} + 500 \text{ grams} + 500 \text{ grams} = 3 \text{ kilograms}$



That's right ! You can measure 3 kilograms using different appropriate combinations of weights.



For Teachers : Teachers should make the above items available to students and give them hands- on experience of measuring weight.

Mirabai Chanu

Mira Chanu is an Indian weightlifter. She won a silver medal in the women's 49 kg weight category at the 2020 Tokyo olympics by lifting 202 kg. weight.

She is the first Indian woman to win a silver medal in weightlifting at the olympics. She was born on 8th August 1994 in East Imphal, Manipur. She was honoured with the Padma Shri award and the Major Dhyanchand Khel Ratna Award by the Government of India in 2018.



I am able to lift kg of weight.



Let's Understand

◎ Conversion of kilograms to grams.

• By addition

You already know, $1 \text{ kg} = 1000 \text{ g}$

$2 \text{ kg} = 1000 \text{ g} + 1000 \text{ g} = 2000 \text{ g}$

So, $1000 \text{ g} + 1000 \text{ g} + 1000 \text{ g} + 1000 \text{ g} + 1000 \text{ g} = \dots\dots\dots \text{ kg}$.

• By multiplication

$4 \text{ kg} = 4 \times 1000 \text{ g} = 4000 \text{ g}$

$3 \text{ kg} = 3 \times \dots\dots\dots \text{ g} = \dots\dots\dots \text{ g}$

$5 \text{ kg} = 5 \times \dots\dots\dots \text{ g} = \dots\dots\dots \text{ g}$



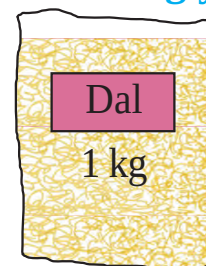
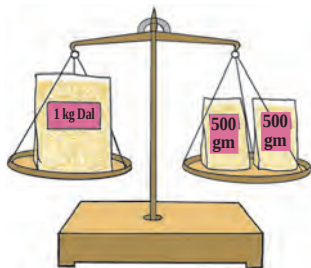
Do the Activity

◎ Game of Weight

In this game, students guess their own weight. Ask your friend to guess your weight. Then you guess your friend's weight. Then measure the actual weights. Find the difference between the estimated weight and the actual weights. Fill all this information in the table. (This activity should be conducted in pairs of students)

Sr. No.	Estimation of your own weight by yourself	Estimation of your weight by your friend	Actual weight	Difference between estimated weight and actual weight.	
				Your's	Friend's
1	28 kg	30 kg	29 kg	less by 1 kg	more by 1 kg
2					
3					
4					
5					

1) Observe the pictures and perform the activity, accordingly.



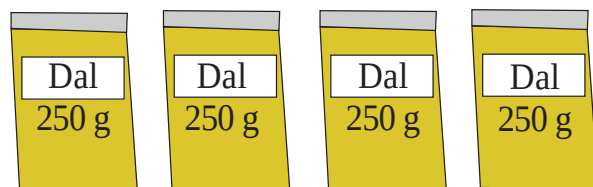
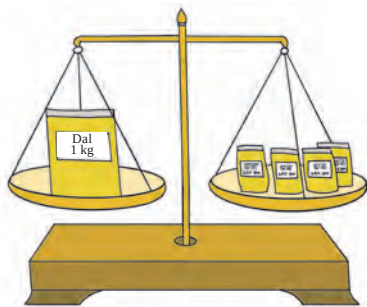
Weight of one packet of 500 g = half kg

Weight of two packets of 500 g = 500 g + 500g = 1000g = 1kg

$$\text{half kg} + \text{half kg} = 1 \text{ kg}$$

- 1 kg dal on weighing pan
weight of four packets of 250 g = 250 g + 250 g + 250 g + 250 g = 1000 g = 1 kg
weight of one packet of 250 g = quarter of a kilogram.

$$250 \text{ g} = \frac{1}{4} \text{ kg} = \text{quarter kg}$$



I Understood

- Quarter kg = 250 g
- half kg = 500 g
- three quarters of a kg = 750 g
- one kg = 1000 g
- one and a quarter kg = 1kg 250 g
- one and half kg = 1 kg 500 g
- one and three quarter kg = 1kg 750 g
- two and quarter kg = 2 kg 250 g

- (9) Two and half kg = 2 kg 500g (10) Two and three quarter kg= 2kg 750g
 (11) Three and a quarter kg=3kg 250g (12) Three and half kg = 3 kg 500g



Do the Activity

1) Help Ankita to bring the grocery.

Ankita take this list of items and ask the storekeeper to keep the grocery ready



List of grocery

- 1) rock sugar - quarter kg
- 2) dried coconut - half kg
- 3) poha - one and quarter kg
- 4) moong dal-one and half kg
- 5) rawa-one and three quarter kg
- 6) sugar- two and quarter kg
- 7) jowar- two and half kg
- 8) rice - two and three fourth kg

I know, a quarter kilo and a half kilo. But, I wonder, how kilo and a quarter, one and half kilo, two and a half kilos or two and three quarter kilos are measured!



Review the grocery list, discuss it and complete the table.
(you may decide the combination of measures)

list of goods	weights/measurements	weights in grams	fractional weights
rock sugar	200 g +50 g	250 g	Quarter kg
dried coconut	200 g +200 g + 100 g		
poha	1 kg + 200 g + 50 g		
moong dal			
rawa			
sugar			
jowar			
rice			



For Teachers : Let's say, if 10 packets of 100 g are weighed, then it is 1000g means 1 kg. Let the students practically experience this. Also in older days, we used traditional units to measure weight. Eg. chatak (छटाक)- a small unit of weight. Adhpav- half of pav.

2) Ask your parents how much grocery does your family need for a month and complete the table.

Item	weight (kg)	Item	weight (kg)
1) wheat flour		4) sugar	
2) rice		5) salt	
3) toor dal		6) ground nut	

3) Estimate the weights of given objects and put a ✓ in the appropriate column. Later write down the actual weights after measuring it.

Sr. No.	fruits and vegetables	estimated weights			actual weight
		more than 1 kg	less than 1 kg	1 kg	
1	7 cucumbers				
2	4 potatoes				
3	8 tomatoes				
4	10 onions				
5	7 lemons				

Exercise

(1) Match the pair of item from Group A to the estimated weight in group B.

A - Group

- cat
- elephant
- bottle filled with 1 litre water
- empty gas cylinder
- pen
- leaf
- wooden chair

B - Group

- 1 g to 5 g
- 1000 g
- 3 kg to 5 kg
- 10 g to 15 g
- 6 kg to 10 kg
- more than 1000 kg
- more than 14 kg



For Teachers : Discuss how an elephant would be weighed.

(2) Complete the table by writing the number of weights required to measure an object of the given weight.

weight	measuring weights		
	200 g	500 g	1 g
1 kg			
2 kg			
3 kg			

(3) Match the following

'A' group (kilogram)		'B' group (g)
a) a quarter kilo	(.....)	1) 1750 g
b) one and a quarter kilo	(.....)	2) 1500 g
c) one and half kilo	(.....)	3) 2500 g
d) two and half kilo	(.....)	4) 1250 g
e) one and three quarter kilo	(.....)	5) 250 g

(4) Convert kilograms into grams

- ⦿ 9 kg =g
- ⦿ 7 kg 500 g =g
- ⦿ 6 kg 250 g =g



Fun with Maths

1) Kamal has a small remote controlled boat that can move forward and backward. She needs to carry these vegetables across a lake: 2 kg brinjal, 4 kg potatoes, 1 kg chillies, 3 kg onions and 1 kg tomatoes. How can she carry all the vegetables to the other side in the minimum number of trips, following the given conditions?

Conditions -

- ⦿ The boat can carry 4 kg at the most at a time.
- ⦿ The boat must have at least 1 kg of weight to make a trip across the lake.



For Teachers : Make weighing scale available for students to measure weight. Let students get hands on experience to measure weights.



Let's Recall

In the previous standards you have organised the data using tally marks. We will use pictures to represent data in this standard.

- ⦿ **A board near the main gate of Adarsh School displays the number of scholarship holding students.**

Scholarship holding students	
Academic year	Number of Students
2020-21	7
2021-22	13
2022-23	9
2023-24	15

- ⦿ **Observe the board and answers the questions below :**

- 1) In 2021-22, the number of scholarship holding students is
- 2) In which year is the number of scholarship holding students the most ?
- 3) By how much is the number of students holding scholarship more in 2022-23 than in the year 2020-21?

Representing Data using Pictures



Let's Understand

- ⦿ First collect the different pictures and information in one place for handling the data.
- ⦿ Symbols should be selected to represent the data in the chart.
- ⦿ Two columns, should be used for this.
- ⦿ In one column, the name of the item should be written and the other column should be used for the pictorial representation, using as many symbols as the quantity indicates.

◎ **Collect information about the trees around your school and make a pictograph for it.**

- ◆ To make a pictograph, first the information has to be collected. Around your school there are 12 neem trees, 8 coconut trees, 2 banayan trees and 14 ashoka trees.
- ◆ First let us make a three column chart. One column for name of the tree, and one for the picture of trees. We will make use of pictures to show the information about trees.
- ◆ Let us use different pictures for each kind of tree. Let's make the pictograph.

Name of tree	Pictures of trees	Numbers of trees
neem		
coconut		
banayan		
ashoka		



For Teachers : Guide students to collect data in different ways for handling. While presenting data how pictures/symbols need to be chosen or drawn, needs to be explained with different examples. Use real life examples for this.

Using scale for making a Pictograph



Let's Understand

A farmer planted 100 guava, 80 orange, 60 pomegranate and 40 custard apple saplings. Prepare a pictograph to represent this information.

Student : To present above information as a pictograph, a lot of pictures would be needed and it will be time consuming.

Teacher : To represent each sapling of different trees without using different symbols, we can indicate 10 saplings using one symbol.

Student : But why does one symbol only mean 10 saplings ?

Teacher : Because 100, 80, 60 and 40 are the numbers divisible by 10. Hence now we don't need to draw 100 symbols for 100 saplings. We will just need to draw at the most 10 symbols, means method by using scale.

Now the following pictograph is completed, using the scale = 10 saplings,

Saplings	Number of saplings
guava	
orange	
pomegranate	
custard apple	

Interpreting Pictograph



Let's Understand

Let's see what information we get from the pictograph.

Example : A pictograph showing sale of the fruit seller for one day is as follows - (scale : 1 symbol () = 5 boxes)

Fruit	Pictorial representation	Number of fruits boxes sold
mango		
chickoo		
grapes		
custard apple		

☉ From the above pictograph, we get the following information about the fruit seller's fruits.

- 1) Which fruits were there for sale?
 - 2) Which fruit boxes were sold the most?
 - 3) Which fruit boxes were sold the least ?
 - 4) How many total boxes of fruits were sold?
- What do you notice from the above information?

- ☉ We can tell that which fruits are sold the most.
- ☉ We can compare the sale of fruits.
- ☉ Representation of data helps to interpret the information.

Exercise

(1) A school has 15 students in first standard, 20 students in second standard, 30 students in third standard and 25 students in fourth standard. Prepare a pictograph for this information.

(Scale :  = 5 students)

Standard	Pictorial representation	Number of students
1 st		
2 nd		
3 rd		
4 th		



For Teachers : Discuss, how to choose a scale when the numbers are large, at such time what precaution is to be taken while representing information in compact form, how to select proper number for scale.

- (2) Sayali has prepared a pictograph showing the vehicles used by society members to go to office. Observe it and answer the questions. (Scale: $\triangle = 5$ people)

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Type of vehicle	Pictorial representation	Total number of people
cycle	$\triangle \triangle \triangle \triangle \triangle$	
two wheeler	$\triangle \triangle \triangle \triangle \triangle \triangle$	
four wheeler	$\triangle \triangle \triangle$	
auto rickshaw	$\triangle \triangle \triangle \triangle$	
bus	$\triangle \triangle \triangle \triangle \triangle$	

- ⦿ How many people use auto rickshaw for going to office ?
 - ⦿ How many people use four-wheeler for going to office ?
 - ⦿ Which vehicle is used the most for going to the office ?
 - ⦿ Which vehicle is used the least for going to the office ?
- (3) Mansi collected information about pets that her classmates have. As per that, 20 classmates have goats, 24 have cows, 32 have buffaloes, 8 of them have horses. Prepare a pictograph for this information. (Use the scale: 1 Picture = 4 pets .)
- (4) In an Art Festival, 20 contestants took part in singing, 30 took part in instrumental, 50 took part in drama and 40 took part in dance competition. Using appropriate scale prepare a pictograph. Which scale would be easy to understand the pictograph 2, 5, or 10? Think about it.
- (5) At a Kid's Fun Fair, the sale at the stalls is as follows :
pohe- 18 plates, upma - 30 plates, sheera 21 plates, bhel - 33 plates. Using this data, make a pictograph using appropriate scale.



For Teachers : Understanding information from a pictograph should be explained by giving opportunities to practise examples. After collecting the data, it should be categorized and a pictograph should be made. Various pictographs must be prepared by collecting data of different events. Use of appropriate scale should be practised.



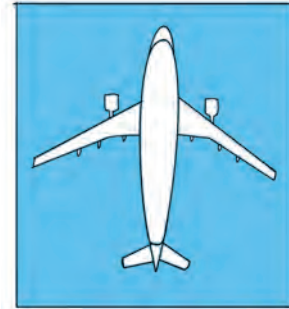
Let's Understand



Side view of the aeroplane
is very beautiful



I liked the front view
of the aeroplane



I liked the top view
of the aeroplane

In the previous standard, we learnt the side view, front view and top view of the objects.

1) Write the type of view for the following.

.....					

2) Match the objects to their views

	side view	
	front view	
	top view	



For Teachers : After showing model of aeroplane, let the students observe front view, side view and top view. Similarly explain with other models.



Do the Activity

- ☉ Draw as instructed in the boxes below, how would the cupboard that you have seen look like.

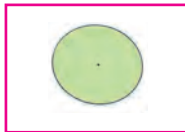
Front View	Side View	Top View



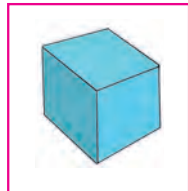
Do the Activity

- ☉ Children have taken different photos of some objects. Match the correct view to its object.

View →



Object →



Do the Activity

- ☉ While playing, Jay and John made buildings of different shapes with front view, top view and side view using blocks as shown below.

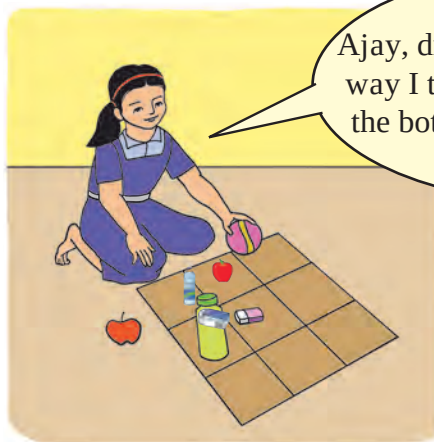


You can also collect blocks from school, similar empty boxes from house or surrounding to make buildings with different shapes. Show your friends the front view, side view and top view.



Do the Activity

1) Grid Game



Ajay, draw the picture in the grid the way I tell you. In the right corner at the bottom draw a picture of bottle kept straight.

This is first row and third column



2) To fill in the adjacent grid, do as Seema instructed.

- ⦿ Draw eraser in top left corner row-3
- ⦿ Draw an apple at the centre of second row second column. row-2
- ⦿ Draw a ball in first row second column. row-1

column-1	column-2	column-3



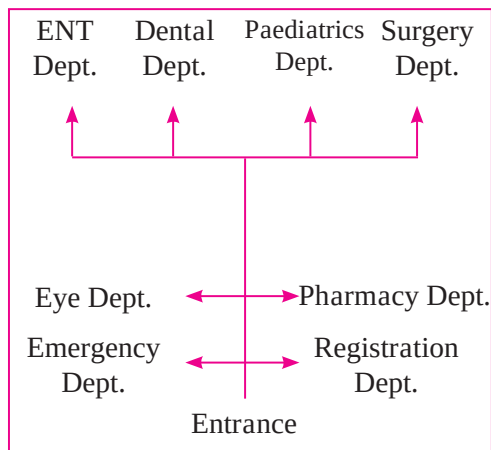
Do the Activity

Play a game with your friend (Scale: 1 box = 1 step) (No slanting move)

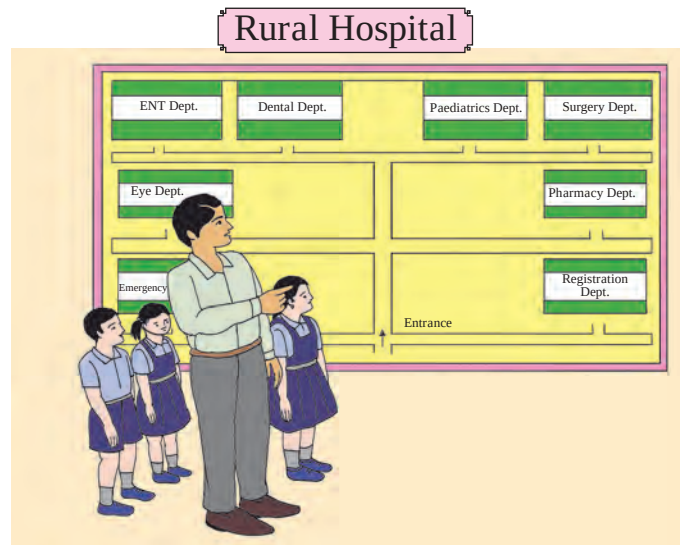
- 1) Show all the routes reaching to the ball from 'Start'? How many steps would be needed to walk each route?
- 2) Instruct to reach the mango tree from the 'Start'.
- 3) From 'Start', using minimum steps, where would you reach, house or banana?

	← left		right →	
4				
3				
2				
1	→ Start			
	A	B	C	D

- Draw a chess board, fix the initial positions of all pieces on it.
- John drew a quick sketch map of a rural hospital during a field visit.



John's Sketch Map



- John has drawn a sketch map of the rural hospital. Observe the sketch map, answer the following questions.

- State the location of the ENT department. After going straight, second left.
- Which department is before Eye Department?
- Write the location of Dental department?



Do the Activity

- Draw a map showing the different sections of your school.

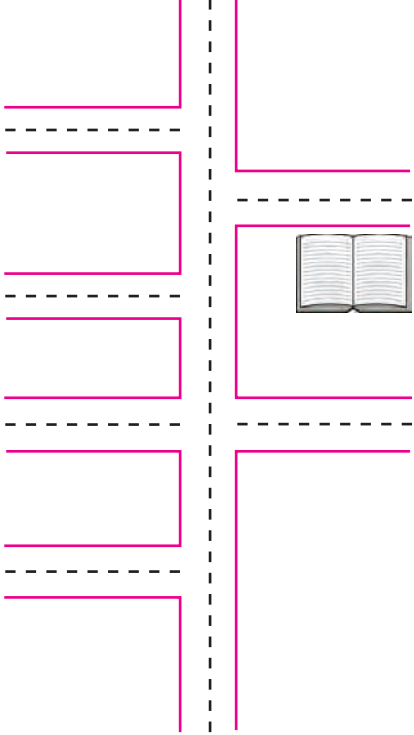
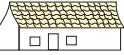
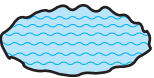


For Teachers : Let students play grid games in class, on the ground or with boards/books, sticks and give them learning experiences like draw objects, locate objects.

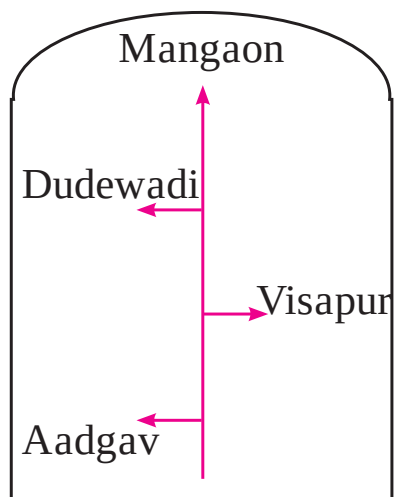
Exercise

(1) List the places you come across on the way from home to your school and draw the map in your note book.

(2) Show the places indicated besides, on the map or write their names.

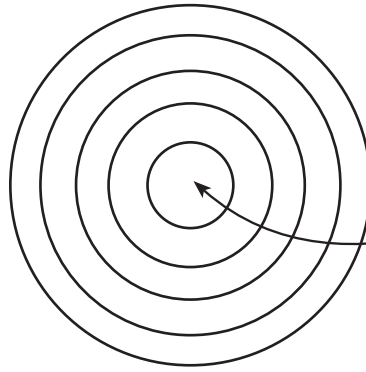
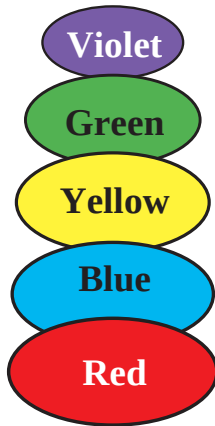
	House	
	School	
	Tree	
	Clinic	
	Post Office	
	Library	
	Lake	

(3) Draw a map as per instructions.



- Have you seen a sketch map like this?
- Draw a map in your notebook showing your village and its nearby villages.

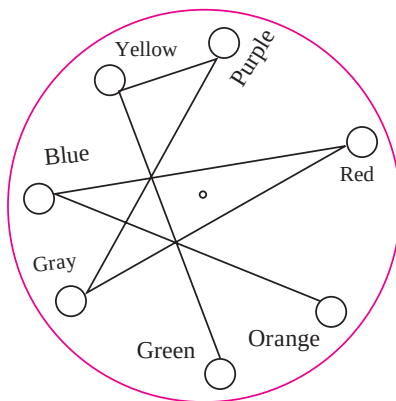
(4) Match size and colour to fill concentric circles.



Which colour will be given here?

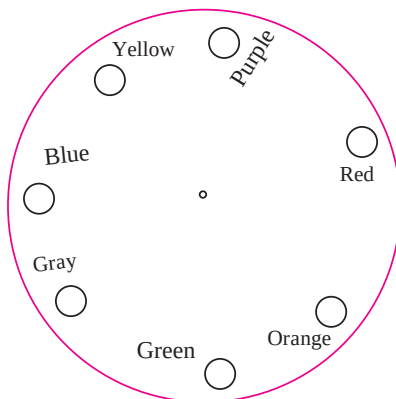
(5) Each number from 1 to 7 is mapped to a colour in the table given below. Observe it. First example is solved.

1	2	3	4	5	6	7
Purple	Red	Orange	Green	Gray	Blue	Yellow



Example : (4,7,1,5,2,6,3)

(6, 2, 5, 1, 4, 7, 3, 6) Follow this order and join the circles.



For Teachers : Take the practice of various examples from students as shown above.



Let's Recall

1) Multiply

- ☐ 8×7
☐ 23×9
☐ 45×17
☐ 135×6

2) Solve

- One kilogram of sugar costs ₹ 45, what is the price for 9 kilograms of sugar?
- An auditorium has 24 chairs in each row. How many chairs would be there in 18 such rows ?
- If 105 boxes of mangoes are kept in each tempo, then how many boxes are kept in 6 tempos in all?



Fun with Maths

- Choose any 3 consecutive numbers. eg. 7, 8, 9
- Multiply the middle number by itself. $8 \times 8 = 64$
- Multiply the first and third number. $7 \times 9 = 63$
- Subtract both the products. $64 - 63 = 1$
- In this way, take any 3 consecutive numbers of your choice and see if the difference is always 1.



Fun with Maths

1	12	18
18	6	2
12	3	6

- Is the product of all 3 numbers in every row and column the same?

Multiplication of a two-digit number by a two-digit number



Let's Understand

☉ Multiply 46×32

◆ Using Lattice Method

$$46 = 40 + 6$$

$$32 = 30 + 2$$

×	40	6
30	$40 \times 30 = 1200$	$6 \times 30 = 180$
2	$40 \times 2 = 80$	12

1200
+ 180
+ 80
+ 12
1472

◆ Using Vertical Method

Th	H	T	U
1	1		
		1	
	×	4	6
		3	2
		9	①2
+	1	①3	①8
	1	4	7
		7	2

carry over after multiplying by tens

carry over after multiplying by units

multiplying by units

multiplying by tens

	4	6
×	3	2
	9	2
+1	3	8
	1	4
	7	2

Practice keeping carry over in mind.

☉ Multiply 39×43

◆ Write the correct numbers in the boxes and complete the multiplication.

×	30	9
40		
3		

+	
+	
+	

Th	H	T	U
	3		
	1	2	
	×	3	9
		4	3
+	1	1	7
			0

Exercise

(1) Multiply.

- ☐ 48×14
 ☐ 57×36
 ☐ 69×29
 ☐ 89×52
 ☐ 66×66

(2) Solve.

- ☐ The cost of 1 notebook is ₹ 75, what would be the cost of 24 such notebooks?
- ☐ One sack can hold 64 coconuts, how many coconuts would be there in 33 such sacks?
- ☐ If the weight of 1 sack of sugar is 53 kg, what is the weight of 46 such sacks ?
- ☐ Janhavi can read 15 pages of a book in a day, then how many pages will she finish reading in 34 days?
- ☐ Which is easy to solve 25×18 or 18×25 ?

Multiplication of a three-digit number by a two-digit number



Let's Understand

- ☐ **Mama bought 25 chairs for his new restaurant. If each chair costs ₹375, then how much money did Mama spend to buy the 25 chairs?**

Method :

Here the cost of one chair is given, from which the total cost of many chairs has to be found. So we need to multiply 375 by 25 and find their product.

Now let us understand how to multiply a three-digit number by a two-digit number.

$$375 \times 25$$

Th	H	T	U
	1	1	
	3	2	
	3	7	5
	×	2	5
1	8	7	5
+ 7	5	0	0
9	3	7	5

carry over after multiplying by tens

carry over after multiplying by units

multiplying 375 by 5 units,
 multiplying 375 by 2 tens,

Amount spent by Mama to buy the chairs is ₹9375

- ⊙ **Costs of one shirt is ₹ 199, what is the cost of 32 such shirts?**

Th	H	T	U
2	2		
	1	1	
	1	9	9
	×	3	2
	3	9	8
+ 5	9	7	0
6	3	6	8

carry over after multiplying by tens

carry over after multiplying by units

multiplying by units

multiplying by tens

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The total cost of 32 shirts is ₹6368

- ⊙ **On her birthday, Anagha decided to distribute fruits to all patients in the village hospital. How much would she spend if she buys 15 kg apples at the rate of ₹185 per kg ?**

Let us multiply 185 by 15 to get the answer.

Th	H	T	U
	1	8	5
	×	1	5
+			

carry over after multiplying by tens

carry over after multiplying by units

multiplying by units

multiplying by tens

Total expenses to buy the apples would be ₹

- ⊙ **The cost of one English Dictionary is ₹ 175. To buy the dictionary, 47 students of a class paid this amount to their class teacher. How much money did she collect ?**

Th	H	T	U
	1	7	5
	×	4	7
+			

carry over after multiplying by tens

carry over after multiplying by units

multiplying by units

multiplying by tens

Total amount collected

Exercise

(1) Multiply.

⊙ 632×23

⊙ 375×12

⊙ 250×30

⊙ 143×43

⊙ 501×14

⊙ 104×51

(2) Solve.

- ⊙ Cost of one mat is Rs. 325, then find the cost of 21 such mats.
- ⊙ A factory can manufacture 243 articles per day, then how many such articles can be produced in 12 days?
- ⊙ The entry ticket for a zoo costs ₹ 120. If 34 people visited the zoo, how much money will be needed to buy the tickets in total?
- ⊙ The cost of sewing a uniform alongwith the cloth is ₹ 315 per student. What would be the cost of uniform for 47 students ?
- ⊙ Write two pairs of numbers whose product is 360.

(3) Frame a multiplication word problem using the given information.

- ⊙ 342 students; rate of notebook ₹ 36
- ⊙ 256 metre of cloth for 1 tent; 29 tents.
- ⊙ 25 boxes; cost per box ₹ 347
- ⊙ Cost of one treeguard ₹ 327; 46 tree guards.
- ⊙ 428 labourers; daily conveyance allowance ₹ 82.

(4) Read the following statements. Think and put a (✓) mark in the correct column.

Sr. No.	Statement	Always true	Sometimes true	Never true
1)	Multiplying a number by 10 gives a '0' in the units place of the product.			
2)	Multiplying any number by 2 gives the answer as an even number.			
3)	Multiplying any number by 5 gives 5 in the units place of the answer			
4)	While counting numbers in sequence, the number just after an odd number is an even number.			
5)	Taking the half of any number gives an even number as the answer			
6)	Adding zero to any number increases the number by 1.			
7)	The product of the double of two numbers is same as the product of the two numbers			



Do the Activity

- At a childrens' Fun Fair in school, Omkar earned ₹ 2855 by selling Bhel. This amount included coins of ₹ 1, ₹ 2 and ₹ 5, also currency notes of ₹ 10, ₹ 20, ₹ 50, ₹ 100, ₹ 200 and ₹ 500. He needs to deposit this amount in the School Savings Bank and has to fill the 'Deposit Slip' for the same. Estimate the number of coins/notes of any denomination that he may have and help him to fill the Deposit Receipt.

S.S.Bank		Deposit Slip																																		
		Date :																																		
		Name: <u>Omkar</u>																																		
Account No. :		<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 20%;">Coins/ Notes</th> <th style="width: 20%;">Number of Coins/ Notes</th> <th style="width: 60%;">Total Amount (Rupees)</th> </tr> </thead> <tbody> <tr><td>500</td><td>3</td><td></td></tr> <tr><td>200</td><td>2</td><td></td></tr> <tr><td>100</td><td>4</td><td></td></tr> <tr><td>50</td><td></td><td></td></tr> <tr><td>20</td><td></td><td></td></tr> <tr><td>10</td><td></td><td></td></tr> <tr><td>5</td><td></td><td></td></tr> <tr><td>2</td><td></td><td></td></tr> <tr><td>1</td><td></td><td></td></tr> <tr><td>Total</td><td></td><td></td></tr> </tbody> </table>		Coins/ Notes	Number of Coins/ Notes	Total Amount (Rupees)	500	3		200	2		100	4		50			20			10			5			2			1			Total		
Coins/ Notes	Number of Coins/ Notes			Total Amount (Rupees)																																
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Total																																				
Amount (in digits): _____																																				
Total Amount (in words): _____																																				

Signature of Depositor																																				



Let's Think

An auditorium has 18 rows of chairs. There are 15 chairs in each row, how many chairs does the auditorium have in all ?

How can all these chairs be arranged in different possible ways of rows and columns ?



For Teachers : For better understanding of mathematical statements like 'always true', 'sometimes true' and 'never true', students must be given the opportunity to discuss them with examples. Students to be motivated for conceptual understanding by asking a variety of analytical questions. Maximum practice to be taken to deepen the understanding of the mathematical concepts, to avoid rote learning and to enhance their reasoning abilities.

Indian Knowledge System.

When multiplying using Ganesh Daivadnya's Kapaatsandhi method, the multiplication is done in the same manner as we multiply using the lattice method. In this the product is written as if two doors of a cupboard are opened. eg. $7 \times 4 = \begin{array}{|c|} \hline 2 \\ \hline 8 \\ \hline \end{array}$

Let's solve the example 357×42 in this way..

3	5	7	
			4
			2

Draw a grid with three columns and two rows. Write the digits 3, 5, 7 on the top of the columns and the digits 4 and 2 on the right.

Divide each box of the grid into two parts.

3	5	7	
		2	4
		8	2

First the product of 4 and 7 is written in the box adjacent to both such that 2 is written above the diagonal line and 8 is written below, as shown in the accompanying box. Similarly, complete the multiplication in the other boxes by writing the product as shown in the box.

	3	5	7	
	1	2	2	4
	2	0	8	
	0	1	1	2
	6	0	4	
1	4	9	9	4

Now add up all the digits that are separated by the extended diagonal lines and write the answers below.

$$357 \times 42 = 14994$$

Do some more multiplications in the same way.

Remember :

- 1) If the digits in a diagonal section add upto 10 or more, then we need to carry over and add the digit to the numbers in the next diagonal section.
- 2) Does this method give the correct answer ? Understand it yourself and explain it to others.



For Teachers : Information about famous Indian Mathematicians can be shared with the students.



Correlation between Multiplication and Division



Let's Understand



Swara and Yogesh, come let's pack these bottles of Kokam sharbat in the boxes today. Remember, we need to place an equal number of bottles in each box.



How many bottles did you pack in each box?

Didi, I packed total 48 bottles in six boxes



$$48 \div 6 = 8.$$

This means I have packed 8 bottles in each box.



$6 \times 8 = 48$ and
 $8 \times 6 = 48$ So,
 $48 \div 6 = 8$
and $48 \div 8 = 6$

Didi, I also packed 48 bottles. But I used 8 boxes. As I packed 6 bottles in each box.

$$48 \div 8 = 6$$



$$5 \times 7 = 35$$

$$35 \div \square = 5 \quad 35 \div \square = 7$$

$$9 \times 8 = 72$$

$$72 \div \square = 9 \quad 72 \div \square = 8$$



Let's Remember

From one multiplication statement two division statements can be obtained.

- ☉ Isha has 42 bottles. She packed 8 bottles in each box. She could pack 5 such boxes and 2 bottles were left unpacked.

Isha : Didi, here also we have used division. $42 \div 8$ gives 5 boxes and 2 bottles remain.

Nisha : Correct, this division can be done by arranging the numbers vertically as shown.

$$\begin{array}{r} 05 \\ 8 \overline{) 42} \\ - 0 \downarrow \\ \hline 42 \\ - 40 \\ \hline 02 \end{array}$$

Dividend	=	42
Divisor	=	8
Quotient	=	5
Remainder	=	2



Let's Recall

- ☉ A company ordered 78 wheels to make autorickshaws. If each autorickshaw is to be fitted with 3 wheels, how many autorickshaws can be fitted with the wheels?

The number of wheels is 78. Each autorickshaw is to be fitted with 3 wheels. To get the number of autorickshaws we need to use division.

$$\begin{array}{r} 3 \overline{) 78} \quad (10 + 10 + 6 = 26 \\ - 30 \\ \hline 48 \\ - 30 \\ \hline 18 \\ - 18 \\ \hline 00 \end{array}$$

$$\begin{array}{r} 26 \\ 3 \overline{) 78} \\ - 6 \downarrow \\ \hline 18 \\ - 18 \\ \hline 00 \end{array}$$

Exercise

- (1) Divide. Write the Dividend, Divisor, Quotient and Remainder.

☉ $6 \overline{) 74}$ ☉ $5 \overline{) 96}$ ☉ $7 \overline{) 71}$ ☉ $8 \overline{) 97}$ ☉ $6 \overline{) 51}$

- (2) Solve.

- ☉ On Children's Day, 96 books were distributed to students in a class. If each child got 4 books, how many children got the books ?
- ☉ This year, Ashok sowed flowering plants in some parts of his field. He sowed 56 plants with 7 plants in each row. How many rows of plants could he sow ?



Let's Remember

While dividing, the divisor is subtracted from the dividend as many times as possible. Therefore, the remainder is always smaller than the divisor.

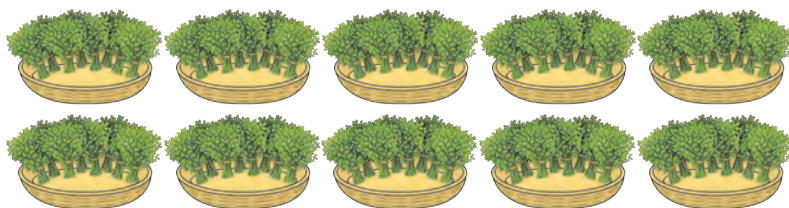
Dividing a three-digit number by a single-digit number

- ☉ **Shobha tied 144 bundles of fenugreek at her farm. To sell these bundles, she arranged 8 bundles in each basket. How many baskets did she need to arrange all the bundles ?**

Nayana : Here we have to divide 144 by 8.

Harsh started solving the division.

First, the bundles were placed in 10 baskets with 8 bundles in each.



$$\begin{array}{r} 8 \overline{) 144} \quad (10 \\ - 80 \\ \hline 64 \end{array}$$

$$\begin{array}{r} 8 \overline{) 144} \quad (10+5 \\ - 80 \\ \hline 64 \\ - 40 \\ \hline 24 \end{array}$$

Then, 5 baskets were arranged with 8 bundles in each



$$\begin{array}{r} 8 \overline{) 144} \quad (10+5+3=18 \\ - 80 \\ \hline 64 \\ - 40 \\ \hline 24 \\ - 24 \\ \hline 00 \end{array}$$

Lastly, 3 baskets were needed with 8 bundles in each basket



Number of baskets = $10+5+3=18$

Dividend = 144 Divisor = 8
Quotient = 18 Remainder = 0

- 144 fenugreek bundles were placed in 18 baskets.

- ☉ **Sanchita has ₹ 749 in the form of 7 notes of ₹100, 4 notes of ₹10 and 9 coins of ₹ 1. How can she divide this amount among 7 people equally? How much money will each person get ?**

$$\begin{array}{r} 107 \\ 7 \overline{) 749} \\ - 7 \downarrow \\ \hline 04 \\ - 0 \downarrow \\ \hline 49 \\ - 49 \\ \hline 00 \end{array}$$

- First the 7 notes of ₹ 100 are equally divided, 1 for each. Each person gets 1 note of ₹100.
- Now 4 notes of ₹ 10 cannot be divided equally among 7 people, hence no one will get the ₹10 note.
- So we would write '0' in the ten's place of the Quotient.
- 4 notes of ₹10 make ₹40 or 40 units of ₹1. $40+9=49$ units of ₹1. 49 rupees are equally divided among 7 people.

Each person will get 7 rupees

$$749 \div 7 = 107$$

Each person will get ₹107

Exercise

(1) Divide.

- | | | | |
|--------------------|---------------------|--------------------|--------------------|
| $\odot 672 \div 6$ | $\odot 840 \div 8$ | $\odot 861 \div 7$ | $\odot 486 \div 9$ |
| $\odot 816 \div 4$ | $\odot 945 \div 15$ | $\odot 245 \div 7$ | $\odot 742 \div 8$ |

(2) Solve.

- $\odot$ On Sports Day, for the flag hoisting ceremony, 161 students were arranged in 7 equal rows. How many students in each row ?
- $\odot$ Meera stitched 207 buttons, with 9 buttons on each shirt. How many shirts could she button?
- $\odot$ If 8 kilograms of rice are filled in one bag, how many bags are needed to fill 496 kilograms of rice ?
- $\odot$ If 6 people share 654 rupees equally, how much money will each person get?

$\odot$ Verification of the Division.



Let's Understand

$$\begin{array}{r}
 036 \\
 7 \overline{) 255} \\
 \underline{- 0 \downarrow} \\
 25 \\
 \underline{- 21 \downarrow} \\
 45 \\
 \underline{- 42} \\
 03
 \end{array}$$

Dividend = 255
 Divisor = 7
 Quotient = 36
 Remainder = 3

$$\begin{array}{r}
 36 \quad \text{Quotient} \\
 \times 7 \quad \text{Divisor} \\
 \hline
 252 \\
 + 3 \quad \text{Remainder} \\
 \hline
 255 \quad \text{Dividend}
 \end{array}$$

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

$\odot$ If ₹738 are distributed equally among 9 people, how much money will each person get ?

$$\begin{array}{r}
 0 \\
 9 \overline{) 738} \\
 \underline{- 0} \\
 7
 \end{array}$$

$$\begin{array}{r}
 08 \\
 9 \overline{) 738} \\
 \underline{- 0 \downarrow} \\
 73 \\
 \underline{- 72} \\
 01
 \end{array}$$

$$\begin{array}{r}
 082 \\
 9 \overline{) 738} \\
 \underline{- 0 \downarrow} \\
 73 \\
 \underline{- 72 \downarrow} \\
 018 \\
 \underline{- 18} \\
 00
 \end{array}$$

Dividend = 738
 Divisor = 9
 Quotient = 82
 Remainder = 0

$$\begin{array}{rclclcl}
 \text{Verification :} & \text{Divisor} & \times & \text{Quotient} & + & \text{Remainder} & = & \text{Dividend} \\
 & 9 & \times & 82 & + & 0 & = & 738
 \end{array}$$

solved by Suraj

$$\begin{array}{r} 18 \\ 5 \overline{) 900} \\ \underline{- 5} \\ 40 \\ \underline{- 40} \\ 00 \end{array}$$

solved by Neeraj

$$\begin{array}{r} 180 \\ 5 \overline{) 900} \\ \underline{- 5} \\ 40 \\ \underline{- 40} \\ 000 \\ \underline{- 0} \\ 0 \end{array}$$

Whose Quotient is correct? How will you decide ?

Exercise

(1) Divide and verify the answer.

- | | | | |
|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| ☐ $768 \div 6$ | ☐ $992 \div 8$ | ☐ $837 \div 9$ | ☐ $875 \div 7$ |
| ☐ $816 \div 6$ | ☐ $800 \div 8$ | ☐ $980 \div 7$ | ☐ $700 \div 5$ |

(2) Frame division word problems and solve.

- ☐ 240 packets of biscuits; each student got 2 packets.

Framed word problem :

If 240 packets of biscuits were distributed such that each student got 2 packets, then how many students got the packets ?

- ☐ 8 bags of salt cost 160



- ☐ 96 bottles in all,
6 bottles in each crate



- ☐ Raghu bought 7 metres of cloth for ₹455



- ☐ 9 friends set off on a train journey. Vicky paid a total of ₹135 for all tickets.
- ☐ Total rakhis 72, each packet contains 6 rakhis.
- ☐ Given information : 5 packets for each, 250 packets in all.



For Teachers : When writing examples of division with symbol, the order of the numbers should be correct. This should be explained with examples. 15 chickoos were equally distributed among 3 people, which is written as $15 \div 3$. Also give practice of examples having '0' in the Ten's place of the Quotient. Maximum practice should given of divide and verify the answer.

☉ Patterns in Division.

How much money will each child get?

Join that amount to the child by drawing lines.

		
		
$3 \div 3 =$	$30 \div 3 =$	$900 \div 3 =$

☉ Observe. Understand the patterns in the given examples and complete.

$6 \div 3 = 2$	$9 \div 3 =$	$5 \div 5 =$	$7 \div 7 =$	$8 \div 8 =$
$60 \div 3 = 20$	$90 \div 3 =$	$50 \div 5 =$	$70 \div 7 =$	$80 \div 8 =$
$600 \div 3 = 200$	$900 \div 3 =$	$500 \div 5 =$	$700 \div 7 =$	$800 \div 8 =$

Dividing by Ten or Multiples of Ten



Let's Understand

A shopkeeper wants to fill bags with 10 kilograms of rice each.

- 1) If he has 60 kilograms of rice, how many bags will he need ?
- 2) If he has 600 kilograms of rice, how many bags will he need ?

$$60 \div 10 = \dots\dots$$

$$600 \div 10 =$$



Let's Think

☉ If a bag of rice weighs 100 kg, how many bags are required to fill 600 kg of rice ?

$$600 \div 100 = \dots\dots\dots$$

- ☉ Siddhu wants to pack 10 tennis balls in each box. If he has to pack 450 tennis balls, how many boxes will he need ?

Here we need to divide 450 by 10

FOR FREE DISTRIBUTION

$$\begin{array}{r} 45 \\ 10 \overline{) 450} \\ \underline{- 40} \\ 050 \\ \underline{- 50} \\ 00 \end{array}$$

So, Siddhu will need 45 boxes to pack the 450 tennis balls.

- ☉ Shreya prepared moravala using 180 gooseberries (*amla*). If she filled the moravala of 20 gooseberries in every jar, how many such jars could she fill ?

Here $180 \div 20 = ?$

$$\begin{array}{r} 9 \\ 20 \overline{) 180} \\ \underline{- 180} \\ 000 \end{array}$$

Shreya could fill 9 jars of moravala.

$20 \times 1 =$	20
$20 \times 2 =$	40
$20 \times 3 =$	60
$20 \times 4 =$	80
$20 \times 5 =$	100
$20 \times 6 =$	120
$20 \times 7 =$	140
$20 \times 8 =$	160
$20 \times 9 =$	180
$20 \times 10 =$	200

- ☉ There are 332 books in a library. If 20 shelves are filled with same number of books in each shelf, how many books are there in each shelf? How many books would remain?

Here we divide 332 by 20.

$$\begin{array}{r} 016 \\ 20 \overline{) 332} \\ \underline{- 0} \\ 33 \\ \underline{- 20} \\ 132 \\ \underline{- 120} \\ 012 \end{array}$$

Each shelf carries 16 books and 12 books are left.



For Teachers : Practice of dividing numbers having '0' in the units or tens place should be taken. Also the need to verify the answer using 'dividend = divisor x quotient + remainder' must be explained.

Exercise

(1) Divide.

- ⊙ $90 \div 10$
- ⊙ $80 \div 20$
- ⊙ $360 \div 10$
- ⊙ $800 \div 10$
- ⊙ $810 \div 90$
- ⊙ $720 \div 40$
- ⊙ $900 \div 60$
- ⊙ $600 \div 40$

(2) Divide and write the remainder.

- ⊙ $66 \div 10$
- ⊙ $73 \div 30$
- ⊙ $258 \div 10$
- ⊙ $420 \div 50$
- ⊙ $725 \div 80$
- ⊙ $442 \div 10$
- ⊙ $824 \div 20$
- ⊙ $392 \div 60$

(3) Solve.

- ⊙ A bakery baked 950 buns. If 10 buns were packed in each packet, how many such packets can be made ?
- ⊙ Sakhumai divided ₹1000 equally among her 10 grand children. How much money did each of them receive ?
- ⊙ There are 72 soaps in a shop. The shopkeeper packed them equally in 12 boxes. How many soaps did he fill in each box ?
- ⊙ At a wedding, 98 chairs were arranged equally around 14 tables. How many chairs could each table have ?
- ⊙ A shopkeeper filled 85 litres of oil equally into 17 bottles. How many litres of oil could he fill in each bottle ?

Dividing a three-digit number by a two-digit number



Let's Understand

- ⊙ A shopkeeper packed 300 pens such that, 1 dozen pens in each packet. How many such packets could he make?

Total pens = 300, 1 dozen pens = 12 pens

$$300 \div 12 = ?$$



Here I will need to prepare the 12 times table



First lets divide 30 by 12.
 $12 \times 2 = 24$.
Now write
 $30 - 24 = 6$



Bring the '0' down. Now the number is 60.
 $12 \times 5 = 60$

$12 \times 1 =$	12
$12 \times 2 =$	24
$12 \times 3 =$	36
$12 \times 4 =$	48
$12 \times 5 =$	60
$12 \times 6 =$	72

$$\begin{array}{r} 25 \\ 12 \overline{) 300} \\ \underline{- 24} \\ 60 \\ \underline{- 60} \\ 00 \end{array}$$

Quotient = 25
Remainder = 0

The shopkeeper made 25 packets with 1 dozen pens in each packet.

- ⊙ **Nayana bought 25 storybooks for ₹400. If the cost of each story-book was the same; what was the cost of each storybook ?**

Here $400 \div 25 = ?$

$25 \times 1 =$	25
$25 \times 2 =$	50
$25 \times 3 =$	75
$25 \times 4 =$	100
$25 \times 5 =$	125
$25 \times 6 =$	150

$$\begin{array}{r} 16 \\ 25 \overline{) 400} \\ \underline{- 25} \\ 150 \\ \underline{- 150} \\ 000 \end{array}$$

Quotient = 16
Remainder = 0

The cost of each storybook is ₹16

Exercise

(1) Divide.

- | | | | |
|-----------------|-----------------|-----------------|-----------------|
| ⊙ $845 \div 13$ | ⊙ $544 \div 17$ | ⊙ $396 \div 17$ | ⊙ $731 \div 19$ |
| ⊙ $816 \div 18$ | ⊙ $840 \div 16$ | ⊙ $981 \div 11$ | ⊙ $264 \div 14$ |

(2) Solve.

- ⊙ A cap seller sold 14 caps for ₹504, the price of each cap being the same. What was the cost of each cap?
- ⊙ If a car needed 12 litres of petrol to cover a distance of 264 km, how much distance could it cover in 1 litre of petrol?
- ⊙ If Soni bought 18 kilograms of sugar for ₹810, what is the rate of sugar per kilogram?
- ⊙ For making sports medals, a ribbon 600 cm long was cut into pieces of 30 cm each. How many pieces of the ribbon will be there?

(3) Calculate the price of one item in the following examples.

- ⦿ The cost of 13 balls is ₹ 312
- ⦿ 19 umbrellas cost ₹988
- ⦿ The cost of 24 erasers is ₹360

Mixed Examples (Multiplication and Division)

- ⦿ Sachin bought 9 pens from a shop for ₹135. How much money would he have to pay to buy 21 such pens?

Method : 9 pens cost ₹135

To get the cost of 1 pen, we need to divide 135 by 9

$$135 \div 9$$

$\begin{array}{r} 015 \\ 9 \overline{) 135} \\ \underline{- 0} \\ 13 \\ \underline{- 9} \\ 45 \\ \underline{- 45} \\ 00 \end{array}$	Quotient = 15 The cost of 1 pen is ₹ 15
--	--

To find the cost of 21 such pens we need to multiply 21 by 15.

$\begin{array}{r} 21 \\ \times 15 \\ \hline 105 \\ + 210 \\ \hline 315 \end{array}$	$21 \times 15 = ₹315$. So, he will have to pay ₹315 to buy 21 pens.
---	---

Exercise

(1) Solve.

- ⦿ 3 kilograms of pomegranates cost ₹ 480, how much would 15 kilograms of pomegranates cost?
- ⦿ If the cost of 5 metres of cloth is ₹ 1250, what is the cost of 12 metres of cloth?
- ⦿ 12 litres of milk costs ₹840, what is the price of 14 litres of milk?
- ⦿ Five friends went on a nature trip. First they travelled by bus. The price of bus ticket was ₹60 per person. Then they arranged a boat ride for 1 hour which cost them ₹1000. How much did each of them spend till now?

(2) Parents purchased various items at the weekly market of Rahatwade school. Solve the following questions based on this information.

- Prajaktatai had 1 note of ₹ 200 and 3 notes of ₹ 50. She shopped worth ₹ 310. How much money is left with her now?
- Ashatai bought 3 matki bhel packets. When she gave a ₹ 200 note to the shopkeeper, he returned ₹ 80. What would be the cost of 1 matki bhel?
If Sagardada wants to buy 2 such matki bhel packets, how much money should he pay?

(3) A school wants to buy 1000 notebooks. The shopkeeper has packed 20 notebooks in each bundle. How many bundles should the school buy ?

(4) Look at the following table. Find examples of multiplication and division and colour them. Write down the examples found from the table and solve them.

Like,

$$250 \times 4 = 1000$$

$$50 \times 20 = 1000$$

$$525 \div 5 = 105$$

134	52	30	31	931	10	93	55
5	20	15	15	250	3	33	101
22	1040	450	0	4	26	104	5555
110	100	50	20	1000	60	16	99
44	104	19	3	6	22	132	7
20	6	950	6	6000	30	200	693
808	624	31	14	1200	8	16	24
35	9	525	5	105	62	3200	78



⦿ Add.

For each letter in the addition example given below, assign any one digit from 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. Do not assign the same digit for two or more letters.

$$\begin{array}{rcccc} & S & E & N & D \\ & \square & \square & \square & \square \\ + & M & O & R & E \\ & \square & \square & \square & \square \\ \hline M & O & N & E & Y \\ \square & \square & \square & \square & \square \end{array}$$

⦿ Fun-filled multiplication - Look for the pattern in the answers.

- | | | | |
|--------------------|--------------------|--------------------|--------------------|
| ⦿ $37 \times 3 =$ | ⦿ $37 \times 6 =$ | ⦿ $37 \times 9 =$ | ⦿ $37 \times 12 =$ |
| ⦿ $37 \times 15 =$ | ⦿ $37 \times 18 =$ | ⦿ $37 \times 21 =$ | ⦿ $37 \times 24 =$ |

⦿ Divide.

In the given division example, the letters a, b, c are used to represent different digits. Identify those digits.

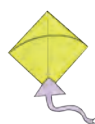
$$\begin{array}{r} \text{bb} \\ a \overline{) \text{bcc}} \\ \underline{- \text{a}} \\ \text{bc} \\ \underline{- \text{a}} \\ \text{b} \end{array}$$



Let's Remember

In the previous standards, we have seen the number of axes that can be drawn in two dimensional shapes. Write how many axes of symmetry are there in the following shapes

Figure:



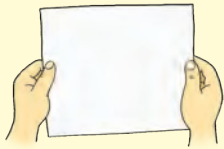
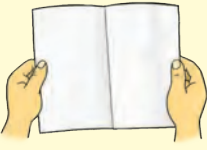
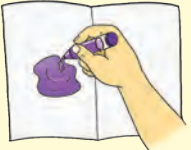
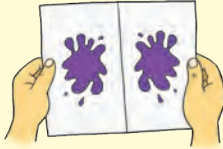
Number
of axes of
symmetry:

Reflective Symmetry



Let's Understand

☉ Do the activity as shown in the picture.

1) Take a blank white paper. 	2) Fold it in the middle. 
3) When the paper is folded in the middle, two equal parts are seen. 	4) Put a drop of ink or paint on one of the parts of the paper. 
5) Now place the other part on top of the first part and press slightly. 	6) Open the paper gently and see. 

The complete figure obtained after unfolding the paper is a symmetrical figure. The fold on the paper is the axis of symmetry.



Let's Discuss

Mirror Image

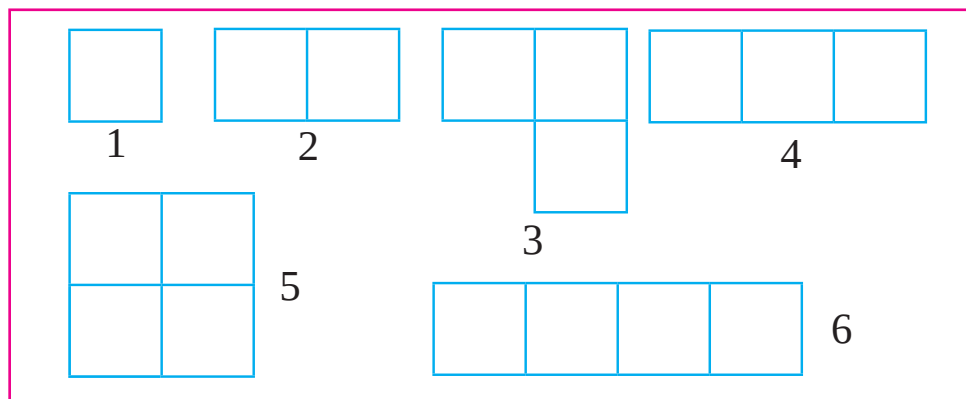
- Once ready for school, Radhika looks into the mirror. Her image in the mirror is as shown in the picture. Does her image look like her? Take experience actually.

The image seen in the mirror is a reflected image of the original object.



Let's Think

- Verify the figure obtained as the object and the image together while mirror is placed at different positions.



- Experience how the mirror image of the picture looks by placing the mirror at different positions.



- Observe the following figures.

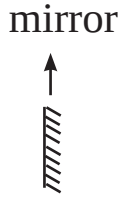


What did you notice after observing the above figures?

The image in the mirror is not necessarily symmetrical to the original image. Why do some mirror images look different from the original objects? Think about it.

☉ How will the mirror image of the figure N look like?

N



Original image

Mirror image

Now draw the mirror image of the number 63

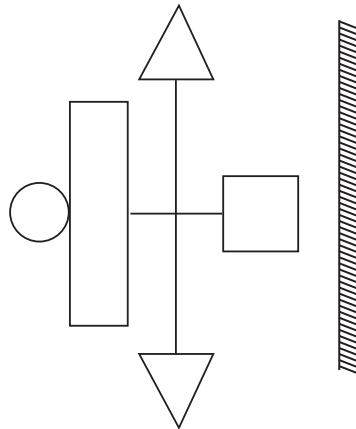
63

Observe the mirror image of this figure.

O

Exercise

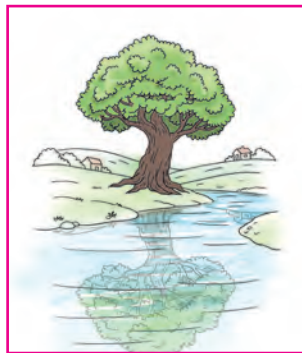
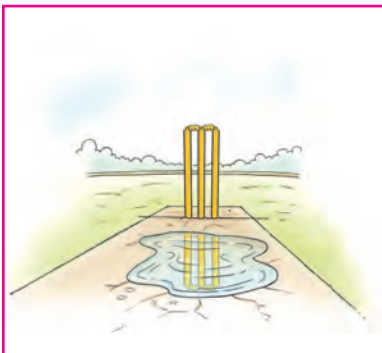
- (1) Take the number cards from 11 to 20 in Marathi and English. Write which of these numbers have mirror images that are the same as the original numbers.
- (2) Draw the mirror image of the following shape.



Let's Understand

Water Reflection

☉ Observe the following examples of water reflection



For teachers : The image seen in the mirror is the reflection of the original self. This should be discussed and brought to the attention of students.

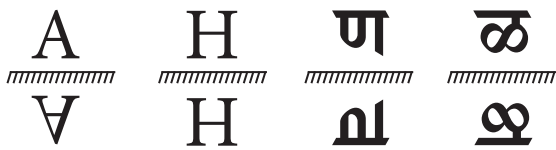


Let's Understand

An object's image can be seen in the mirror. An object's reflection can be seen in the water. The image of an object in water is called its water reflection.

When an object's image is reflected in the water, the axis of symmetry is horizontal.

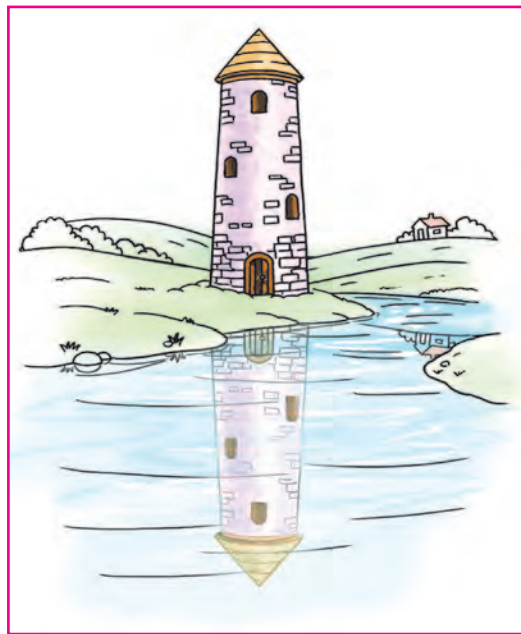
Look at the reflection of letters in water.



Look at the water reflection of words.



Observe the water reflection in nature.



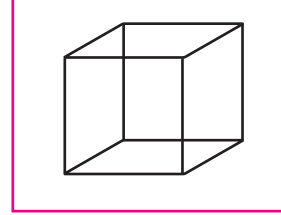
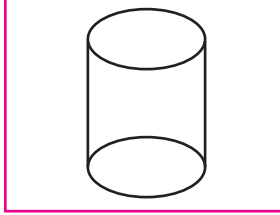
For Teachers : 1) While looking at mirror images, the changes that occur in the image prominently, should be explained in an entertaining way by practically demonstrating the use of mirrors. 2) Examples of water reflection should be demonstrated. 3) Images obtained by increasing the size of letters should be experienced.

Exercise

(1) Draw the water reflections of the following letters in your notebook.

ण	त	थ	द	ध	न	प
---	---	---	---	---	---	---

(2) Draw the water reflections of following shapes in your notebook.



Let's Think

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Why are the letters (AMBULANCE) written on an ambulance in this manner?



For Teachers : Students should be encouraged to observe water reflections of different figures. They should observe the differences between water reflections and original figures and have a discussion based on it.



Do the Activity



Radhika: I want to put flower garlands on the edges of the blackboard. But what should the length of the garland be? What can be done?

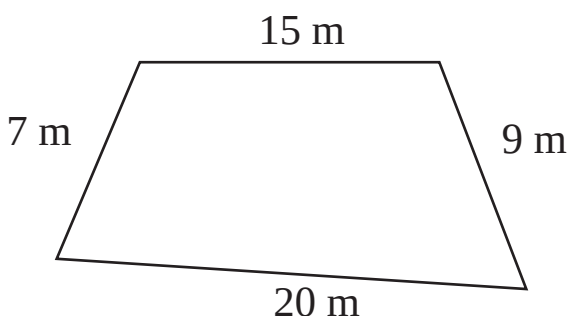
Rohan: To know this, the length of the edges of the board needs to be measured using measuring tape and a garland of that length should be made.

Let us measure the length of the edges of the board and write them below.

.....cm + cm +cm +cm

A garland of totalcm will have to be made.

The sum of lengths of all edges of the board is the perimeter of the board. The total length of the garland is also equal to the perimeter of the board.



For Teacher : Observe whether students are able to measure the lengths of the edges of the blackboard correctly . Similarly, the concept of perimeter should be explained by measuring the perimeter of various objects in the classroom.

A farmer wants to build a fence using wire mesh to protect his field from animals. What should be the length of the wire mesh?

Add up the lengths of all the sides of the farmer's field.

....m +m +.... m +.... m =metre will be the length of the wire mesh that he will need to buy. This indicates that,

☉ **The sum of the measures of all sides of a closed figure is the perimeter of that figure.**

Exercise

(1) Calculate and write the perimeter of the following figures.

Perimeter = cm	Perimeter = cm	Perimeter = cm	Perimeter = cm

(2) Calculate the perimeter of the given figures using dots:

(Assume the distance between two adjacent dots to be 1 cm)

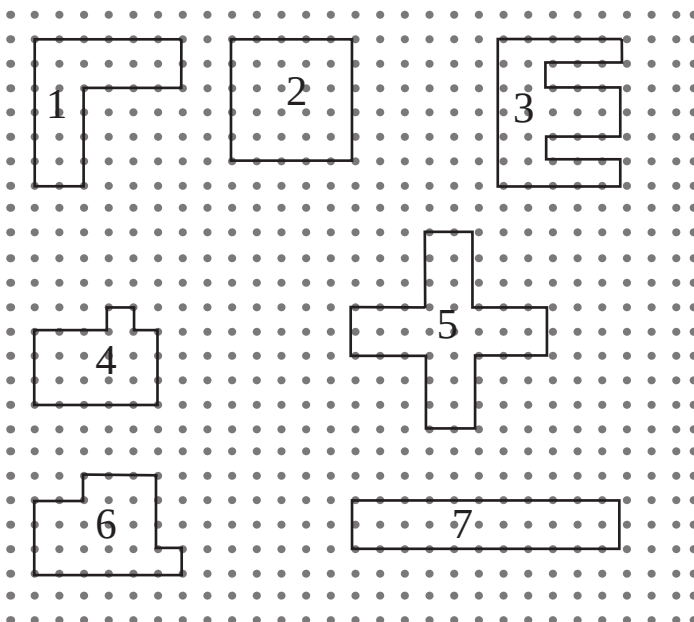


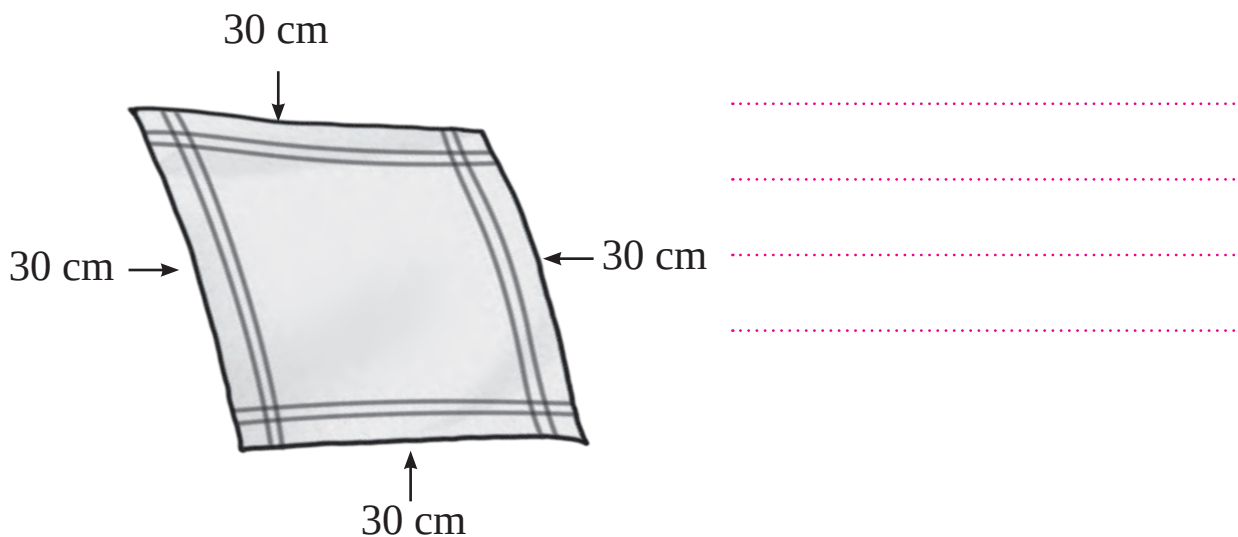
Figure	Perimeter
1)	24 cm
2)	 cm
3)	 cm
4)	 cm
5)	 cm
6)	 cm
7)	 cm

(3) Estimate the perimeter of following objects in the classroom and verify by actual measurement.

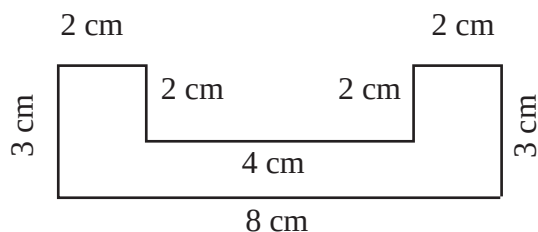
Sr.No.	Name of object	Approximate perimeter	Actual perimeter
1	cover of mathematics textbook		
2	window		
3	surface of table		

(4) Namita wants to lace all the edges of the handkerchief shown in the picture. What would be the length of the lace that she needs to buy?

☉ **What would be the length of the lace needed for 3 such handkerchiefs?**



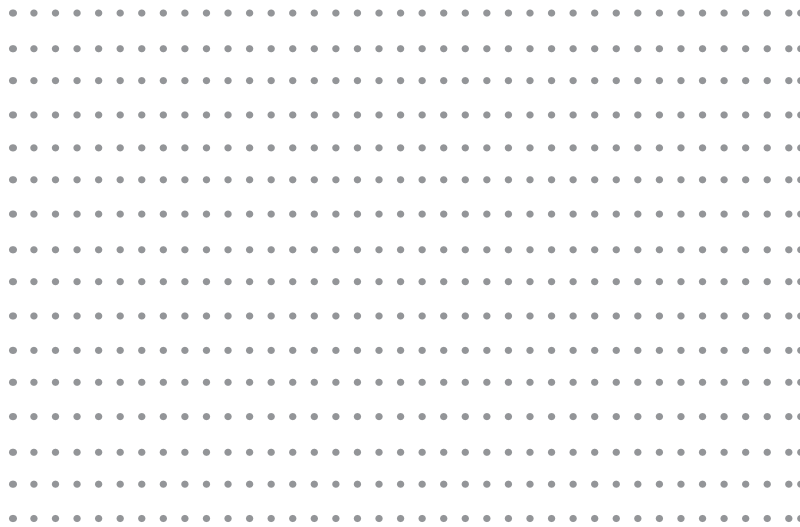
(5) Calculate and write the perimeter of the figure below.



Perimeter = cm.

(6) Draw two figures using dots given below whose perimeter is 20 cm.

(The distance between two adjacent dots is 1 cm)



Let's Understand

Area



Student: Sir, why have we brought so many tiles?

Teacher: To be placed on the classroom floor.

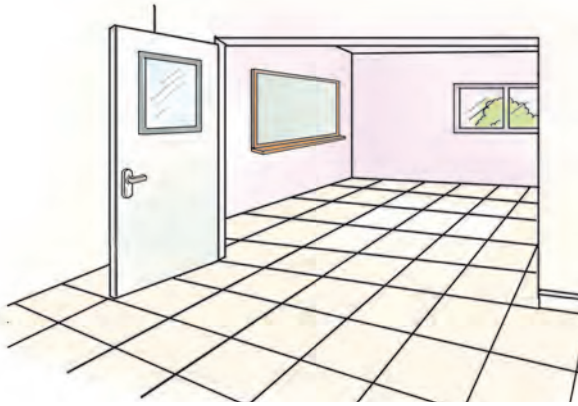
Student: Will these many tiles be enough to cover the classroom floor? How can we decide this?

Teacher: We need to measure the surface of the classroom floor.

Measurement of the surface of the floor means area of the floor.



For Teachers : Discuss about measuring the surface. Discuss about different surfaces to be measured using different measuring instruments. Guide accordingly.



Measurement of the space covered by the surface of any figure is the **area** of that figure.

☉ **Measure the surfaces of the following figures with the help of measuring tools.**

Name of Surface	Measuring tool	Approximate number	Actual number of such tools that cover up the surface
surface of a table top	notebook	 notebooks	
surface of an iron trunk	compass box		
surface of a book			

In the above activity, we have measured various surfaces using a variety of tools. Now try measuring the same surface with different tools.

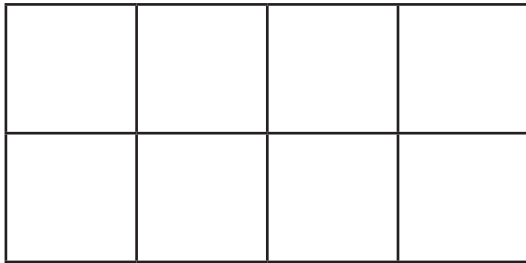
☉ **Measure the surface of the classroom table using different measuring tools.**

Name of the tool	Approximate Area	Actual number of such tools that cover up the surface
notebook	 notebooks	
textbook	 textbooks	
compass box	 compass boxes	

Every time we get different answers of measurement. Just as the standard unit for measuring length is decided, similarly the standard unit to measure area also needs to be decided.

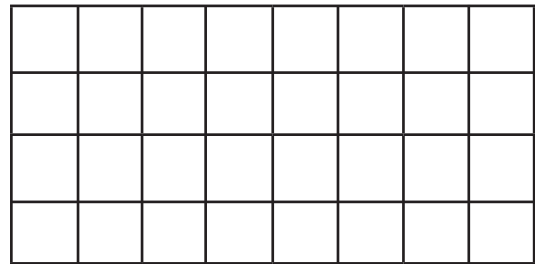
⦿ **Now we will measure the surface area of the table using a square.**

The teacher asked Anil and Aditi to cover the surface of the table in the classroom with squares of different sizes.



Area obtained after covering the surface of the table with squares used by Anil

Area = squares



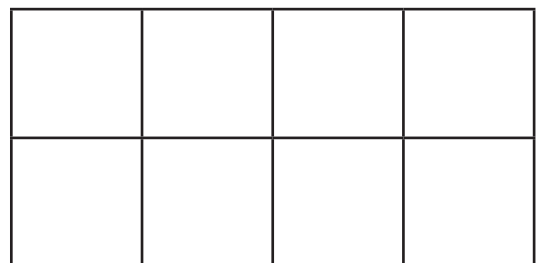
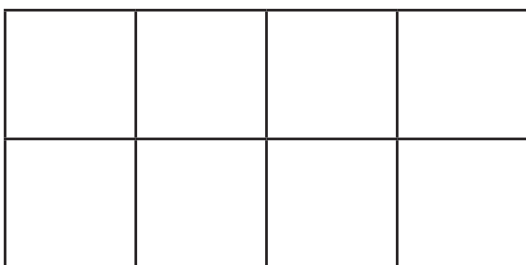
Area obtained after covering the table surface with squares used by Aditi

Area = squares

Both of them got different measurements when they measured the surface of the same table with squares of different sizes.

To ensure that everyone's measurement of area of any figure is the same, a square with side of 1 unit is used as a standard unit for measuring area.

⦿ **Now both of them measured the surface of the table using the same sized squares.**



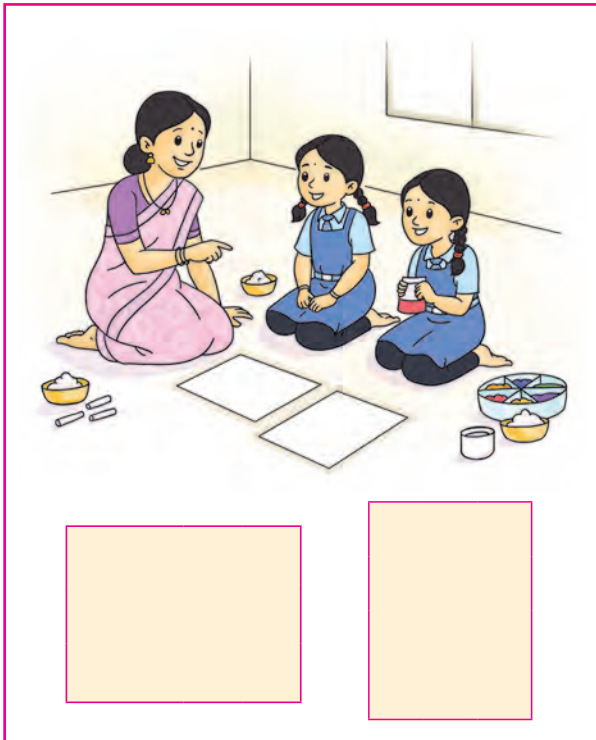
So both of them got the same surface area of the table.

⦿ **The square unit is the standard unit used to measure surface area.**

When measuring various surfaces, we can measure the area using squares with sides 1 cm, 1 metre, 1 kilometre or 1 foot as needed.

Neeta and Sayali participated on behalf of the class in the rangoli competition organised at school. They were asked to draw a rangoli at a designated place.

🕒 **Observe the picture below.**



Neeta: Teacher, the space allotted to me to draw the rangoli is less than the space allotted to Sayali.

Teacher: I don't think so.

Teacher: Come, let us measure the spaces allotted to both, using squares of same size.

Count the number of squares.

Tell the number of squares that cover the space allotted to

Neeta = squares.

Number of squares that cover the

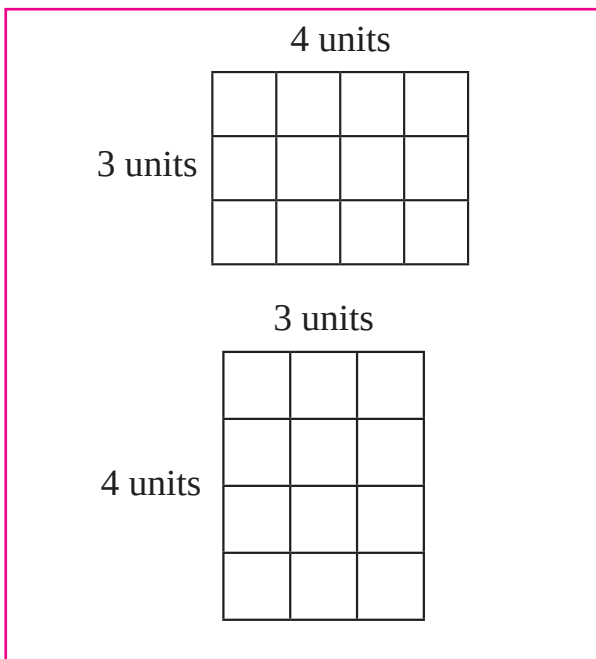
space allotted to Sayali = squares

What did you notice?

Sayali: The number of squares used to cover the spaces allotted to both is the same.

Neeta: That is, the area of the spaces for rangolis allotted to both is the same.

The area of spaces allotted to each of them is 12 square units.



For Teachers : Students attention should be brought to note that if the length and breadth of the surface is given in cm. then the area is expressed in square cm.

Exercise

(1) Observe and write the areas of the figures.

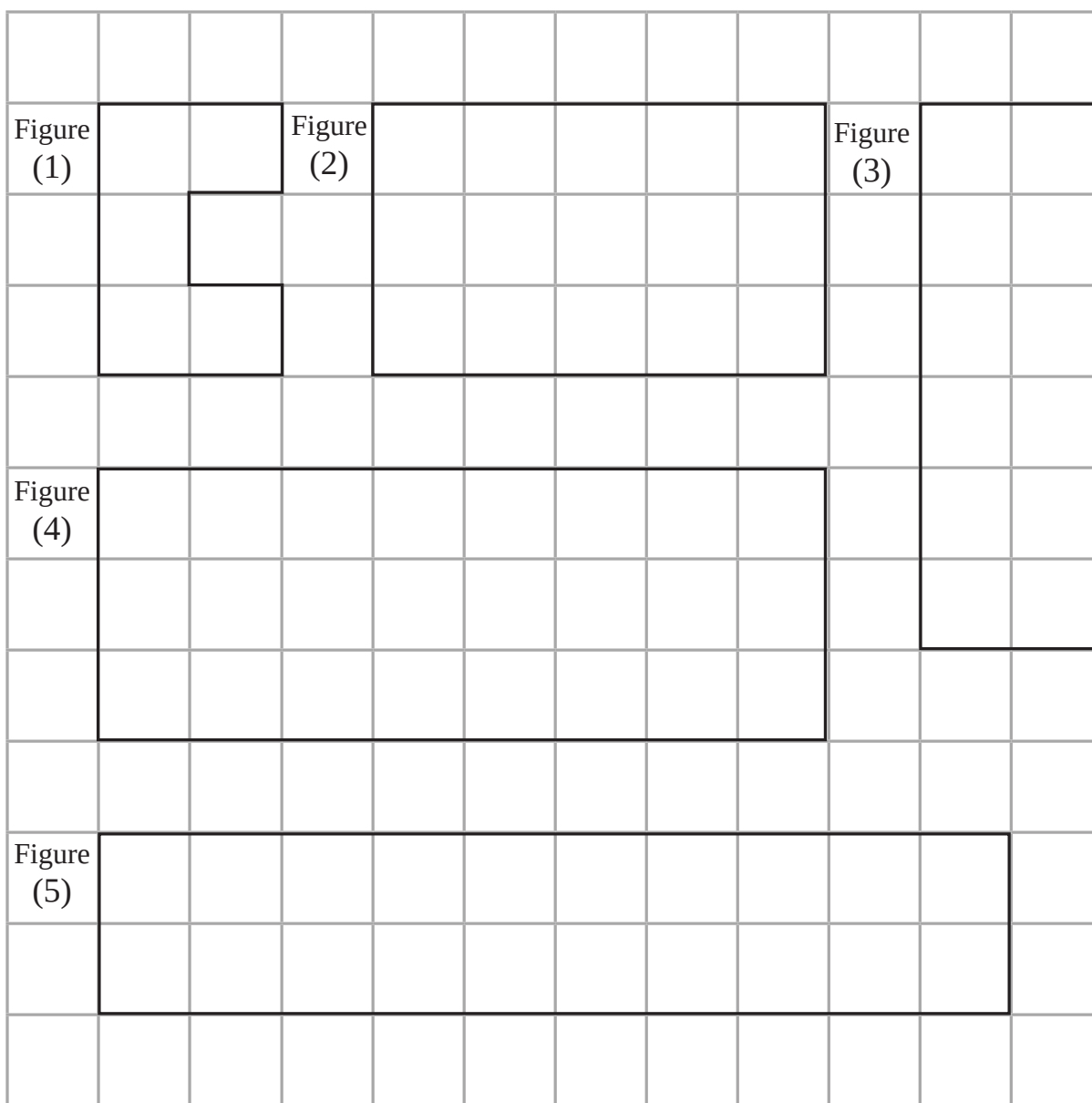


Figure (1) $\frac{5}{2}$ square units

Figure (2) 10 square units

Figure (3) 4 square units

Figure (4) 16 square units

Figure (5) 10 square units



For Teachers : Students should be asked to draw figures having same area but different perimeter and same perimeter but different area. Such practice should be given. Discuss in the class whether area of a triangle can be obtained using square units, demonstrate to find the area of triangle. Explain to students, why a triangle, circle, rectangle or irregular shapes cannot be used to measure the area occupied by the surface.

(2) Observe the figures given below and complete the table. Answer the questions that follow.

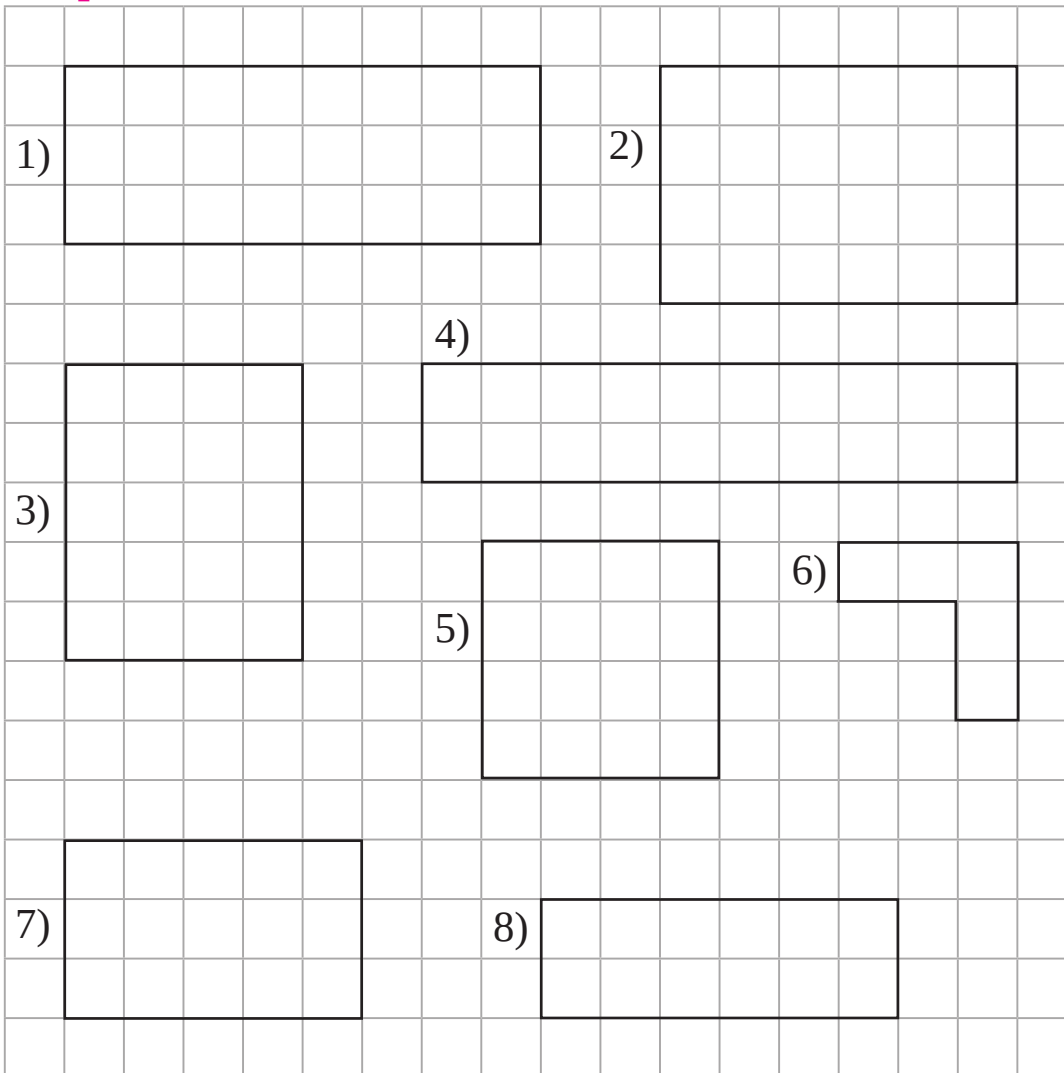


Figure No.	Perimeter	Area
1)	----- cm	----- square cm
2)	----- cm	----- square cm
3)	----- cm	----- square cm
4)	-----	-----
5)	-----	-----
6)	-----	-----
7)	-----	-----
8)	-----	-----

- Write the figures having same area but different perimeter.
- Which figures have same perimeter but different areas?.....
- Which figure has the same area and perimeter ?.....

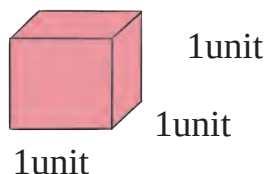


Let's Understand

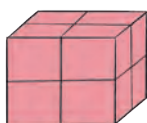
Volume

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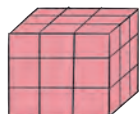
☉ Observe the cube.



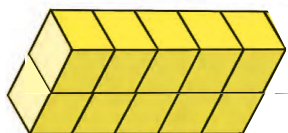
- This is a cube. Each side of it is 1 unit.
- This means that length, width and height of this cube is the same.
- The volume of this cube is 1 cubic unit.



- Here are 8 cubes of volume 1 cubic unit each.
- The volume of the resulting block is 8 cubic units.



- Here are 27 cubes of volume 1 cubic unit each.
- The volume of the resulting block is 27 cubic units.



- Here are 10 cubes of volume 1 cubic unit each.
- The volume of the resulting block is 10 cubic units.

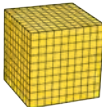
The volume of any 3D shape means the number of cubes of measure 1 unit, it holds.

The volume of solid is measured in cubic units.

Exercise

(1) Observe the following figures and fill in the blanks.

☉ Volume of  = cubic units

☉ Volume of  = cubic units

(2) Join together the cubes from the Maths Kit to make various structures and measure the volume of the resulting shapes.



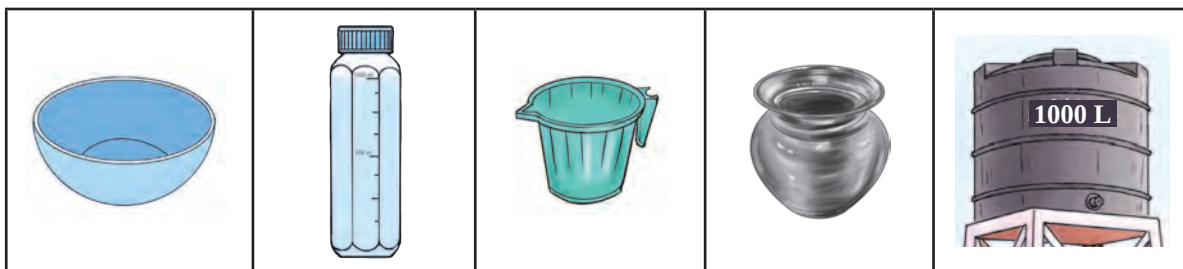
For Teachers : • The concept of volume should be explained using cubes and various cuboids from the Maths Kit. • Every animal requires to drink different amounts of water. Based on this fact the concept of water drinking capacity must be explained to students.



Let's Remember

The amount of water needed to fill a container upto the brim is the **capacity of the container**. The units used to measure capacity are litres and millilitres.

🕒 **Write the approximate capacity of the following objects.**



Water from a bottle when poured into any vessel, whichever vessel fills to the brim, has the same **volume** as that of the bottle. In the last picture, the capacity of the tank is 1000 litres. If 200 litres of water is poured into empty tank, then the volume of the water in the tank is 200 litres.

The volume of liquids is measured in litres.

In short, litres is written as (l) and millilitres as (ml)

Exercise

1) **Write the capacity of the objects listed in the table. For actual measurements you can use 500 ml, 200 ml, 100 ml, 50 ml, 10 ml.**

Object	Estimated capacity	Actual capacity	Difference between estimated and actual capacity
Glass			
Bowl			
Jug			
Mug			
Spoon			

🕒 **Read and understand.**

At the hospital, the doctor examined Vijay, and gave an injection  of 2 ml and liquid medicine  of 100 ml. From this he was advised to take a dose of 5 ml  twice a day.



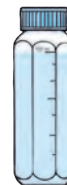
Let's Understand

◎ Conversion of Units of Measurement.

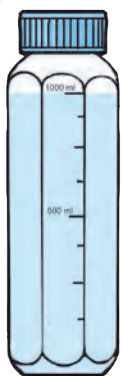
- 1) How many bottles of 500 ml are required to fill a 1 litre bottle with water completely?

$$500 \text{ ml} + 500 \text{ ml} = 1 \text{ litre.} \quad \text{So, } 500 \text{ ml} = \frac{1}{2} \text{ (half litre)}$$

$$\text{half litre} + \text{half litre} = 1 \text{ litre}$$



- 2) How many bottles of 250 ml are needed to fill a 1 litre bottle completely?



$$250 \text{ ml} = \frac{1}{4} \text{ (quarter) litre.}$$

$$250 \text{ ml} + 250 \text{ ml} + 250 \text{ ml} + 250 \text{ ml} = 1 \text{ litre}$$

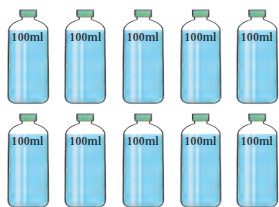
- 3) How many 250 ml bottles are needed to fill three quarters of a 1 litre bottle?

$$250 \text{ ml} + 250 \text{ ml} + 250 \text{ ml} = 750 \text{ ml}$$

$$750 \text{ ml} = \frac{3}{4} \text{ (3 quarters) litre.}$$



- 4) How many bottles of 100 ml are needed to fill 1 litre bottle completely?



$$100 \text{ ml} + 100 \text{ ml} + 100 \text{ ml} + 100 \text{ ml} + 100 \text{ ml} +$$

$$100 \text{ ml} + 100 \text{ ml} + 100 \text{ ml} + 100 \text{ ml} + 100 \text{ ml} = 1000 \text{ ml}$$

$$(10 \times 100 \text{ ml} = 1000 \text{ ml} = 1 \text{ litre})$$

◎ One and quarter litre = 1 litre + quarter litre = $1000 \text{ ml} + 250 \text{ ml} = 1250 \text{ ml}$

◎ One and a half litre = 1 litre + half litre = $1000 \text{ ml} + 500 \text{ ml} = 1500 \text{ ml}$

◎ One and three quarters of a litre = 1 litre + three quarters of a litre =
 $1000 \text{ ml} + 750 \text{ ml} = 1750 \text{ ml}$

⊙ Conversion of litres to millilitres ⊙ Conversion of millilitres to litre

- (a) 1 litre = $1 \times 1000 \text{ ml} = 1000 \text{ ml}$ $1000 \text{ ml} = 1 \text{ litre}$
 (b) 2 litres = $2 \times 1000 \text{ ml} = 2000 \text{ ml}$ $5000 \text{ ml} = 5 \text{ litres}$
 (c) 3 litres = $3 \times 1000 \text{ ml} = \boxed{} \text{ ml}$ $4000 \text{ ml} = \boxed{} \text{ litres}$
 (d) 4 litres = $4 \times \boxed{} \text{ ml} = \boxed{} \text{ ml}$ $\boxed{} \text{ ml} = 8 \text{ litres}$

1 litre = 1000 millilitres

⊙ **3450 ml** = $3000 \text{ ml} + 450 \text{ ml} = 3 \text{ litres} + 450 \text{ ml}$

Exercise

(1) Convert millilitres to litres.

- 1) $6000 \text{ ml} = \boxed{}$
 2) $7000 \text{ ml} = \boxed{}$

(2) Fill in the boxes below.

- ⊙ $3250 \text{ ml} = \boxed{3} \text{ l} + \boxed{250} \text{ ml}$
 ⊙ $4750 \text{ ml} = \boxed{} \text{ l} + \boxed{} \text{ ml}$
 ⊙ $3025 \text{ ml} = \boxed{} \text{ l} + \boxed{} \text{ ml}$
 ⊙ $100 \text{ ml} = \boxed{50} \text{ ml} + \boxed{} \text{ ml}$
 ⊙ $100 \text{ ml} = \boxed{} \text{ ml} + \boxed{} \text{ ml}$

(3) Fill in the boxes below.

- ⊙ How many millilitres in two and a quarter litres? $\boxed{} \text{ ml}$
 ⊙ How many millilitres in two and three fourth litres? $\boxed{} \text{ ml}$
 ⊙ How many millilitres in three and a quarter litres? $\boxed{} \text{ ml}$
 ⊙ How many millilitres in three and a half litres? $\boxed{} \text{ ml}$



For Teachers : Students should actually perform the above activity. Teachers should explain the concepts of three-fourths, one and one-fourth and one and a half.

(4) Mother poured oil in the pan to fry puris. After frying, 2 ml of oil is left on each puri.

- ⦿ If you ate five such puris, how much oil did you consume?
- ⦿ How much oil will be left on 30 such puris after mother fries them?

(5) Perform the following activity and record your observations.



Picture 1



Picture 2



Picture 3



Picture 4

How many such  of water will you have to pour to fill every vessel to its brim in the above pictures?

- ⦿ How many cups of water is needed to fill the vessel in picture 1?
- ⦿ How many cups of water is needed to fill the vessel in picture 2?
- ⦿ How many cups of water is needed to fill the vessel in picture 3?
- ⦿ How many cups of water is needed to fill the vessel in picture 4?
- ⦿ Which vessel from picture 3 or 4 can hold more water?

(6) Observe and write the correct answers in the box.

Capacity	How many measures of 1000 ml?	How many measures of 500 ml?	How many measures of 200 ml?	How many measures of 100 ml?	How many measures of 50 ml?
For 1 litre	01	02			
For 2 litres					
For 3 litres					

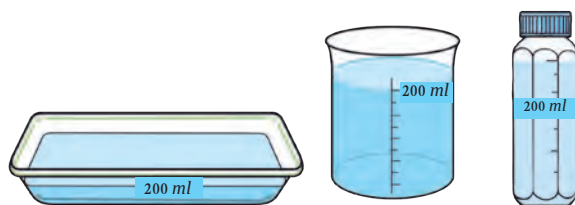


For Teachers : Students should be asked to visit their nearest shop with their parents. Also ask the students to classify the items into 10 ml, 100 ml, 250 ml, 500 ml and 1 litre capacities.



Do the Activity

- ⊙ Measure the water from the glass tray and glass jar by pouring it into a 200 ml bottle.



Even if the water in the glass tray appears to be less and water in the glass jar appears to be more, the water in both the vessels is the same when measured using a measuring bottle.

The volume of the liquid does not change even if the container is changed.



Fun with Maths

- 1) In a garden, there are two empty vessels, one of 5 litre capacity and other is of 3 litre capacity. Also, there is a tank filled with water. If you want 4 litres of water from it, how will you get it?
- 2) A barrel is filled with 15 litres of cooking oil. Measuring jars of 2 litres and 1 litre capacity are kept along side. Also there are three empty cans. You need to fill 5 litres of oil in every can. What is the least number of times you will use the measuring jars?
- 3) A milk seller has 12 litres of milk in a can. He has empty vessels of 3 litre, 5 litre and 7 litre capacity. Help the milk seller to distribute 4 litres of milk equally among 3 people using these vessels.



Project

List the places from your surroundings where water gets wasted. Estimate how much water is wasted and suggest solutions.

Make the following table in your notebook.

Sr. No.	Place	Approximately how many litres of water is wasted	Solution
---------	-------	--	----------



For Teachers : Explain the mathematics in the above activities by practically performing them.



Let's Observe



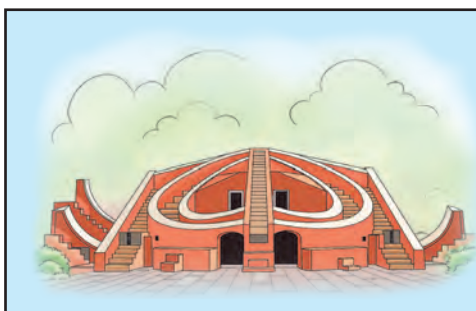
Digital clock



Analog clock

- A clock that displays time using only digits is called a **'digital clock.'**
- A clock that uses hands to show time in hours and minutes is an **'analog clock.'**

You can see these types of clocks everywhere. In ancient times what kind of devices would have been used to tell time? Gather information from friends, parents, teachers and the internet such as:



In the 18th century, the Rajput King Sawai Jai Singh of Jaipur, in Rajasthan built the 'Bruhat Samrat Yantra'. It is a huge sundial, that uses the shadow cast by the sun to indicate accurate time.

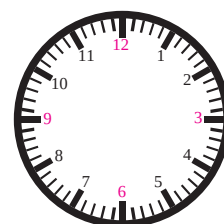
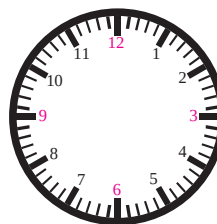
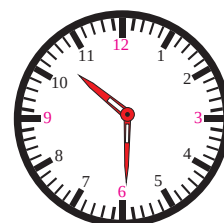
Measuring time.

Pawan left for school at 15 minutes past 10 in the morning. This is written as 10:15. It took him 15 min to reach school. If 15 minutes are added to 10:15, Pawan reached school at 10:30.

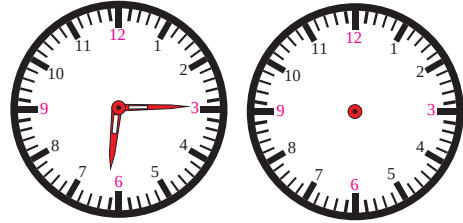


Do the Activity

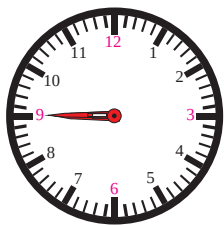
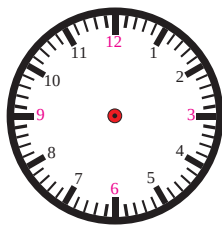
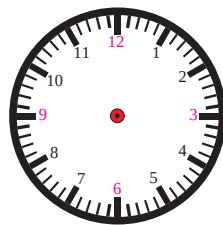
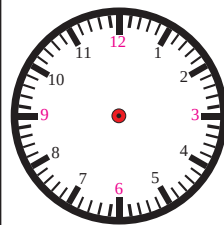
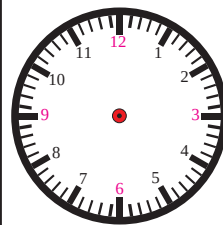
- 1) Everyday Naman starts doing Yoga at 6:10 and stops at 6:40 in the morning. Show both these timings of doing Yoga by drawing hands on the clock.



- 2) Shrawani started reading a story from her book at 6:15 in the evening. She finished reading the story in 45 minutes. Draw clock hands to show the time on the clock.

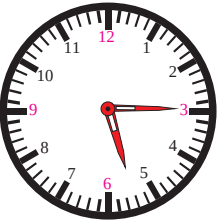


- 3) Raju leaves home at 9:45 in the morning to go to school. Show the time on the clock for his various activities.

After 15 minutes		After 20 minutes		After 40 minutes		After 35 minutes	
							
Raju left 9:45 for school in the morning	He reached school at this time 10:00	 The School Assembly	 First period in school	 Second period in school starts			

Exercise

- (1) Three friends read the time on the clock. Who told the accurate time? What mistakes did others make while reading the time? Discuss with your friends.

Friends reading the time on the clock	Time shown on the clock		
			
Sachin	12:03	2:05	5:03
Neeta	12:15	5:10	3:05
Ojas	3:00	2:25	5:15

- (2) Riya said that 2 hours 30 minutes is more than 150 minutes, is she correct?



For Teachers : Practice of showing different times using clock from the Maths Kit should be taken. Also give practice of solving puzzles and examples.

- (3) Ganesh studied for 2 hours. If he finished studying at 1:00 in the afternoon, at what time did he start studying?
- (4) Record the number of times the minute hand crosses over the hour hand in 12 hours by actually moving the hands of the clock.

Great Indian Mathematician - Shrinivas Ramanujan



Shrinivas Ramanujan was born on 22 December 1887 at Erode in Tamil Nadu State. His father was employed in a cloth shop. The family conditions were very poor. When little Shrinivas was sent to school at the age of five, his brilliant intellect became evident. He completed his primary education in just two years and at the age of seven he joined the Town High School in Kumbhakonam. He had great interest in Mathematics since childhood. His memory was also very sharp. He could verbally state the values of the square root of 2 and π the irrational number upto any decimal place. While still in school, he calculated the length of the Earth's equator using mathematics. It was a surprise that the length he calculated was almost accurate. At the age of sixteen he passed the High School examination. He got a scholarship.

He joined a college for further education, but since he was not interested in any subject other than mathematics, he neglected other subjects like English. The result was as expected, he failed, scholarship was discontinued. Poverty forced him to stop his education. Now, in order to stand on his own feet and help his parents by reducing their hardships, he started working as a treasury clerk at the Madras Port Trust. He spent 10-12 hours a day at work. The rest of his free time was spent studying mathematics. Math was like his breath. His mathematical genius found a suitable direction, The authorities tried to ensure that his research reached renowned mathematicians. Some of Ramanujan's research writings reached Professor Hardy a mathematician at Cambridge University in England.

Prof. Hardy was very impressed by his writings. Ramanujan is a precious cut diamond. He realised this and immediately responded to the University of Madras. The University made a provision to give his family an amount sufficient to support their livelihood and to cover Ramanujan's expenses to England. He was invited to Cambridge University for further studies. He reached Cambridge University in 1914. Under the guidance of Prof. Hardy, he started studying with single-minded concentration. Within two years, many of the mathematical theories he had proven were recognised world wide. In recognition of his research work, the Royal Society of London awarded him a fellowship. He was the first Indian to receive this honour and at that time he was only 31 years old. But later he fell ill due to unsuitable climate of England and got treated at the Cambridge Nursing Home. Despite his illness, his mathematical studies did not stop. Properties of numbers were at the tip of his tongue. Once Prof. Hardy went to meet him. His taxi number was 1729. Prof Hardy said to Ramanujan, "1729 seems like a very common number, doesn't it?" Ramanujan replied in a moment, Ah! This is a special number. It can be written in two ways as the sum of two numbers 10^3+9^3 . and 12^3+1^3 . Hearing this Prof. Hardy was surprised and he was convinced of the statement once made by Prof. Littlewood, "Every whole number is Ramanujan's best friend." Unfortunately Ramanujan's illness continued to worsen longing for home and motherland, he returned to India in February 1919. Finally he passed away on April 27, 1920.

Prof. Hardy had said about his student, "I could teach him a little but I could learn a lot from him."



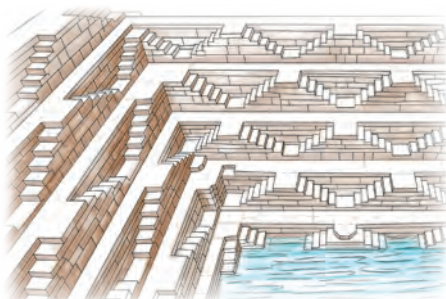
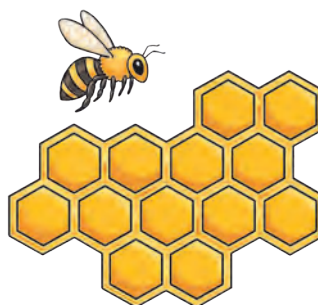
Let's Remember

In the previous standards, we have studied growing patterns formed by using **two-dimensional** shapes. Now let us understand the growing patterns formed by using **three-dimensional** shapes.

Patterns found in three-dimensional shapes



Let's Understand



After observing the above pictures, we notice that the pineapple, the honeycomb, the stepwell in Jaipur, the lantern, the house of cards, the peacock feathers, all have some specific styles in which repeating patterns are seen. Look out for other patterns like above.

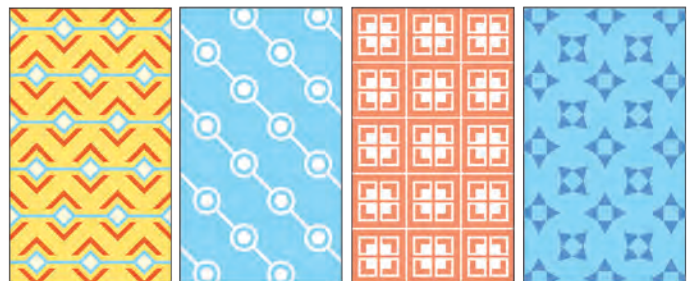
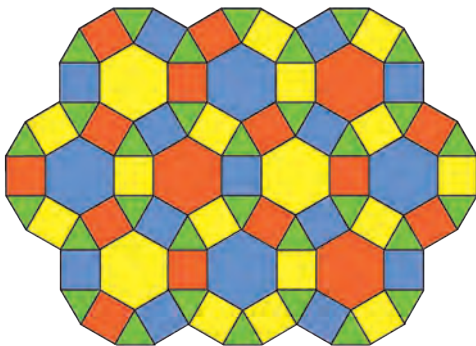
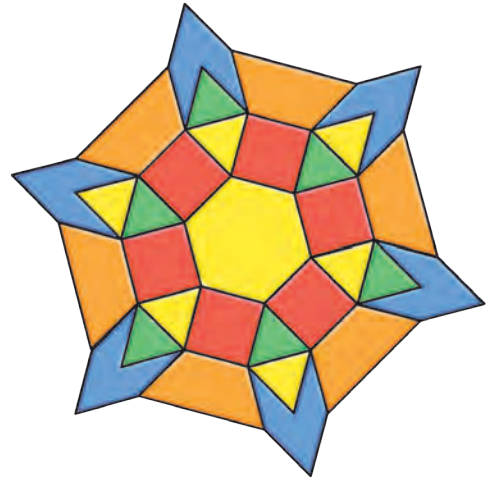
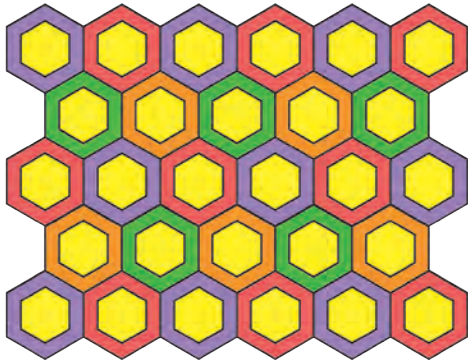
Patterns that grow with the help of objects and pictures



Let's Understand

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Growing patterns can be created by using the objects and pictures. Following are such examples.



Which geometric shapes have been used in the above designs ?



Do the Activity

- Dip the slices of lady's finger in paint and make beautiful designs on paper.

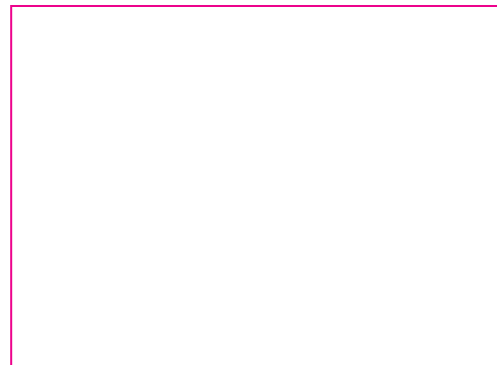
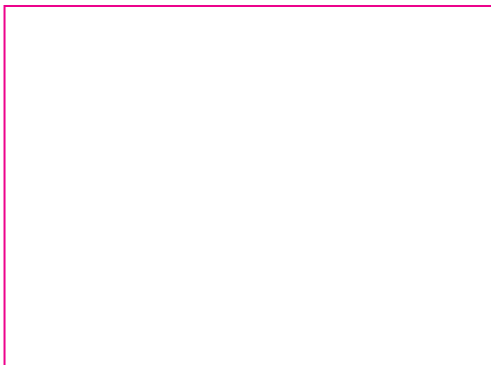
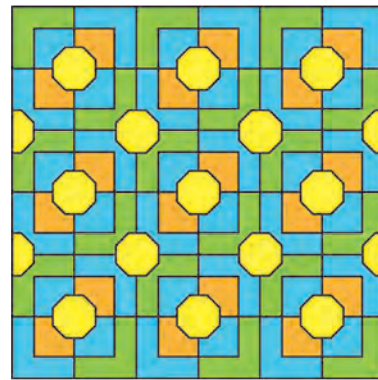
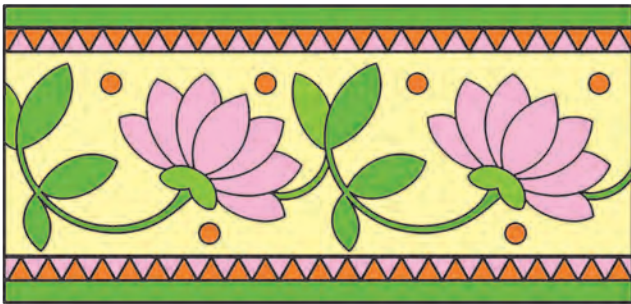
Designs : Creating designs using different shapes



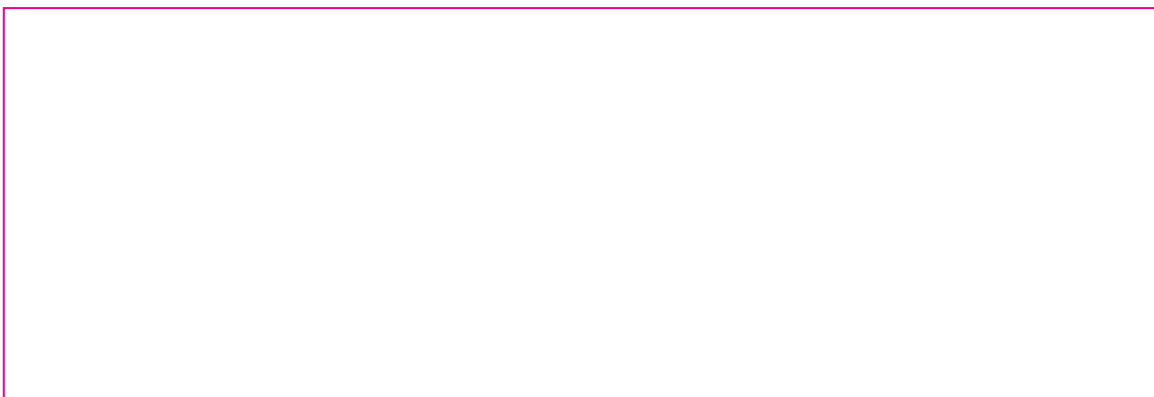
Let's Understand

A design is an attractive pattern made using various regular and irregular shapes.

For example, here are some designs of leaves and flowers made using geometric shapes on the pallu of mother's saree and on the walls of the house. Look for such motifs in you surroundings and make similar designs.



◎ Create a pattern using geometric shapes in the given frame.



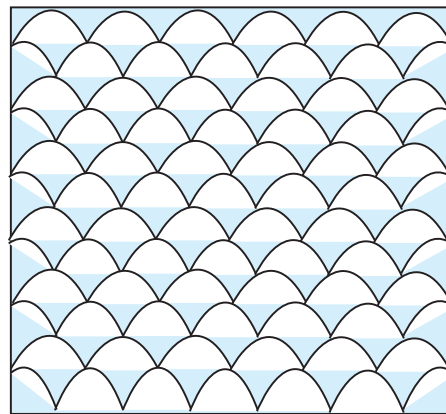
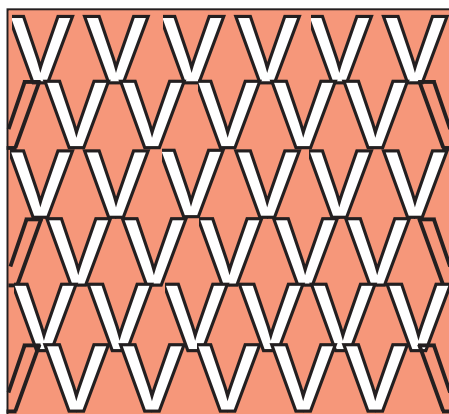
For Teachers : Get various designs created from students using a variety of objects, shapes and stamps.

Tiling



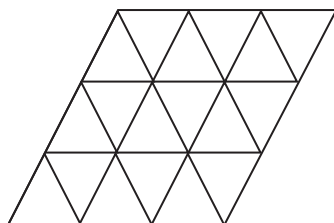
Let's Understand

In this, pieces of tiles of different shapes are arranged one after the other in such a way that there are no gaps or empty spaces left between them and the tiles do not overlap each other. Understand this from the following example.

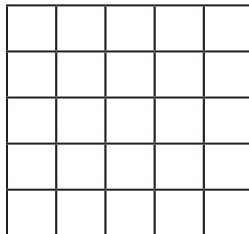


☉ Do the tiling by filling colours of your choice in the following:

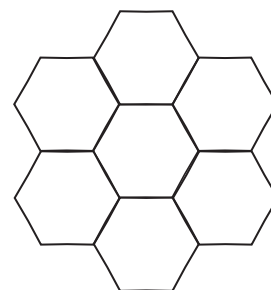
A



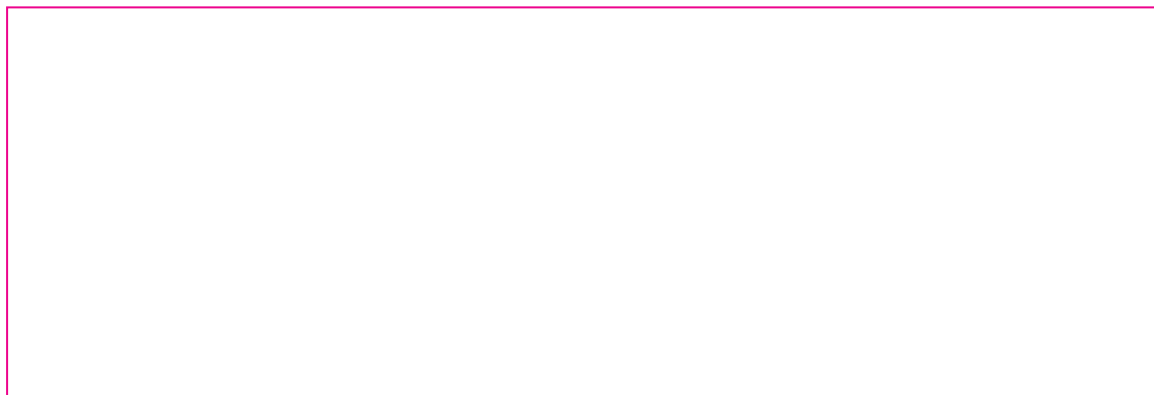
B



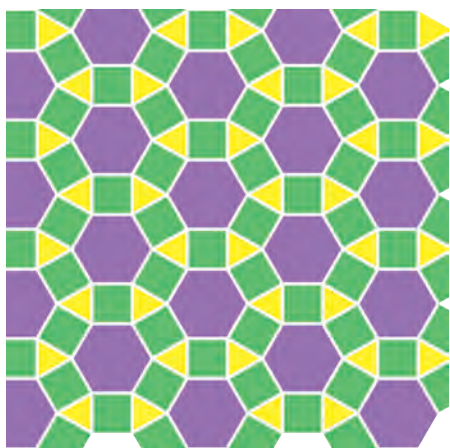
C



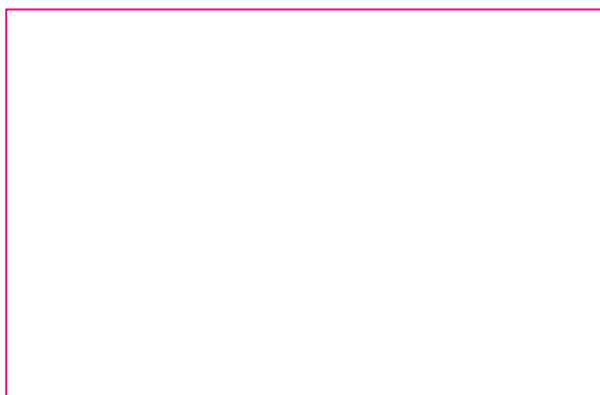
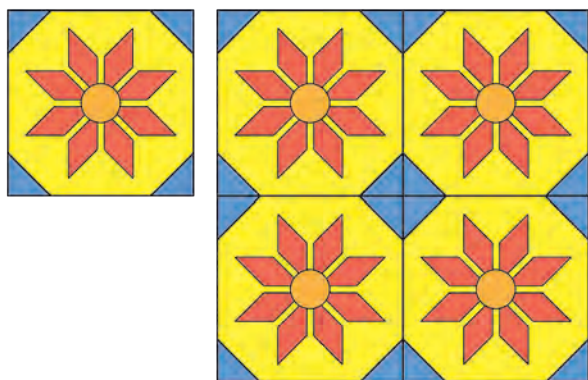
☉ Do the tiling in the area given below using two different shapes.



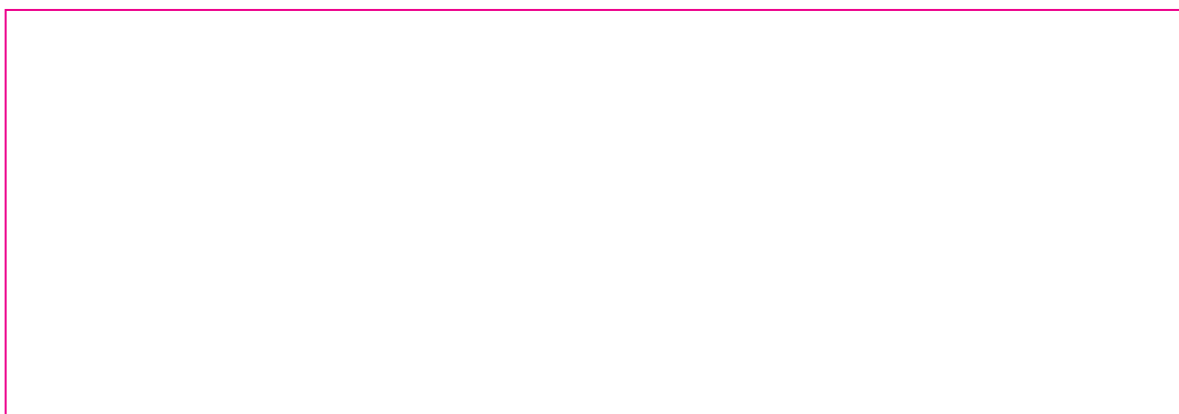
☉ Do the tiling in the adjoining area as shown in the figure below.



☉ Do the tiling in the area as shown in the figure below.



☉ Do the tiling using triangles in the area given below.



Number Patterns



Let's Understand

☉ Complete the pattern.

1)	Pattern of Counting Numbers
2)	Pattern of Triangular Numbers
3)	Pattern of Even Numbers
4)	Pattern of Odd Numbers

☉ Observe the following 'times tables' and make a pattern for the digits in the units place of all the numbers that occur in pattern.

1) 2, 4, 6, 8, 10, 12, 14, 16, 18, 20

--	--	--	--	--	--	--	--	--	--

2) 4, 8, 12, 16, 20, 24, 28, 32, 36, 40

--	--	--	--	--	--	--	--	--	--



For Teachers : Concept of patterns in odd-even numbers should be explained by pairing objects, pictures, dots.

- 3) Write the 8 times table in your notebook and then use it to write the pattern of digits in the unit's place.

--	--	--	--	--	--	--	--	--	--

- 4) Write the 3 times table in the notebook and use it to write the pattern of digits in the unit's place.

--	--	--	--	--	--	--	--	--	--	--	--

- 5) Write the 7 times table in the notebook and use it to write a pattern of digits in the unit's place

--	--	--	--	--	--	--	--	--	--	--	--	--



Let's Think

- Are all the numbers in the 3 times table odd?
- Asmita has prepared the following number series. Identify whether they are of odd numbers or even numbers.

3	5	7	9	11	13	15	17	Odd numbers
---	---	---	---	----	----	----	----	-------------

4	8	12	16	20	24	28	32	
---	---	----	----	----	----	----	----	--

3	13	23	33	43	53	63	73	
---	----	----	----	----	----	----	----	--

10	22	34	46	58	70	82	94	
----	----	----	----	----	----	----	----	--



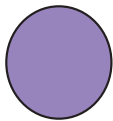
For Teachers : Make pairs of two students. They should be given different number strips. Practice of identifying patterns should be taken. One child can call out a number and other can identify the number as odd or even.



Let's Solve

- ⦿ Make a grid of numbers from 301 to 400. Colour the boxes with even numbers in green and odd numbers in blue.

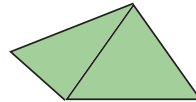
- ⦿ Using the figures below, form a pattern according to style AB, ABB, ABC, DEFG, HGED,



A



B



C



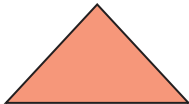
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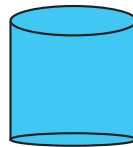
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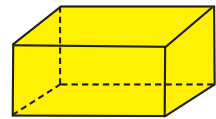
E



F



G



H

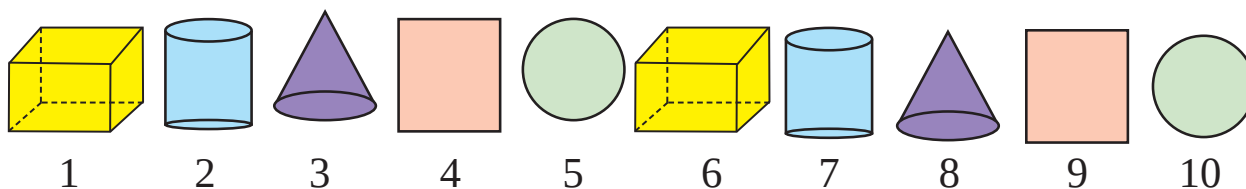
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- ☉ In the following pattern, every shape has been placed in the given order. They are numbered as 1, 2, 3, 4, ...



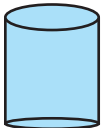
- Observe the above shapes and their order and form the following pattern.
- For  the pattern is 4, 9, ... Complete this  pattern.

4	9				
---	---	--	--	--	--

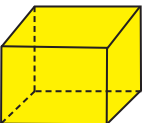
- Write the pattern for this shape 

--	--	--	--	--	--

- Observe the above order and form a pattern of shapes that come in the given order 2, 4, 6, 8, 10, 12, 14.

					
---	--	--	--	--	--

- Form a pattern of shapes that come at positions 1, 3, 5, 7, 9, 11, 15

					
---	---	--	--	--	--

-  Write the pattern for the position of this shape

--	--	--	--	--	--



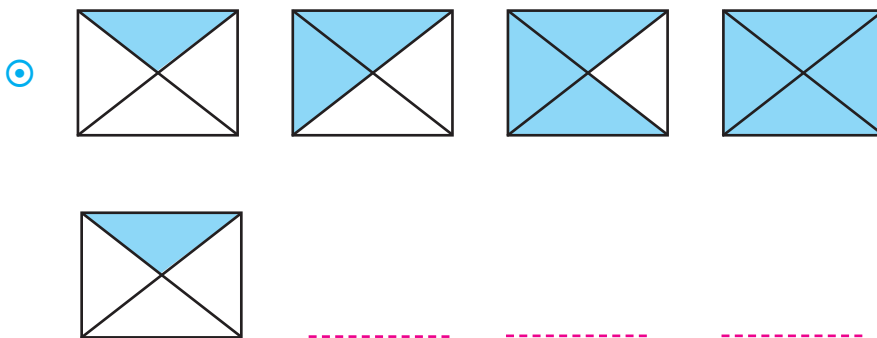
For Teachers : Explain concept of pattern formation using various geometric shapes in the Maths Kit. Students should create patterns using geometric shapes. Activity of numbering the shapes and creating patterns of sequences should be performed. Students should be encouraged to ask questions based on the topic.

Exercise

(1) Solve.

- ⦿ Can you create patterns of only odd numbers? If you can, create one.
- ⦿ If a number pattern starts with an even number and increases by 2 every time, will you get an odd number ? Justify.
- ⦿ Using the first 10 counting numbers, form the patterns of odd and even numbers.
- ⦿ Which type of number pattern is this 3, 6, 9, 12, ...?

(2) Complete the following patterns.



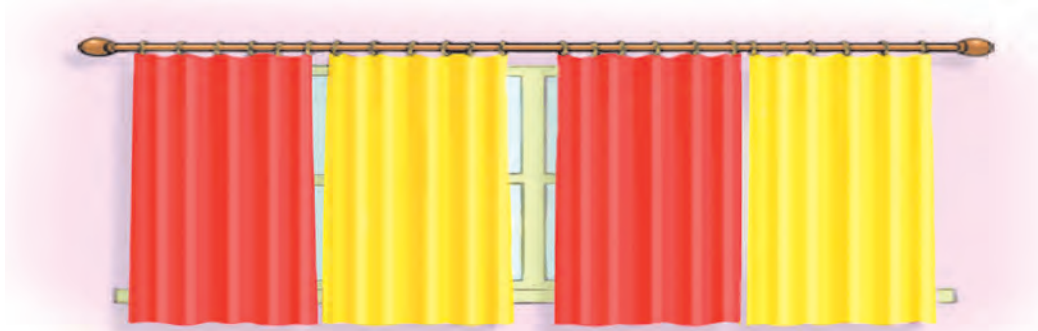


Let's Understand

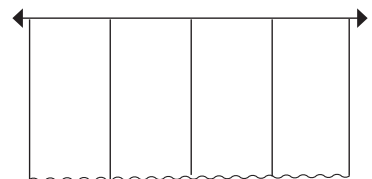
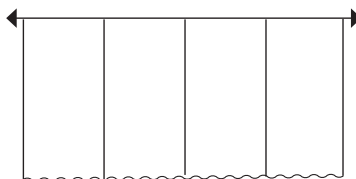
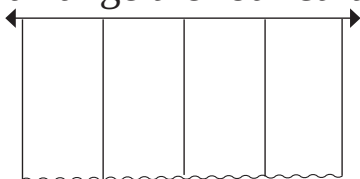


The shopkeeper showed mother some curtains of different colours. She wants to buy curtains of two colours. Think and tell about which colours she can buy.

Mother bought two red and two yellow curtains and hung them on the large window of the house in the following way.



Colour the following picture and show in which other ways can you arrange the four curtains.



What other arrangements can you make for these curtains? Discuss with your friends. Draw the picture in your notebook and colour it.

- A girl wants to wear 2 red and 2 green bangles on one hand, brought by her mother. She has worn them as shown below. Try out other ways in which she can wear them.



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- For a school program, it was decided to set up a drill using dumbbells, lezim and rings. Discuss the different ways in which students can be lined up on the ground for the drill.



with dumbbells



with lezim



with rings

Possibility of various arrangements in our surroundings.

- Neeta bagged three different prizes for three games. How can she arrange those prizes in her cupboard at home? Make a list of all the possible arrangements.
- Can Neeta arrange the three prizes in her cupboard in any other way than shown below? Discuss.



1



2



3



4

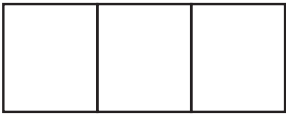


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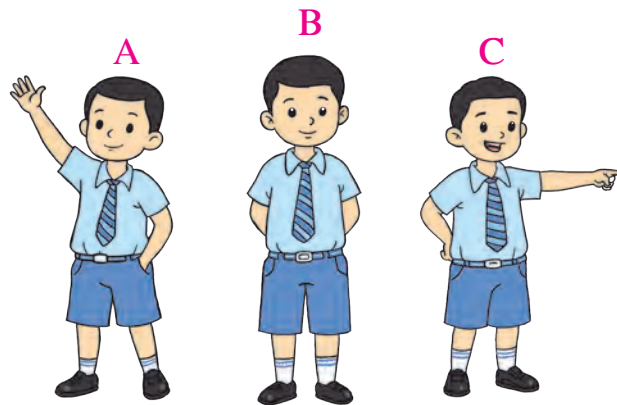
6

- Form various rangoli patterns in your notebook by connecting three dots at a time in a continuous straight line.
- Use the colours yellow, red and green only once to colour the wall shown below in different ways.



- Think about what are the other different ways in which the wall can be painted.

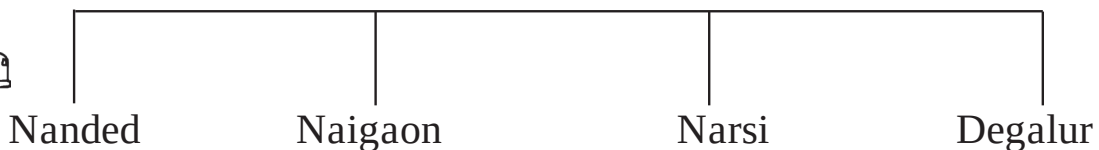
Exercise



- (1) Think of all the possible ways in which Vihaan, Suraj and Sam can be arranged at the given positions A, B, C and write all the arrangements using the given letters

.....

- (2) While travelling from Nanded to Degalur, two main stops are made at Naigaon and Narsi. If a express bus travelling from Nanded to Degalur, halts only at Naigaon and Narsi, how many varieties of tickets will have to be made available for this bus?

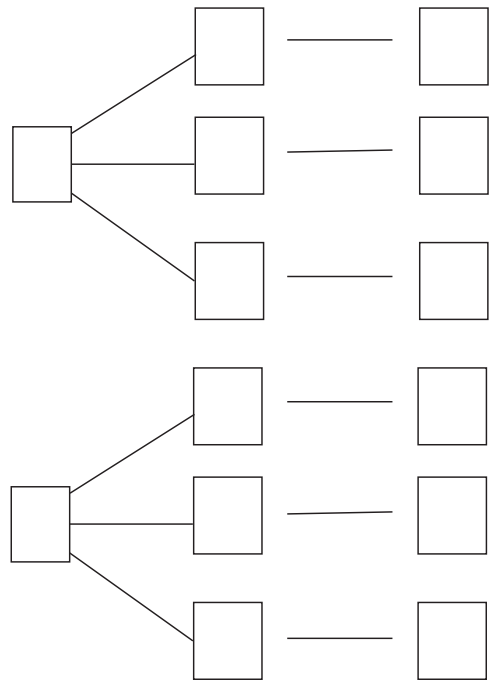
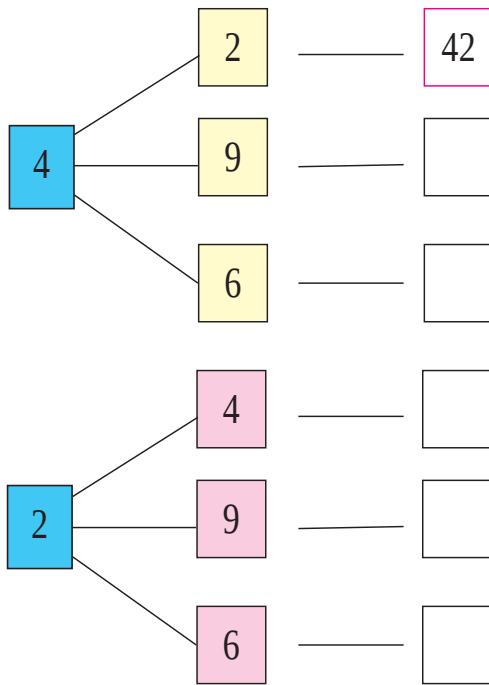


How many types of tickets will need to be made available for return journey as well?

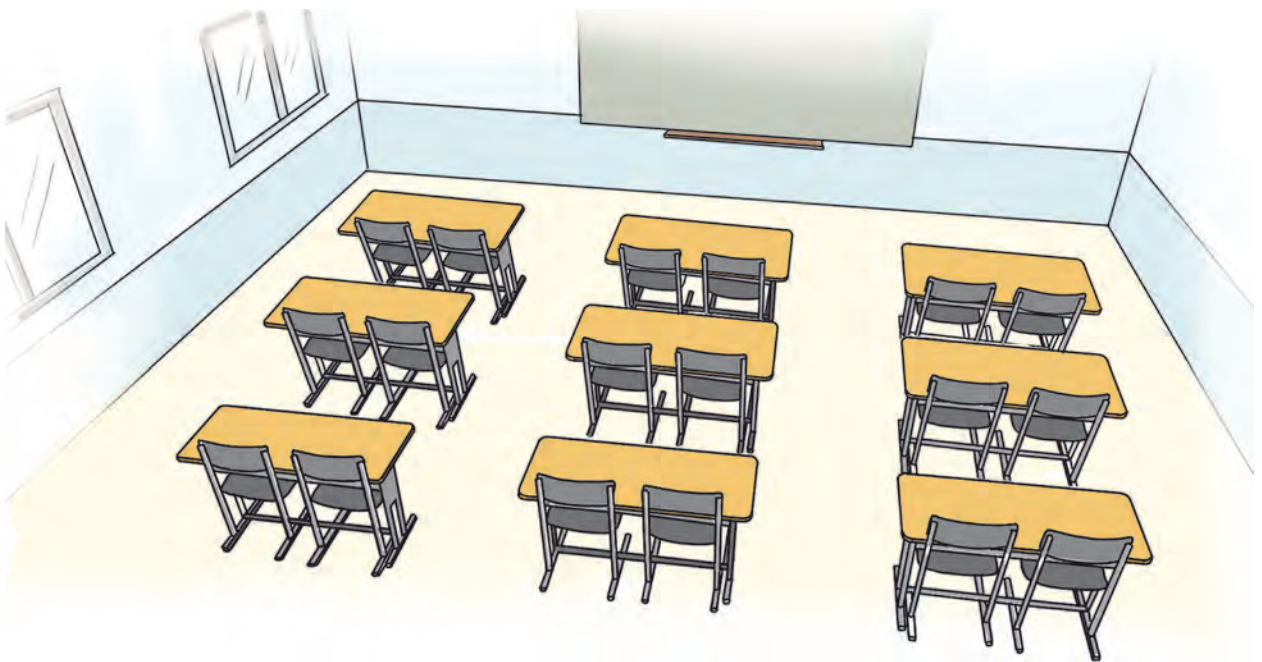
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(3) How many two-digit numbers can be formed using the digits 4, 2, 9, 6 only once? Think and write.



(4) In the classroom, every vertical row has three desks. The desks need to be tagged with stickers A, B, C in different sequences. How can the teacher tag them? Show the different ways in the picture that follows :



Combinations

Formed based on the given conditions



Let's Understand

- The 'Asia Hockey Cup' is organised into two groups (Pools) 'A' and 'B.' Pool 'A' includes the countries India, China, Japan and Kazakhstan. So, observe how many matches the countries in pool 'A' will have to play to reach the semi-finals.

Asia Hockey Cup	Pool 'A'
	India
	China
	Japan
	Kazakhstan

- | | |
|----------|------------|
| 1) India | China |
| 2) India | Japan |
| 3) India | Kazakhstan |
| 4) China | Japan |
| 5) China | Kazakhstan |
| 6) Japan | Kazakhstan |

Exercise

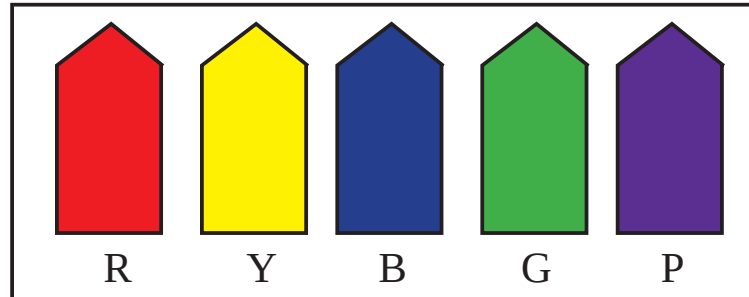
- (1) Three school students decided to play ring game together. The school has rings of four different colours red, green, yellow and blue. Write down all possible combinations of three coloured rings that the three students can choose from a set of four rings.



- (2) Five students, Rama(R), Gautam (G), Ishita (I) , Sameer (S), Jacob(J) have won the school council elections. Two of them have to be selected so that one can be made the Head and the other the

Vice Head. Using the given letters write all the possible names of two students that you can select.

- (3) Arya decided to colour the picture she had drawn using only two colours. Write down all the pairs of colours that she can choose out of 5, to colour the picture.



- (4) Mother bought some fruits like pineapple, oranges, apples, pomegranates and grapes from the market. Jane said to mother, "Mamma, today I want mixed fruit juice. Choose any two types of fruits." Mother started thinking about which two fruits to make the juice from. Help mother by suggesting different combinations of two fruits.
- (5) On the occasion of National Nutrition Week, mother packed Sonali's tiffin with salad, moong sprouts, rajgira laddoo, pulao and thalipeeth. But Sonali said, "Today I will eat only three items." Write down all the combinations of items that she can choose to eat.
- (6) A farmer is planning to plant any two crops from wheat, gram, jowar and maize in his field during the Rabi season. Write all the possible pairs of crops that he can plant in the field.
- (7) Salma went to an ice cream parlour with her father. There were three flavours of ice creams at the shop - pistachio, mango and strawberry. The ice cream vendor said, "Dear, today we have the 'Buy One Get One Free Offer'. You can also have two different ice cream flavours together." Salma thinks... should I take strawberry and mango or pistachio and strawberry? In how many ways can she choose the ice cream? Write down these pairs.

(8) Daddy has forgotten the password for his computer, but he only remembers that his password contains the digits 1 and 2 and the symbols @, # which he has used only once. If the password starts with the symbol @ then try which passwords he can enter.

1	2	@	#
@	1	2	#



Fun with Maths

Find and write the number that will fill the blank.

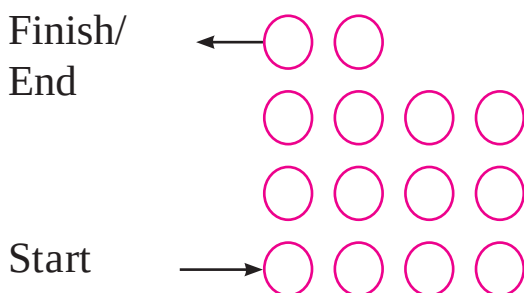
Find the missing number in the grid below.

7	6	9
8	4	5
15	10	

Find the missing number in the grid below.

2		6	15
	5		15
4		8	15
15	15	15	

From the start to reach the end, you need to go by the straight path. The condition is that you cannot reverse your path or follow a diagonal path. No circle should be skipped. Try it.



There are five guests on a dias. When they got on the dias, they greeted each other by shaking hands. How many handshakes were there in total ? Try out this activity with your friends to find out.

How many different combinations of notes can be created using the four notes of music, Sa, Re, Ga, Ma only once ? Prepare a table of all these combinations.

Hello Children!

If you feel uneasy about someone touching you inappropriately, you should not keep quiet. You must :

1. Not blame yourself.
2. Tell someone whom you trust.
3. You can also inform National Commission for Protection of Child Rights through the **POCSO e-box**.

When you get an unsafe touch, you may feel bad,
confused and helpless.

You need not feel “bad” because it’s not your fault.



Press This Button

POCSO e-box available at ncpcr@gov.in

If you are below 18 years of age, and are troubled or confused
or abused or in distress or know some other child who is...

*Call 1098...because some numbers are good!
They change lives!!!*



CHILDLINE 1098 - a national 24 hours toll free
emergency phone service for children in distress is an
initiative of CHILDLINE India Foundation supported
by Ministry of Women & Child Development.



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